Observability (OBS)

Operations & Reliability

- Observability (OBS)
- Analytics & Telemetry (ANL)
- Logging & Tracing Standards (LTS)
- Monitoring & Alerting (ALRT)
- SLOs/SLAs & Error Budgets (SLO)
- Performance & Scalability (PERF)
- Load/Stress Planning (LOAD)
- Capacity & Cost Modeling (COST)
- Performance Budgets & Profiling (PBP)
- Quality & Testing (QA)

Operations & Reliability — Subcategories + Template Set (Locked v1)

Observability (OBS)
OBS-01 Observability Overview (scope, principles, toolchain)
OBS-02 Telemetry Schema Standard (common fields, correlation IDs)
OBS-03 Metrics Catalog (by metric_id)
OBS-04 Trace Instrumentation Standard (spans, propagation)
OBS-05 Log Event Catalog (by log_event_id)
OBS-06 Redaction & Privacy Rules for Telemetry (PII/secret handling)
OBS-07 Dashboards Inventory (by dashboard_id)
OBS-08 Alert Routing & Ownership (teams, oncall, escalation)
OBS-09 Sampling & Cardinality Policy (limits, hashing, budgets)
OBS-10 Observability Gates (required signals per surface)

Analytics & Telemetry (ANL)
ANL-01 Analytics Overview (goals, constraints)
ANL-02 Event Taxonomy (by event_name)
ANL-03 Event Schema Spec (payload rules, versioning)
ANL-04 Identity/Attribution Model (anon/user/tenant)
ANL-05 Consent & Opt-Out Enforcement (ties to PRIV-04)
ANL-06 Data Quality Rules (completeness, duplication)
ANL-07 Funnels & KPIs Definition (by kpi_id)
ANL-08 Experimentation Support (flags/assignments)
ANL-09 Retention & Deletion for Analytics (ties to PRIV-05)
ANL-10 Analytics Access Controls (who can query/export)

Logging & Tracing Standards (LTS)
LTS-01 Logging Standard (levels, structure, fields)
LTS-02 Tracing Standard (trace/span IDs, propagation)
LTS-03 Error Logging Standard (error shapes, stack rules)

LTS-04 PII/Secret Redaction Standard (denylist, masking)
LTS-05 Log Routing & Storage (sinks, retention classes)
LTS-06 Client Logging Standard (mobile/web constraints)
LTS-07 Audit vs App Logs Boundary (what goes where)
LTS-08 Log Volume Controls (sampling, rate limits)
LTS-09 Logging Verification (tests, scanners)
LTS-10 Logging Runbooks (common issues)

Monitoring & Alerting (ALRT)
ALRT-01 Alerting Overview (principles, severity model)
ALRT-02 Alert Catalog (by alert_id)
ALRT-03 Alert Rule Spec (condition, threshold, window)
ALRT-04 Oncall & Escalation Policy (rota, paging)
ALRT-05 Noise Reduction Rules (dedupe, suppression)
ALRT-06 Incident Triggers (alert → IRP runbook)
ALRT-07 Customer Impact Rules (status page, comms)
ALRT-08 Alert Testing & Drills (synthetics, fire drills)
ALRT-09 Alert Dashboards (what to watch)
ALRT-10 Alert Change Control (who can edit rules)

SLOs/SLAs & Error Budgets (SLO)
SLO-01 SLO/SLI Overview (definitions, scope)
SLO-02 Service SLO Catalog (by service_id)
SLO-03 Endpoint SLO Catalog (by endpoint_id)
SLO-04 Error Budget Policy (burn rates, actions)
SLO-05 SLA Commitments (external, contracts)
SLO-06 Measurement Rules (windows, aggregation)
SLO-07 Alerting on Burn (ties to ALRT)
SLO-08 Review Cadence & Ownership (monthly reviews)
SLO-09 Exceptions & Adjustments (temporary changes)
SLO-10 Reporting (dashboards, exec summaries)

Performance & Scalability (PERF)
PERF-01 Performance Overview (targets, constraints)
PERF-02 Critical Path Inventory (user journeys)
PERF-03 Latency Budgets (per surface)
PERF-04 Throughput Targets (RPS, concurrency)
PERF-05 Scaling Strategy (horizontal/vertical/caching)
PERF-06 Database Performance Policy (indexes, query limits)
PERF-07 Cache Strategy (tiers, invalidation)
PERF-08 Queue/Async Performance (jobs, backpressure)
PERF-09 Performance Regression Gates (CI thresholds)
PERF-10 Performance Runbooks (triage playbooks)

Load/Stress Planning (LOAD)
LOAD-01 Load Testing Overview (goals, scope)
LOAD-02 Workload Models (traffic shapes, personas)
LOAD-03 Test Scenarios Catalog (by scenario_id)
LOAD-04 Data & Environment Setup (fixtures, isolation)
LOAD-05 Tooling & Harness (runner, metrics capture)
LOAD-06 Pass/Fail Criteria (SLO ties)
LOAD-07 Capacity Findings Report (results template)
LOAD-08 Bottleneck Analysis Template (what failed)
LOAD-09 Chaos/Failure Injection Plan (optional)
LOAD-10 Re-run Cadence & Change Triggers

Capacity & Cost Modeling (COST)
COST-01 Cost Model Overview (dimensions, owners)
COST-02 Cost Drivers Catalog (compute, storage, egress)
COST-03 Forecast Model (monthly/annual)
COST-04 Budget & Alerts (thresholds, routing)
COST-05 Unit Economics (cost per user/action)
COST-06 Optimization Backlog (ranked opportunities)
COST-07 Showback/Chargeback Policy (by team/tenant)
COST-08 Cost Attribution Tags Standard (required tags)
COST-09 Cost Reporting (dashboards, cadence)
COST-10 Incident Cost Review (post-incident spend)

Performance Budgets & Profiling (PBP)
PBP-01 Budget Overview (web/mobile budgets)
PBP-02 Web Budget Spec (TTFB, LCP, CLS, bundle)
PBP-03 Mobile Budget Spec (startup, frame drops)
PBP-04 API Budget Spec (p50/p95 per endpoint)
PBP-05 Profiling Plan (tools, cadence)
PBP-06 Perf CI Gates (thresholds, block rules)
PBP-07 Profiling Findings Template (report)
PBP-08 Optimization Guidelines (what to do)
PBP-09 Third-Party Impact Policy (SDK budgets)
PBP-10 Perf Regression Runbooks

Quality & Testing (QA)
QA-01 Quality Strategy Overview (layers, goals)
QA-02 Test Pyramid Policy (unit/integration/e2e)
QA-03 Test Coverage Rules (min %, exceptions)
QA-04 Test Case Catalog (by test_id)
QA-05 Defect Taxonomy (severity, root cause codes)
QA-06 Bug Triage Workflow (intake → fix → verify)
QA-07 Release Quality Gates (must-pass suites)
QA-08 Test Reporting (dashboards, cadence)

QA-09 Flaky Test Policy (detect, quarantine)
QA-10 QA Runbooks (common failures)

OBS-01

OBS-01 — Observability Overview (scope, principles, toolchain)
Header Block
template_id: OBS-01
title: Observability Overview (scope, principles, toolchain)
type: observability_overview
template_version: 1.0.0
output_path: 10_ops/observability/OBS-01_Observability_Overview.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["SEC-06", "AUDIT-02", "LTS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SEC-06", "AUDIT-02", "LTS-01", "ALRT-01", "SLO-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of observability for the system: what is measured, why, where telemetry lives, and the principles and toolchain used to achieve reliable operations. This document anchors the Ops/Observability set and must align with security monitoring and audit schema requirements.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Security monitoring baseline: {{xref:SEC-06}} | OPTIONAL
Audit schema/correlation IDs: {{xref:AUDIT-02}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Alerting overview: {{xref:ALRT-01}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Observability scope (what services/surfaces covered)
Telemetry pillars (logs/metrics/traces + optional profiles)
Primary goals (detect, diagnose, measure user impact)
Core principles (cardinality, cost awareness, privacy)
Toolchain inventory (metrics/logs/traces vendors/tools)
Environment coverage (dev/stage/prod)
Ownership model (who owns instrumentation/alerts)
Data governance rules (PII/secret redaction refs)
Oncall/alerting linkage (ALRT refs)
SLO linkage (what SLOs depend on)
Completeness definition (what “instrumented” means)

Optional Fields
Migration/rollout notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Do not log secrets or raw PII; reference redaction standards.
Every claim should be traceable to inputs or marked UNKNOWN.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Scope
   In-scope surfaces:
   {{scope.in[0]}}
   {{scope.in[1]}}
   Out-of-scope (explicit):
   {{scope.out[0]}} | OPTIONAL
   {{scope.out[1]}} | OPTIONAL
2. Telemetry Pillars
   pillers_enabled: {{pillars.enabled}} (logs/metrics/traces/profiles)
   notes: {{pillars.notes}} | OPTIONAL
3. Goals
   {{goals[0]}}
   {{goals[1]}}
   {{goals[2]}} | OPTIONAL
4. Principles
   {{principles[0]}}
   {{principles[1]}}
   {{principles[2]}}
5. Toolchain
   metrics_tool: {{tools.metrics}}
   logs_tool: {{tools.logs}}
   traces_tool: {{tools.traces}}
   profiling_tool: {{tools.profiling}} | OPTIONAL
6. Environment Coverage
   dev: {{env.dev}}
   stage: {{env.stage}}
   prod: {{env.prod}}
7. Ownership
   instrumentation_owner: {{owners.instrumentation_owner}}
   alert_owner: {{owners.alert_owner}}
   dashboard_owner: {{owners.dashboard_owner}} | OPTIONAL
8. Data Governance
   redaction_ref: {{xref:LTS-04}} | OPTIONAL

audit_linkage_ref: {{xref:AUDIT-02}} | OPTIONAL
pii_rule: {{governance.pii_rule}}
secret_rule: {{governance.secret_rule}}

9. Alerting & Oncall Linkage
   alerting_overview_ref: {{xref:ALRT-01}} | OPTIONAL
   routing_policy_ref: {{xref:ALRT-04}} | OPTIONAL
10. SLO Linkage
    slo_overview_ref: {{xref:SLO-01}} | OPTIONAL
    critical_services: {{slo.critical_services}} | OPTIONAL
11. Completeness Definition
    instrumented_definition: {{complete.instrumented_definition}}
    minimum_required_signals: {{complete.minimum_signals}}
12. References
    Telemetry schema: {{xref:OBS-02}} | OPTIONAL
    Metrics catalog: {{xref:OBS-03}} | OPTIONAL
    Dashboards: {{xref:OBS-07}} | OPTIONAL

Cross-References
Upstream: {{xref:SEC-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OBS-02}}, {{xref:ALRT-02}}, {{xref:SLO-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define scope, pillars, toolchain, and completeness definition (UNKNOWN allowed where needed).
intermediate: Required. Define ownership and governance rules and alert/SLO linkage.
advanced: Required. Add migration notes and strict traceability pointers and cost-aware principles.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, out-of-scope items, pillars notes, profiling tool, optional owners, slo.critical_services, migration/rollout notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scope.in is UNKNOWN → block Completeness Gate.
If pillars.enabled is UNKNOWN → block Completeness Gate.
If tools.metrics/logs/traces are UNKNOWN → block Completeness Gate.
If complete.instrumented_definition is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
scope_defined == true
pillars_and_toolchain_defined == true

governance_defined == true
completeness_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-02

OBS-02 — Telemetry Schema Standard (common fields, correlation IDs)
Header Block
template_id: OBS-02
title: Telemetry Schema Standard (common fields, correlation IDs)
type: telemetry_schema_standard
template_version: 1.0.0
output_path: 10_ops/observability/OBS-02_Telemetry_Schema_Standard.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["AUDIT-02", "LTS-01", "SEC-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "AUDIT-02", "LTS-01", "LTS-02", "SEC-06", "OBS-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical telemetry schema used across logs, metrics events, and traces: required common fields, correlation identifiers, naming conventions, and redaction rules. This standard enables consistent querying and incident correlation across services and clients.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Audit schema (correlation fields): {{xref:AUDIT-02}} | OPTIONAL
Logging standard (structure/fields): {{xref:LTS-01}} | OPTIONAL
Tracing standard (propagation): {{xref:LTS-02}} | OPTIONAL
Security monitoring baseline: {{xref:SEC-06}} | OPTIONAL
Observability overview: {{xref:OBS-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Schema version (semver)
Required common fields list (service, env, timestamp)
Correlation fields (request_id, trace_id, span_id)
Actor/context fields (user_id hash, tenant_id)
Client context fields (platform, app_version)
Error fields standard (error_code, stack policy ref)
Naming conventions (metric names, event names)
Cardinality constraints (allowed high-card fields)
Redaction/denylist fields (tokens, secrets, raw PII)
Validation rule (schema enforced in code/CI)
Telemetry requirements (schema validation failures)

Optional Fields
Sampling fields (sample_rate) | OPTIONAL
Extra dimensions allowlist | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Never emit secrets/tokens; hash or drop sensitive identifiers per redaction rules.
Correlation fields must be present for request-based telemetry where possible.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Schema Version
   schema_version: {{schema.version}}
2. Required Common Fields
   fields: {{schema.required_fields}}
3. Correlation
   request_id_field: {{corr.request_id_field}}
   trace_id_field: {{corr.trace_id_field}}
   span_id_field: {{corr.span_id_field}} | OPTIONAL
4. Actor / Tenant Context
   user_id_hash_field: {{ctx.user_id_hash_field}}
   tenant_id_field: {{ctx.tenant_id_field}} | OPTIONAL
5. Client Context
   platform_field: {{client.platform_field}}
   app_version_field: {{client.app_version_field}} | OPTIONAL
6. Error Fields
   error_code_field: {{err.error_code_field}}
   stack_policy_ref: {{xref:LTS-03}} | OPTIONAL
7. Naming Conventions
   metric_naming_rule: {{names.metric_rule}}
   event_naming_rule: {{names.event_rule}} | OPTIONAL
8. Cardinality Constraints
   high_cardinality_fields: {{card.fields}}
   limits_rule: {{card.limits_rule}} | OPTIONAL
9. Redaction
   denylist_fields: {{redact.denylist_fields}}
   hashing_rule: {{redact.hashing_rule}} | OPTIONAL
10. Validation
    enforcement_rule: {{validate.enforcement_rule}}
    ci_gate_ref: {{validate.ci_gate_ref}} | OPTIONAL
11. Telemetry
    schema_validation_fail_metric: {{telemetry.schema_validation_fail_metric}}

12. References
    Logging standard: {{xref:LTS-01}} | OPTIONAL
    Tracing standard: {{xref:LTS-02}} | OPTIONAL
    Audit correlation: {{xref:AUDIT-02}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OBS-03}}, {{xref:LTS-01}}, {{xref:ALRT-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define required fields, correlation fields, denylist, and validation rule.
intermediate: Required. Define naming/cardinality rules and client context fields.
advanced: Required. Add CI gate details, extra dimension allowlists, and strict enforcement language.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, span id field, tenant/app version fields, event naming, stack policy ref, limits rule, hashing rule, ci gate ref, sampling fields, dimension allowlist, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If schema.version is UNKNOWN → block Completeness Gate.
If schema.required_fields is UNKNOWN → block Completeness Gate.
If corr.request_id_field or corr.trace_id_field is UNKNOWN → block Completeness Gate.
If redact.denylist_fields is UNKNOWN → block Completeness Gate.
If telemetry.schema_validation_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
schema_and_correlation_defined == true
redaction_defined == true
validation_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-03

OBS-03 — Metrics Catalog (by metric_id)
Header Block
template_id: OBS-03
title: Metrics Catalog (by metric_id)
type: metrics_catalog
template_version: 1.0.0
output_path: 10_ops/observability/OBS-03_Metrics_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["OBS-01", "SLO-02", "ALRT-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-01", "OBS-02", "SLO-02", "SLO-03", "ALRT-02", "API-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of metrics used by the system, indexed by metric_id, including definition, type, labels, owners, and where each metric is emitted. This catalog supports SLOs, alerts, dashboards, and cost controls.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Observability overview: {{xref:OBS-01}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Endpoint SLO catalog: {{xref:SLO-03}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
API endpoint catalog (surface refs): {{xref:API-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Metric registry (metric_id list)
metric_id (stable identifier)
Metric name (exported name)
Type (counter/gauge/histogram/summary/UNKNOWN)
Unit (ms, bytes, count, %)
Definition (what it measures)
Emit location (service/surface)
Labels/dimensions (allowed labels list)
Cardinality constraints (max label values)
Owner (team/service)
Primary use (SLO/alert/dashboard/cost)

Retention class (short/standard/long/UNKNOWN)
Telemetry requirements (metric emission failures)

Optional Fields
SLO bindings (slo_id list) | OPTIONAL
Alert bindings (alert_id list) | OPTIONAL
Example query | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-OBS-CARDINALITY]}} | OPTIONAL
Metrics must have clear definitions and units; no ambiguous names.
Labels must be bounded; avoid unbounded user_id/session_id labels.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_metrics: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Metrics (repeat per metric_id)
   Metric
   metric_id: {{metrics[0].metric_id}}
   name: {{metrics[0].name}}
   type: {{metrics[0].type}}
   unit: {{metrics[0].unit}}
   definition: {{metrics[0].definition}}
   emit_location: {{metrics[0].emit_location}}
   labels: {{metrics[0].labels}}
   cardinality_rule: {{metrics[0].cardinality_rule}}
   owner: {{metrics[0].owner}}
   primary_use: {{metrics[0].primary_use}}
   retention_class: {{metrics[0].retention_class}}
   slo_bindings: {{metrics[0].slo_bindings}} | OPTIONAL
   alert_bindings: {{metrics[0].alert_bindings}} | OPTIONAL
   example_query: {{metrics[0].example_query}} | OPTIONAL
   telemetry_metric: {{metrics[0].telemetry_metric}}
   open_questions:
   {{metrics[0].open_questions[0]}} | OPTIONAL

(Repeat per metric.)

3. References
   SLOs: {{xref:SLO-02}} | OPTIONAL, {{xref:SLO-03}} | OPTIONAL

Alerts: {{xref:ALRT-02}} | OPTIONAL
Dashboards: {{xref:OBS-07}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-03}}, {{xref:SLO-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define metric_id/name/type/unit/definition and emit location and owner.
intermediate: Required. Define labels + cardinality and primary use and retention class.
advanced: Required. Add SLO/alert bindings and example queries and tighter bounded-label rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, slo/alert bindings, example query, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If metrics[*].metric_id is UNKNOWN → block Completeness Gate.*
*If metrics[*].name is UNKNOWN → block Completeness Gate.
If metrics[*].type is UNKNOWN → block Completeness Gate.*
*If metrics[*].unit is UNKNOWN → block Completeness Gate.
If metrics[*].definition is UNKNOWN → block Completeness Gate.*
*If metrics[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
metric_registry_defined == true
definitions_and_units_defined == true
labels_and_cardinality_defined == true
ownership_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-04

OBS-03 — Metrics Catalog (by metric_id)
Header Block
template_id: OBS-03
title: Metrics Catalog (by metric_id)
type: metrics_catalog
template_version: 1.0.0
output_path: 10_ops/observability/OBS-03_Metrics_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["OBS-01", "SLO-02", "ALRT-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-01", "OBS-02", "SLO-02", "SLO-03", "ALRT-02", "API-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of metrics used by the system, indexed by metric_id, including definition, type, labels, owners, and where each metric is emitted. This catalog supports SLOs, alerts, dashboards, and cost controls.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Observability overview: {{xref:OBS-01}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Endpoint SLO catalog: {{xref:SLO-03}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
API endpoint catalog (surface refs): {{xref:API-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Metric registry (metric_id list)
metric_id (stable identifier)
Metric name (exported name)
Type (counter/gauge/histogram/summary/UNKNOWN)
Unit (ms, bytes, count, %)
Definition (what it measures)
Emit location (service/surface)
Labels/dimensions (allowed labels list)
Cardinality constraints (max label values)
Owner (team/service)
Primary use (SLO/alert/dashboard/cost)

Retention class (short/standard/long/UNKNOWN)
Telemetry requirements (metric emission failures)

Optional Fields
SLO bindings (slo_id list) | OPTIONAL
Alert bindings (alert_id list) | OPTIONAL
Example query | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-OBS-CARDINALITY]}} | OPTIONAL
Metrics must have clear definitions and units; no ambiguous names.
Labels must be bounded; avoid unbounded user_id/session_id labels.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_metrics: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Metrics (repeat per metric_id)
   Metric
   metric_id: {{metrics[0].metric_id}}
   name: {{metrics[0].name}}
   type: {{metrics[0].type}}
   unit: {{metrics[0].unit}}
   definition: {{metrics[0].definition}}
   emit_location: {{metrics[0].emit_location}}
   labels: {{metrics[0].labels}}
   cardinality_rule: {{metrics[0].cardinality_rule}}
   owner: {{metrics[0].owner}}
   primary_use: {{metrics[0].primary_use}}
   retention_class: {{metrics[0].retention_class}}
   slo_bindings: {{metrics[0].slo_bindings}} | OPTIONAL
   alert_bindings: {{metrics[0].alert_bindings}} | OPTIONAL
   example_query: {{metrics[0].example_query}} | OPTIONAL
   telemetry_metric: {{metrics[0].telemetry_metric}}
   open_questions:
   {{metrics[0].open_questions[0]}} | OPTIONAL

(Repeat per metric.)

3. References
   SLOs: {{xref:SLO-02}} | OPTIONAL, {{xref:SLO-03}} | OPTIONAL

Alerts: {{xref:ALRT-02}} | OPTIONAL
Dashboards: {{xref:OBS-07}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-03}}, {{xref:SLO-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define metric_id/name/type/unit/definition and emit location and owner.
intermediate: Required. Define labels + cardinality and primary use and retention class.
advanced: Required. Add SLO/alert bindings and example queries and tighter bounded-label rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, slo/alert bindings, example query, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If metrics[*].metric_id is UNKNOWN → block Completeness Gate.*
*If metrics[*].name is UNKNOWN → block Completeness Gate.
If metrics[*].type is UNKNOWN → block Completeness Gate.*
*If metrics[*].unit is UNKNOWN → block Completeness Gate.
If metrics[*].definition is UNKNOWN → block Completeness Gate.*
*If metrics[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
metric_registry_defined == true
definitions_and_units_defined == true
labels_and_cardinality_defined == true
ownership_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-05

OBS-05 — Log Event Catalog (by log_event_id)
Header Block
template_id: OBS-05
title: Log Event Catalog (by log_event_id)
type: log_event_catalog
template_version: 1.0.0
output_path: 10_ops/observability/OBS-05_Log_Event_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["LTS-01", "LTS-03", "SKM-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LTS-01", "LTS-03", "SKM-09", "OBS-02", "API-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of application log events, indexed by log_event_id, including what each event means, required fields, severity level, and redaction constraints. This catalog drives consistent logging, alerting, and debugging across services.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Error logging standard: {{xref:LTS-03}} | OPTIONAL
Secrets/log redaction rules: {{xref:SKM-09}} | OPTIONAL
Telemetry schema: {{xref:OBS-02}} | OPTIONAL
Error/status code policy: {{xref:API-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Log event registry (log_event_id list)
log_event_id (stable identifier)
Event name (human-readable)
Log level (debug/info/warn/error)
Trigger condition (what causes it)
Required fields (schema fields)
Optional fields (safe fields only)
Correlation fields required (request_id/trace_id)
Redaction rule (denylist refs)
Owner (team/service)
Primary use (debugging/alerting/audit-adjacent)
Telemetry requirements (log emit rate, missing fields)

Optional Fields
Example log line (redacted) | OPTIONAL
Alert bindings (alert_id list) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Never log tokens, secrets, raw PII; apply denylist/redaction standards.
Every error-level log event must include correlation fields for traceability.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_log_events: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Log Events (repeat per log_event_id)
   Log Event
   log_event_id: {{events[0].log_event_id}}
   name: {{events[0].name}}
   level: {{events[0].level}}
   trigger: {{events[0].trigger}}
   required_fields: {{events[0].required_fields}}
   optional_fields: {{events[0].optional_fields}} | OPTIONAL
   correlation_required: {{events[0].correlation_required}}
   redaction_ref: {{events[0].redaction_ref}} (expected: {{xref:SKM-09}}/{{xref:LTS-04}}) | OPTIONAL
   owner: {{events[0].owner}}
   primary_use: {{events[0].primary_use}}
   alert_bindings: {{events[0].alert_bindings}} | OPTIONAL
   example_log: {{events[0].example_log}} | OPTIONAL
   telemetry_metric: {{events[0].telemetry_metric}}
   open_questions:
   {{events[0].open_questions[0]}} | OPTIONAL

(Repeat per log event.)

3. References
   Logging standard: {{xref:LTS-01}} | OPTIONAL
   Redaction standard: {{xref:LTS-04}} | OPTIONAL
   Alert catalog: {{xref:ALRT-02}} | OPTIONAL

Cross-References
Upstream: {{xref:LTS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:ALRT-02}}, {{xref:OBS-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define log_event_id, level, trigger, required fields, correlation, owner.
intermediate: Required. Define redaction refs, primary use, and telemetry metric per event.
advanced: Required. Add example redacted logs and alert bindings and stricter schema enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, optional fields, redaction ref, alert bindings, example log, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If events[*].log_event_id is UNKNOWN → block Completeness Gate.*
*If events[*].level is UNKNOWN → block Completeness Gate.
If events[*].trigger is UNKNOWN → block Completeness Gate.*
*If events[*].required_fields is UNKNOWN → block Completeness Gate.
If events[*].correlation_required is UNKNOWN → block Completeness Gate.*
*If events[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
log_event_registry_defined == true
correlation_defined == true
redaction_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-06

OBS-06 — Redaction & Privacy Rules for Telemetry (PII/secret handling)
Header Block
template_id: OBS-06
title: Redaction & Privacy Rules for Telemetry (PII/secret handling)
type: telemetry_redaction_privacy_rules
template_version: 1.0.0
output_path: 10_ops/observability/OBS-06_Telemetry_Redaction_Privacy_Rules.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["PRIV-02", "SKM-09", "LTS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRIV-02", "PRIV-03", "SKM-09", "LTS-04", "OBS-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical privacy and redaction rules for all telemetry (logs, traces, metrics dimensions, analytics events): what must be denied, what may be hashed, what may be stored, and how enforcement/verification works. This template must align with PII tiering and secrets redaction rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PII classification model: {{xref:PRIV-02}} | OPTIONAL
Data minimization rules: {{xref:PRIV-03}} | OPTIONAL
Secrets/log redaction rules: {{xref:SKM-09}} | OPTIONAL
PII/secret redaction standard (logging): {{xref:LTS-04}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Telemetry types covered (logs/traces/metrics/analytics)
Global denylist fields (tokens, secrets, auth headers)
PII handling rules (hash/mask/drop by tier)
Allowed identifiers policy (user_id hashed, tenant_id allowed?)
Trace attribute rules (what can be in spans)
Metric label rules (no unbounded identifiers)
Analytics event privacy constraints (consent tie-in)
Retention constraints for telemetry (short/standard)
Export/privacy controls (who can query/export)
Verification/enforcement rule (tests/scanners)
Telemetry requirements (redaction violations, dropped fields)

Optional Fields
Per-surface overrides (admin vs public) | OPTIONAL
Incident handling for telemetry leaks | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Default is minimize: if not needed for operations, do not emit.
Never emit secret material; redaction must occur before data leaves the process.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Covered Telemetry
   types: {{cover.types}}
2. Denylist
   denylist_fields: {{deny.fields}}
3. PII Handling
   tier_rules: {{pii.tier_rules}}
   hashing_rule: {{pii.hashing_rule}} | OPTIONAL
4. Allowed Identifiers
   allowed_ids: {{ids.allowed}}
   prohibited_ids: {{ids.prohibited}} | OPTIONAL
5. Traces
   span_attribute_allowlist: {{traces.allowlist}}
   span_attribute_denylist: {{traces.denylist}} | OPTIONAL
6. Metrics Labels
   allowed_labels: {{metrics.allowed_labels}}
   label_cardinality_rule: {{metrics.cardinality_rule}} | OPTIONAL
7. Analytics Privacy
   consent_ref: {{xref:PRIV-04}} | OPTIONAL
   analytics_minimization_rule: {{analytics.min_rule}}
8. Retention
   retention_class_rule: {{retention.class_rule}}
   default_retention_days: {{retention.default_days}}
9. Access / Exports
   query_access_rule: {{access.query_rule}}
   export_access_rule: {{access.export_rule}} | OPTIONAL
10. Enforcement
    enforcement_rule: {{enforce.rule}}
    tooling_ref: {{enforce.tooling_ref}} | OPTIONAL
11. Telemetry
    redaction_violation_metric: {{telemetry.violation_metric}}
    dropped_field_metric: {{telemetry.dropped_field_metric}} | OPTIONAL

12. References
    Secrets redaction: {{xref:SKM-09}} | OPTIONAL
    PII classification: {{xref:PRIV-02}} | OPTIONAL
    Logging redaction: {{xref:LTS-04}} | OPTIONAL

Cross-References
Upstream: {{xref:PRIV-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OBS-09}}, {{xref:ALRT-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define denylist, allowed identifiers, retention rule, and enforcement rule.
intermediate: Required. Define tier-based PII handling and trace/metric rules and telemetry metrics.
advanced: Required. Add per-surface overrides and incident handling and strict access/export controls.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, hashing rule, prohibited ids, trace denylist, cardinality rule, consent ref, export rule, tooling ref, optional metrics, overrides/incidents, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If deny.fields is UNKNOWN → block Completeness Gate.
If pii.tier_rules is UNKNOWN → block Completeness Gate.
If analytics.min_rule is UNKNOWN → block Completeness Gate.
If retention.default_days is UNKNOWN → block Completeness Gate.
If telemetry.violation_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
denylist_and_pii_rules_defined == true
retention_defined == true
enforcement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-07

OBS-07 — Dashboards Inventory (by dashboard_id)
Header Block
template_id: OBS-07
title: Dashboards Inventory (by dashboard_id)
type: dashboards_inventory
template_version: 1.0.0
output_path: 10_ops/observability/OBS-07_Dashboards_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["OBS-03", "ALRT-02", "SLO-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-03", "ALRT-02", "SLO-02", "PERF-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical inventory of operational dashboards, indexed by dashboard_id, including what each dashboard is for, required panels, owners, and linked alerts/SLOs. This template ensures dashboard coverage and reduces “tribal knowledge” during incidents.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Dashboard registry (dashboard_id list)
dashboard_id (stable identifier)
Dashboard name
Purpose (what questions it answers)
Primary audience (oncall/dev/product)
Panels list (panel_id/name + metric refs)
Required filters (env, service)
Owner (team/role)
SLO bindings (slo_id list)
Alert bindings (alert_id list)
Link to runbooks (IRP/DOC refs)
Telemetry requirements (dashboard usage/staleness)

Optional Fields
Screenshot link/pointer | OPTIONAL
Drill-down links (to traces/logs) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-OBS-CARDINALITY]}} | OPTIONAL
Dashboards must reference existing metric_ids (OBS-03) and not invent metrics.
Each SLO-critical service must have at least one primary dashboard.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_dashboards: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Dashboards (repeat per dashboard_id)
   Dashboard
   dashboard_id: {{dashboards[0].dashboard_id}}
   name: {{dashboards[0].name}}
   purpose: {{dashboards[0].purpose}}
   audience: {{dashboards[0].audience}}
   required_filters: {{dashboards[0].required_filters}}
   panels: {{dashboards[0].panels}}
   owner: {{dashboards[0].owner}}
   slo_bindings: {{dashboards[0].slo_bindings}}
   alert_bindings: {{dashboards[0].alert_bindings}} | OPTIONAL
   runbook_refs: {{dashboards[0].runbook_refs}} | OPTIONAL
   screenshot_ref: {{dashboards[0].screenshot_ref}} | OPTIONAL
   drilldowns: {{dashboards[0].drilldowns}} | OPTIONAL
   telemetry_metric: {{dashboards[0].telemetry_metric}}
   open_questions:
   {{dashboards[0].open_questions[0]}} | OPTIONAL

(Repeat per dashboard.)

3. References
   Metrics catalog: {{xref:OBS-03}} | OPTIONAL
   Alerting: {{xref:ALRT-02}} | OPTIONAL
   Incident runbooks: {{xref:IRP-10}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-09}}, {{xref:SLO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dashboards with panels and owners and required filters.
intermediate: Required. Add SLO bindings, alert bindings, and runbook refs.
advanced: Required. Add drilldowns, usage/staleness telemetry, and strict metric_id traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, alert bindings, runbook refs, screenshot ref, drilldowns, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dashboards[*].dashboard_id is UNKNOWN → block Completeness Gate.
If dashboards[*].panels is UNKNOWN → block Completeness Gate.
If dashboards[*].owner is UNKNOWN → block Completeness Gate.
If dashboards[*].slo_bindings is UNKNOWN → block Completeness Gate.
If dashboards[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
dashboard_registry_defined == true
panels_and_filters_defined == true
ownership_defined == true
slo_linkage_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-08

OBS-08 — Alert Routing & Ownership (teams, oncall, escalation)
Header Block
template_id: OBS-08
title: Alert Routing & Ownership (teams, oncall, escalation)
type: alert_routing_ownership
template_version: 1.0.0
output_path: 10_ops/observability/OBS-08_Alert_Routing_Ownership.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["ALRT-04", "ALRT-02", "IRP-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ALRT-02", "ALRT-04", "IRP-03", "OBS-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical routing and ownership model for operational alerts: which teams own which alerts/services, paging/escalation behavior, and how ownership is maintained. This template reduces misrouted pages and improves incident response speed.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Oncall/escalation policy: {{xref:ALRT-04}} | OPTIONAL
Incident roles & responsibilities: {{xref:IRP-03}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Ownership model (service owner vs alert owner)
Team registry (team_id list)
Oncall targets (pager schedules)
Routing rules (alert_id → team_id)
Escalation ladder (primary → secondary → manager)
Handoff rules (when to re-route)
Coverage expectations (24/7 vs business hours)
Runbook linkage rule (alerts must link to runbooks)
Maintenance/change control rule (who can edit routing)
Telemetry requirements (misroutes, ack time)

Optional Fields
Follow-the-sun notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Every paging alert must have an owning team and an escalation path.
Routing changes must be audited or recorded.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Ownership Model
   model: {{own.model}}
   definitions: {{own.definitions}} | OPTIONAL
2. Teams
   teams: {{teams.list}}
3. Oncall Targets
   targets: {{oncall.targets}}
4. Routing
   routes:
   alert_id: {{routes[0].alert_id}}
   team_id: {{routes[0].team_id}}
   oncall_target: {{routes[0].oncall_target}} | OPTIONAL
   coverage: {{routes[0].coverage}} | OPTIONAL

(Repeat per route.)

5. Escalation
   ladder: {{escalation.ladder}}
   handoff_rule: {{escalation.handoff_rule}} | OPTIONAL
6. Runbooks
   runbook_link_rule: {{runbooks.link_rule}}
   runbook_index_ref: {{xref:IRP-10}} | OPTIONAL
7. Change Control
   who_can_edit: {{change.who_can_edit}}
   approval_rule: {{change.approval_rule}} | OPTIONAL
8. Telemetry
   misroute_metric: {{telemetry.misroute_metric}}
   ack_time_metric: {{telemetry.ack_time_metric}} | OPTIONAL
9. References
   Alert catalog: {{xref:ALRT-02}} | OPTIONAL
   Escalation policy: {{xref:ALRT-04}} | OPTIONAL
   Incident roles: {{xref:IRP-03}} | OPTIONAL

Cross-References
Upstream: {{xref:ALRT-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define teams, routing rules, and escalation ladder.
intermediate: Required. Define coverage expectations and change control and telemetry.
advanced: Required. Add follow-the-sun and strict audit/change governance and handoff rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, definitions, oncall target, coverage, handoff rule, approval rule, optional metrics, follow-the-sun, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If teams.list is UNKNOWN → block Completeness Gate.
If routes[*].alert_id or routes[*]*.team_id is UNKNOWN → block Completeness Gate.
If escalation.ladder is UNKNOWN → block Completeness Gate.
If runbooks.link_rule is UNKNOWN → block Completeness Gate.
If telemetry.misroute_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
teams_and_routes_defined == true
escalation_defined == true
runbook_linkage_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-09

OBS-09 — Sampling & Cardinality Policy (limits, hashing, budgets)
Header Block
template_id: OBS-09
title: Sampling & Cardinality Policy (limits, hashing, budgets)
type: sampling_cardinality_policy
template_version: 1.0.0
output_path: 10_ops/observability/OBS-09_Sampling_Cardinality_Policy.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["OBS-02", "OBS-06", "COST-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-02", "OBS-06", "LTS-08", "COST-02", "ALRT-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for sampling and cardinality control across telemetry: which signals can be sampled, how sample rates are chosen, how high-cardinality dimensions are handled, and how budgets are enforced to control cost and reliability.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Telemetry redaction/privacy rules: {{xref:OBS-06}} | OPTIONAL
Log volume controls: {{xref:LTS-08}} | OPTIONAL
Cost drivers: {{xref:COST-02}} | OPTIONAL
Noise reduction rules: {{xref:ALRT-05}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Sampling scope (logs/traces/metrics/events)
Signals that MUST NOT be sampled (security/audit/privileged)
Default sampling rates (by signal type)
Override rules (incident mode, debug mode)
Cardinality limits policy (per label/attribute)
Prohibited high-cardinality fields (user_id raw, request bodies)
Hashing/anonymization rules for identifiers
Budget model (max volume per service/env)
Enforcement mechanism (drop, aggregate, throttle)
Telemetry requirements (dropped events, cardinality violations)

Optional Fields
Per-team budgets | OPTIONAL

Adaptive sampling notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Never sample away audit/privileged/security-critical events.
High-cardinality controls must be enforceable (bounded labels only).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Sampling Scope
   scope: {{sampling.scope}}
2. Non-Sampled Signals
   must_not_sample: {{sampling.must_not_sample}}
3. Default Rates
   logs_default_rate: {{rates.logs}}
   traces_default_rate: {{rates.traces}}
   events_default_rate: {{rates.events}} | OPTIONAL
4. Overrides
   override_rules: {{override.rules}}
5. Cardinality Limits
   limits_rule: {{card.limits_rule}}
   max_label_values: {{card.max_label_values}} | OPTIONAL
6. Prohibited Fields
   prohibited_fields: {{card.prohibited_fields}}
7. Hashing
   hashing_rule: {{hash.rule}}
   salt_key_ref: {{hash.salt_key_ref}} | OPTIONAL
8. Budgets
   budget_model: {{budget.model}}
   per_service_budget: {{budget.per_service}} | OPTIONAL
9. Enforcement
   enforcement_actions: {{enforce.actions}}
   fail_open_or_closed_rule: {{enforce.fail_rule}} | OPTIONAL
10. Telemetry
    dropped_telemetry_metric: {{telemetry.dropped_metric}}
    cardinality_violation_metric: {{telemetry.cardinality_violation_metric}} | OPTIONAL
11. References
    Cost drivers: {{xref:COST-02}} | OPTIONAL
    Redaction/privacy: {{xref:OBS-06}} | OPTIONAL
    Log volume controls: {{xref:LTS-08}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:OBS-10}}, {{xref:COST-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define must-not-sample list, default rates, prohibited fields, and budgets.
intermediate: Required. Define override rules, enforcement actions, and telemetry metrics.
advanced: Required. Add per-team budgets and adaptive sampling with strict incident-mode guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, events default rate, max label values, salt key ref, per-service budgets, fail rule, optional metric, per-team/adaptive notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If sampling.must_not_sample is UNKNOWN → block Completeness Gate.
If rates.logs or rates.traces is UNKNOWN → block Completeness Gate.
If card.prohibited_fields is UNKNOWN → block Completeness Gate.
If budget.model is UNKNOWN → block Completeness Gate.
If telemetry.dropped_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
sampling_and_rates_defined == true
cardinality_controls_defined == true
budget_and_enforcement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OBS-10

OBS-10 — Observability Gates (required signals per surface)
Header Block
template_id: OBS-10
title: Observability Gates (required signals per surface)
type: observability_gates
template_version: 1.0.0
output_path: 10_ops/observability/OBS-10_Observability_Gates.md
compliance_gate_id: TMP-05.PRIMARY.OBS
upstream_dependencies: ["OBS-01", "OBS-02", "ALRT-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-01", "OBS-02", "OBS-03", "OBS-07", "ALRT-01", "SLO-01", "PERF-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical “observability completeness” gates: what telemetry must exist for each surface (API endpoint, background job, websocket, client screen), what evidence proves it, and what blocks release if missing. This template operationalizes observability as a requirement, not an afterthought.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Observability overview: {{xref:OBS-01}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Alerting overview: {{xref:ALRT-01}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Perf regression gates: {{xref:PERF-09}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Surface types covered (api/job/ws/client/admin)
Gate registry (gate_id list)
gate_id (stable identifier)
Surface selector (endpoint_id/job_id/route_id)
Required signals list (logs/metrics/traces)
Required metric_ids (OBS-03 refs)
Required log_event_ids (OBS-05 refs)
Required trace coverage rule (OBS-04)
Required dashboards (OBS-07 refs)
Required alerts (ALRT refs)

Evidence rule (how to prove present)
Release block rule (what fails the gate)
Telemetry requirements (gate failures count)

Optional Fields
Grace period rules (for early MVP) | OPTIONAL
Exceptions process ref (COMP-08/SEC-08) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Do not invent metric_ids/log_event_ids; reference existing catalogs.
Critical surfaces must have at least one alert and one dashboard.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Covered Surfaces
   surfaces: {{cover.surfaces}}
2. Gate Registry Summary
   total_gates: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
3. Gates (repeat per gate_id)
   Gate
   gate_id: {{gates[0].gate_id}}
   surface_type: {{gates[0].surface_type}}
   surface_selector: {{gates[0].surface_selector}}
   required_signals: {{gates[0].required_signals}}
   required_metric_ids: {{gates[0].required_metric_ids}}
   required_log_event_ids: {{gates[0].required_log_event_ids}} | OPTIONAL
   trace_coverage_rule: {{gates[0].trace_coverage_rule}}
   required_dashboards: {{gates[0].required_dashboards}} | OPTIONAL
   required_alert_ids: {{gates[0].required_alert_ids}} | OPTIONAL
   evidence_rule: {{gates[0].evidence_rule}}
   release_block_rule: {{gates[0].release_block_rule}}
   telemetry_metric: {{gates[0].telemetry_metric}}
   grace_period: {{gates[0].grace_period}} | OPTIONAL
   exceptions_ref: {{gates[0].exceptions_ref}} | OPTIONAL
   open_questions:
   {{gates[0].open_questions[0]}} | OPTIONAL

(Repeat per gate.)

4. References
   Metrics catalog: {{xref:OBS-03}} | OPTIONAL

Log event catalog: {{xref:OBS-05}} | OPTIONAL
Trace standard: {{xref:OBS-04}} | OPTIONAL
Alerting: {{xref:ALRT-01}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:SLO-02}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define surface types and gate registry and required signals.
intermediate: Required. Bind gates to metric/log/trace rules and release block rule.
advanced: Required. Add dashboards/alerts requirements, grace/exception governance, and strict evidence rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, required log ids, required dashboards/alerts, grace period, exceptions ref, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If cover.surfaces is UNKNOWN → block Completeness Gate.
If gates[*].gate_id is UNKNOWN → block Completeness Gate.
If gates[*].required_signals is UNKNOWN → block Completeness Gate.
If gates[*].trace_coverage_rule is UNKNOWN → block Completeness Gate.
If gates[*].evidence_rule is UNKNOWN → block Completeness Gate.
If gates[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OBS
Pass conditions:
required_fields_present == true
gate_registry_defined == true
signal_requirements_defined == true
evidence_and_block_rules_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Analytics & Telemetry (ANL)

ANL-01

ANL-01 — Analytics Overview (goals, constraints)
Header Block
template_id: ANL-01
title: Analytics Overview (goals, constraints)
type: analytics_overview
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-01_Analytics_Overview.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["PRIV-04", "OBS-02", "PRIV-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-02", "PRIV-01", "PRIV-03", "PRIV-04", "ANL-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of product analytics and telemetry: why analytics exist, what questions they answer, what constraints apply (privacy, cost, performance), and how analytics differs from logs/metrics/traces. This document anchors the ANL set.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Data inventory: {{xref:PRIV-01}} | OPTIONAL
Data minimization rules: {{xref:PRIV-03}} | OPTIONAL
Consent model: {{xref:PRIV-04}} | OPTIONAL
Event taxonomy: {{xref:ANL-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Analytics goals (top questions/KPIs)
In-scope surfaces (web/mobile/backend events)
Out-of-scope statement (explicit)
Privacy constraints (consent, minimization)
Identity/attribution constraints (anon vs logged-in)
Event quality principles (naming/versioning)
Toolchain inventory (analytics provider/warehouse)
Access controls (who can query)
Retention/deletion linkage (PRIV-05)
Completeness definition (what “instrumented” means)
Telemetry requirements (data drop rate, schema violations)

Optional Fields
Migration/rollout notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Do not collect PII in analytics unless explicitly allowed by PRIV rules and consent model.
Every claim should be traceable to inputs or marked UNKNOWN.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Goals
   {{goals[0]}}
   {{goals[1]}}
   {{goals[2]}} | OPTIONAL
2. Scope
   in_scope:
   {{scope.in[0]}}
   {{scope.in[1]}}
   out_of_scope:
   {{scope.out[0]}} | OPTIONAL
   {{scope.out[1]}} | OPTIONAL
3. Privacy & Consent
   minimization_ref: {{xref:PRIV-03}} | OPTIONAL
   consent_ref: {{xref:PRIV-04}} | OPTIONAL
   privacy_rule: {{privacy.rule}}
4. Identity & Attribution
   identity_model_ref: {{xref:ANL-04}} | OPTIONAL
   constraints: {{identity.constraints}}
5. Event Quality
   taxonomy_ref: {{xref:ANL-02}} | OPTIONAL
   schema_ref: {{xref:ANL-03}} | OPTIONAL
   quality_principles: {{quality.principles}}
6. Toolchain
   provider: {{tools.provider}}
   warehouse: {{tools.warehouse}} | OPTIONAL
7. Access Controls
   query_access_rule: {{access.query_rule}}
   export_access_rule: {{access.export_rule}} | OPTIONAL
8. Retention / Deletion
   retention_ref: {{xref:PRIV-05}} | OPTIONAL
   analytics_retention_rule: {{retention.rule}}

9. Completeness Definition
   instrumented_definition: {{complete.instrumented_definition}}
   minimum_required_events: {{complete.minimum_events}}
10. Telemetry
   event_drop_metric: {{telemetry.event_drop_metric}}
   schema_violation_metric: {{telemetry.schema_violation_metric}} | OPTIONAL
11. References
   Event taxonomy: {{xref:ANL-02}} | OPTIONAL
   Funnels/KPIs: {{xref:ANL-07}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-02}}, {{xref:ANL-03}}, {{xref:ANL-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define goals, scope, privacy rule, toolchain, and completeness definition.
intermediate: Required. Define identity constraints, access controls, and retention rule and telemetry metrics.
advanced: Required. Add rollout notes and strict traceability and quality governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional goals, out-of-scope items, identity model ref, export access, warehouse, optional telemetry metric, rollout notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If goals[0] is UNKNOWN → block Completeness Gate.
If scope.in is UNKNOWN → block Completeness Gate.
If privacy.rule is UNKNOWN → block Completeness Gate.
If tools.provider is UNKNOWN → block Completeness Gate.
If complete.instrumented_definition is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
goals_and_scope_defined == true
privacy_and_access_defined == true
retention_defined == true
completeness_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-02

ANL-02 — Event Taxonomy (by event_name)
Header Block
template_id: ANL-02
title: Event Taxonomy (by event_name)
type: analytics_event_taxonomy
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-02_Event_Taxonomy.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["ANL-01", "OBS-02", "PRIV-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ANL-01", "OBS-02", "PRIV-04", "PRIV-02", "FE-04", "MOB-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical taxonomy of analytics events, indexed by event_name, including naming conventions, when each event fires, and ownership. This taxonomy is the source of truth for analytics instrumentation and must align with privacy/consent rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Analytics overview: {{xref:ANL-01}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Consent model: {{xref:PRIV-04}} | OPTIONAL
PII classification: {{xref:PRIV-02}} | OPTIONAL
Data binding rules: {{xref:FE-04}} | OPTIONAL
Mobile screen bindings: {{xref:MOB-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Event registry (event_name list)
event_name (stable identifier)
Category (screen, action, system, funnel)
Trigger definition (when fired)
Source surface (web/mobile/backend)
Required properties list (property names)
Allowed properties list (safe fields)
Consent requirement (consent_id or NONE/UNKNOWN)
PII rule (must not include tier-X data)
Owner (team)
Primary KPI/funnel linkage (kpi_id) | OPTIONAL
Telemetry requirements (event drop rate per event)

Optional Fields
Example payload (redacted) | OPTIONAL
Deprecation rule (rename/version) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Event names must follow a consistent convention (e.g., noun_verb or screen.view).
No raw PII in event properties; use allowed properties only.
If consent is required, the event must not fire without it.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Naming Convention
   event_name_rule: {{names.event_name_rule}}
   examples: {{names.examples}} | OPTIONAL
2. Events (repeat per event_name)
   Event
   event_name: {{events[0].event_name}}
   category: {{events[0].category}}
   trigger: {{events[0].trigger}}
   surface: {{events[0].surface}}
   required_properties: {{events[0].required_properties}}
   allowed_properties: {{events[0].allowed_properties}}
   consent_requirement: {{events[0].consent_requirement}}
   pii_rule: {{events[0].pii_rule}}
   owner: {{events[0].owner}}
   kpi_link: {{events[0].kpi_link}} | OPTIONAL
   example_payload: {{events[0].example_payload}} | OPTIONAL
   deprecation_rule: {{events[0].deprecation_rule}} | OPTIONAL
   telemetry_metric: {{events[0].telemetry_metric}}
   open_questions:
   {{events[0].open_questions[0]}} | OPTIONAL

(Repeat per event.)

3. References
   Event schema spec: {{xref:ANL-03}} | OPTIONAL
   Identity model: {{xref:ANL-04}} | OPTIONAL
   Consent model: {{xref:PRIV-04}} | OPTIONAL

Cross-References
Upstream: {{xref:ANL-01}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:ANL-03}}, {{xref:ANL-07}}, {{xref:ANL-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define event registry and triggers and required/allowed properties.
intermediate: Required. Define consent requirements and PII rules and owners.
advanced: Required. Add deprecation/versioning policy and KPI linkage and example payloads.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, examples, kpi link, example payload, deprecation rule, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If events[*].event_name is UNKNOWN → block Completeness Gate.*
*If events[*].trigger is UNKNOWN → block Completeness Gate.
If events[*].required_properties is UNKNOWN → block Completeness Gate.*
*If events[*].allowed_properties is UNKNOWN → block Completeness Gate.
If events[*].pii_rule is UNKNOWN → block Completeness Gate.*
*If events[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
event_registry_defined == true
properties_defined == true
privacy_and_consent_defined == true
ownership_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-03

ANL-03 — Event Schema Spec (payload rules, versioning)
Header Block
template_id: ANL-03
title: Event Schema Spec (payload rules, versioning)
type: analytics_event_schema_spec
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-03_Event_Schema_Spec.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["ANL-02", "OBS-02", "PRIV-08"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ANL-02", "OBS-02", "PRIV-02", "PRIV-08", "ANL-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical schema rules for analytics events: payload structure, required envelope fields, property typing, versioning, and backward compatibility. This spec enforces consistency across events defined in ANL-02 and supports reliable analytics queries.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Event taxonomy: {{xref:ANL-02}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
PII classification: {{xref:PRIV-02}} | OPTIONAL
Anonymization rules: {{xref:PRIV-08}} | OPTIONAL
Analytics data quality rules: {{xref:ANL-06}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Envelope fields list (event_name, ts, user/anon ids)
Property typing rules (string/number/bool/object)
Required property rule (from taxonomy)
Optional property rule (allowed list only)
Versioning scheme (schema_version per event)
Backward compatibility rules (additive only)
Deprecation rules (rename, sunset)
Validation/enforcement rule (client/server validation)
Redaction/denylist fields (no PII beyond allowed)
Telemetry requirements (schema validation failures)

Optional Fields
Batching rules | OPTIONAL

Compression rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
No raw PII in properties; apply PRIV-08 transformation rules when needed.
Event payloads must validate against schema before send/store.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Envelope
   required_fields: {{env.required_fields}}
   id_fields: {{env.id_fields}} | OPTIONAL
2. Properties
   typing_rules: {{props.typing_rules}}
   required_properties_rule: {{props.required_rule}}
   optional_properties_rule: {{props.optional_rule}}
3. Versioning
   version_field: {{ver.version_field}}
   compatibility_rule: {{ver.compatibility_rule}}
   deprecation_rule: {{ver.deprecation_rule}} | OPTIONAL
4. Validation
   validation_location: {{validate.location}} (client/server/both/UNKNOWN)
   enforcement_rule: {{validate.enforcement_rule}}
   error_handling_rule: {{validate.error_handling_rule}} | OPTIONAL
5. Redaction
   denylist_fields: {{redact.denylist_fields}}
   transform_ref: {{xref:PRIV-08}} | OPTIONAL
6. Telemetry
   schema_validation_fail_metric: {{telemetry.schema_validation_fail_metric}}
   drop_metric: {{telemetry.drop_metric}} | OPTIONAL
7. Optional Transport
   batching_rule: {{transport.batching_rule}} | OPTIONAL
   compression_rule: {{transport.compression_rule}} | OPTIONAL
8. References
   Event taxonomy: {{xref:ANL-02}} | OPTIONAL
   Data quality: {{xref:ANL-06}} | OPTIONAL
   Consent enforcement: {{xref:ANL-05}} | OPTIONAL

Cross-References
Upstream: {{xref:ANL-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-06}}, {{xref:ANL-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define envelope, typing rules, versioning, and validation location.
intermediate: Required. Define compatibility/deprecation and denylist/transform linkage and telemetry.
advanced: Required. Add batching/compression and strict enforcement language + example validation failure handling.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, id fields, deprecation rule, error handling, optional transport rules, optional metrics, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If env.required_fields is UNKNOWN → block Completeness Gate.
If props.typing_rules is UNKNOWN → block Completeness Gate.
If ver.version_field is UNKNOWN → block Completeness Gate.
If validate.enforcement_rule is UNKNOWN → block Completeness Gate.
If telemetry.schema_validation_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
envelope_and_properties_defined == true
versioning_defined == true
validation_defined == true
redaction_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-04

ANL-04 — Identity/Attribution Model (anon/user/tenant)
Header Block
template_id: ANL-04
title: Identity/Attribution Model (anon/user/tenant)
type: analytics_identity_attribution_model
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-04_Identity_Attribution_Model.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["IAM-03", "PRIV-08", "PRIV-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IAM-03", "PRIV-04", "PRIV-08", "ANL-02", "ANL-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical identity and attribution model used for analytics: anonymous identifiers, logged-in user identifiers, tenant/org attribution, sessionization, and identity merge rules. This model must align with privacy minimization and consent and must not expose raw user identifiers.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Session management policy: {{xref:IAM-03}} | OPTIONAL
Consent model: {{xref:PRIV-04}} | OPTIONAL
Pseudonymization rules: {{xref:PRIV-08}} | OPTIONAL
Event taxonomy: {{xref:ANL-02}} | OPTIONAL
Schema spec: {{xref:ANL-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Identifier types (anon_id, user_id_hash, device_id?)
Generation rules (when created)
Storage rules (client storage limits)
Session model (session_id, timeout)
User login attribution rule (anon→user merge)
Tenant attribution rule (tenant_id field)
Cross-device attribution rule (if any)
PII minimization rule (no raw identifiers)
Deletion/retention interaction (PRIV-05)
Telemetry requirements (merge success, id missing)

Optional Fields
GDPR/consent interaction notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Do not send raw user IDs in analytics; use hashed/pseudonymized identifiers.
Merge rules must be deterministic and documented (avoid double-counting).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Identifier Types
   types: {{ids.types}}
2. Generation
   anon_id_generation_rule: {{gen.anon_rule}}
   user_id_hash_rule: {{gen.user_hash_rule}} | OPTIONAL
3. Storage
   client_storage_rule: {{store.client_rule}}
   ttl_rule: {{store.ttl_rule}} | OPTIONAL
4. Sessions
   session_id_rule: {{session.id_rule}}
   session_timeout_minutes: {{session.timeout_minutes}}
5. Login Attribution
   merge_rule: {{merge.rule}}
   merge_window_rule: {{merge.window_rule}} | OPTIONAL
6. Tenant Attribution
   tenant_field: {{tenant.field}}
   tenant_rule: {{tenant.rule}} | OPTIONAL
7. Cross-Device
   supported: {{xdev.supported}}
   rule: {{xdev.rule}} | OPTIONAL
8. Privacy / Retention
   minimization_rule: {{privacy.minimization_rule}}
   retention_ref: {{xref:PRIV-05}} | OPTIONAL
9. Telemetry
   id_missing_metric: {{telemetry.id_missing_metric}}
   merge_success_metric: {{telemetry.merge_success_metric}} | OPTIONAL
10. References
    Consent model: {{xref:PRIV-04}} | OPTIONAL
    Pseudonymization: {{xref:PRIV-08}} | OPTIONAL
    Session policy: {{xref:IAM-03}} | OPTIONAL

Cross-References
Upstream: {{xref:IAM-03}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:ANL-05}}, {{xref:ANL-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define identifier types, session timeout, and merge rule.
intermediate: Required. Define storage rules, tenant attribution, and telemetry.
advanced: Required. Add cross-device support and consent/GDPR notes and strict retention linkage.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, user hash rule, ttl/merge window rules, tenant rule, xdev rules, optional metrics, consent notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ids.types is UNKNOWN → block Completeness Gate.
If session.timeout_minutes is UNKNOWN → block Completeness Gate.
If merge.rule is UNKNOWN → block Completeness Gate.
If privacy.minimization_rule is UNKNOWN → block Completeness Gate.
If telemetry.id_missing_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
ids_and_sessions_defined == true
merge_and_tenant_defined == true
privacy_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-05

ANL-05 — Consent & Opt-Out Enforcement (ties to PRIV-04)
Header Block
template_id: ANL-05
title: Consent & Opt-Out Enforcement (ties to PRIV-04)
type: analytics_consent_optout_enforcement
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-05_Consent_OptOut_Enforcement.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["PRIV-04", "ANL-02", "ROUTE-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRIV-04", "ANL-02", "ANL-04", "ROUTE-04", "CSec-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical enforcement of analytics consent and opt-out: where consent is captured, how it is stored, how instrumentation checks consent before emitting, and how opt-out affects identity and retention. This must align with PRIV-04 and must be enforceable (not just UI).

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Consent model: {{xref:PRIV-04}} | OPTIONAL
Event taxonomy: {{xref:ANL-02}} | OPTIONAL
Identity model: {{xref:ANL-04}} | OPTIONAL
Guard rules (role/auth gating): {{xref:ROUTE-04}} | OPTIONAL
Client data protection: {{xref:CSec-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Consent signals used (consent_id list)
Default state rule (opt-in/opt-out)
Where consent stored (client/server)
Enforcement points (client SDK, backend pipeline)
Opt-out behavior (stop emit, delete/stop processing)
Identity behavior on opt-out (anon_id reset?)
Scope of opt-out (analytics only vs telemetry)
Audit requirements (consent changes logged)
Telemetry requirements (events blocked by consent)
Testing/verification rule (ensure no events leak)

Optional Fields
Per-jurisdiction overrides | OPTIONAL

Do-not-track handling | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
If an event requires consent, it must not be emitted without consent.
Opt-out must be honored promptly across clients and backend pipelines.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Consent Signals
   consents: {{consents.list}}
2. Defaults
   default_state: {{defaults.state}}
3. Storage
   storage_model: {{store.model}} (client/server/both/UNKNOWN)
   fields_stored: {{store.fields}} | OPTIONAL
4. Enforcement
   points: {{enforce.points}}
   sdk_rule: {{enforce.sdk_rule}} | OPTIONAL
   backend_rule: {{enforce.backend_rule}} | OPTIONAL
5. Opt-Out Behavior
   stop_emit_rule: {{opt.stop_emit_rule}}
   delete_or_stop_processing_rule: {{opt.delete_or_stop_processing_rule}} | OPTIONAL
6. Identity on Opt-Out
   identity_rule: {{opt.identity_rule}}
7. Scope
   scope_rule: {{scope.rule}}
8. Audit
   audit_required: {{audit.required}}
   audit_ref: {{xref:AUDIT-01}} | OPTIONAL
9. Telemetry
   blocked_event_metric: {{telemetry.blocked_event_metric}}
   opt_out_rate_metric: {{telemetry.opt_out_rate_metric}} | OPTIONAL
10. Verification
    test_rule: {{verify.test_rule}}
    ci_gate_ref: {{verify.ci_gate_ref}} | OPTIONAL

Cross-References
Upstream: {{xref:PRIV-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-09}}, {{xref:ANL-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define defaults, enforcement points, stop-emit rule, and verification rule.
intermediate: Required. Define storage model, opt-out identity rule, and telemetry metrics.
advanced: Required. Add jurisdiction/DNT handling and strict audit/event-block evidence.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, fields stored, sdk/backend rules, delete/stop processing, opt-out rate metric, ci gate ref, jurisdiction/DNT notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If consents.list is UNKNOWN → block Completeness Gate.
If defaults.state is UNKNOWN → block Completeness Gate.
If enforce.points is UNKNOWN → block Completeness Gate.
If opt.stop_emit_rule is UNKNOWN → block Completeness Gate.
If telemetry.blocked_event_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
consent_and_defaults_defined == true
enforcement_defined == true
opt_out_behavior_defined == true
verification_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-06

ANL-06 — Data Quality Rules (completeness, duplication)
Header Block
template_id: ANL-06
title: Data Quality Rules (completeness, duplication)
type: analytics_data_quality_rules
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-06_Data_Quality_Rules.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["ANL-03", "ANL-02", "OBS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ANL-02", "ANL-03", "ANL-04", "OBS-02", "ANL-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical data quality rules for analytics: required-field completeness, schema validation, duplication control, ordering constraints (where relevant), and how quality is monitored and remediated. This template ensures analytics can be trusted.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Event taxonomy: {{xref:ANL-02}} | OPTIONAL
Event schema spec: {{xref:ANL-03}} | OPTIONAL
Identity model: {{xref:ANL-04}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Funnels/KPIs: {{xref:ANL-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Quality dimensions list (completeness, validity, dedupe)
Completeness rules (required properties present)
Validity rules (types/enums)
Duplication rules (idempotency keys/event_id)
Late arrival handling (events arriving late)
Ordering assumptions (if any)
Monitoring rules (dashboards/alerts)
Remediation workflow (fix instrumentation, backfill?)
Ownership (who fixes quality issues)
Telemetry requirements (invalid events %, dup rate)

Optional Fields
Backfill policy | OPTIONAL

Sampling effects notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Data quality rules must be measurable (metrics) and enforceable (validation).
If required fields missing, event should be dropped or marked invalid per rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Quality Dimensions
   dimensions: {{dims.list}}
2. Completeness
   rules: {{complete.rules}}
   drop_or_invalid_rule: {{complete.drop_or_invalid_rule}} | OPTIONAL
3. Validity
   rules: {{valid.rules}}
   enum_policy: {{valid.enum_policy}} | OPTIONAL
4. Duplication
   event_id_rule: {{dedupe.event_id_rule}}
   idempotency_window_minutes: {{dedupe.window_minutes}} | OPTIONAL
5. Late Arrivals
   late_event_rule: {{late.rule}}
   max_lateness_hours: {{late.max_hours}} | OPTIONAL
6. Ordering
   ordering_assumptions: {{order.assumptions}}
7. Monitoring
   dashboards_ref: {{xref:OBS-07}} | OPTIONAL
   alert_ref: {{xref:ALRT-02}} | OPTIONAL
   monitoring_rule: {{monitor.rule}}
8. Remediation
   workflow_steps: {{remed.steps}}
   backfill_policy: {{remed.backfill_policy}} | OPTIONAL
9. Ownership
   owner: {{owner.team}}
10. Telemetry
    invalid_event_rate_metric: {{telemetry.invalid_rate_metric}}
    duplicate_rate_metric: {{telemetry.dup_rate_metric}} | OPTIONAL
    late_event_rate_metric: {{telemetry.late_rate_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:ANL-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-07}}, {{xref:ANL-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define completeness and validity and monitoring rule.
intermediate: Required. Define dedupe and late handling and remediation workflow and telemetry metrics.
advanced: Required. Add backfill policy and sampling notes and strict ownership + SLA for fixes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, drop/invalid rule, enum policy, window minutes, lateness hours, dashboards/alert refs, backfill policy, sampling notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dims.list is UNKNOWN → block Completeness Gate.
If complete.rules is UNKNOWN → block Completeness Gate.
If valid.rules is UNKNOWN → block Completeness Gate.
If monitor.rule is UNKNOWN → block Completeness Gate.
If telemetry.invalid_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
quality_rules_defined == true
monitoring_defined == true
remediation_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-07

ANL-07 — Funnels & KPIs Definition (by kpi_id)
Header Block
template_id: ANL-07
title: Funnels & KPIs Definition (by kpi_id)
type: analytics_funnels_kpis
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-07_Funnels_KPIs_Definition.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["ANL-02", "ANL-04", "PRD-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRD-01", "ANL-02", "ANL-04", "ANL-06", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical set of funnels and KPIs for the product, indexed by kpi_id, including definitions, event mappings, segmentation rules, and ownership. This template is the bridge between product goals and analytics events.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Product overview: {{xref:PRD-01}} | OPTIONAL
Event taxonomy: {{xref:ANL-02}} | OPTIONAL
Identity model: {{xref:ANL-04}} | OPTIONAL
Data quality rules: {{xref:ANL-06}} | OPTIONAL
Metrics catalog (for hybrid KPIs): {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
KPI registry (kpi_id list)
kpi_id (stable identifier)
Name/description
Type (kpi/funnel/guardrail)
Definition (exact formula)
Event mapping (event_name list)
Inclusion/exclusion rules
Segmentation dimensions (allowed only)
Reporting cadence (daily/weekly/UNKNOWN)
Owner (team/role)
Data quality dependency (ANL-06 refs)
Telemetry requirements (kpi freshness, missing data)

Optional Fields
Target/goal value | OPTIONAL
Dashboard reference | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-OBS-CARDINALITY]}} | OPTIONAL
KPIs must reference existing events from ANL-02; do not invent event names.
Segmentation must use allowed dimensions and respect privacy constraints.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_kpis: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. KPIs (repeat per kpi_id)
   KPI
   kpi_id: {{kpis[0].kpi_id}}
   name: {{kpis[0].name}}
   type: {{kpis[0].type}}
   definition: {{kpis[0].definition}}
   event_mapping: {{kpis[0].event_mapping}}
   rules: {{kpis[0].rules}}
   segments: {{kpis[0].segments}}
   cadence: {{kpis[0].cadence}}
   owner: {{kpis[0].owner}}
   data_quality_refs: {{kpis[0].data_quality_refs}}
   target_value: {{kpis[0].target_value}} | OPTIONAL
   dashboard_ref: {{kpis[0].dashboard_ref}} | OPTIONAL
   telemetry_metric: {{kpis[0].telemetry_metric}}
   open_questions:
   {{kpis[0].open_questions[0]}} | OPTIONAL

(Repeat per KPI.)

3. References
   Event taxonomy: {{xref:ANL-02}} | OPTIONAL
   Data quality: {{xref:ANL-06}} | OPTIONAL
   Access controls: {{xref:ANL-10}} | OPTIONAL

Cross-References
Upstream: {{xref:ANL-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-10}}, {{xref:ANL-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define KPI list and event mappings and owners and cadence.
intermediate: Required. Define formulas and segmentation and data quality refs and telemetry.
advanced: Required. Add targets and dashboard links and strict inclusion/exclusion rules + freshness SLAs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, target value, dashboard ref, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If kpis[*].kpi_id is UNKNOWN → block Completeness Gate.
If kpis[*].definition is UNKNOWN → block Completeness Gate.
If kpis[*].event_mapping is UNKNOWN → block Completeness Gate.
If kpis[*].owner is UNKNOWN → block Completeness Gate.
If kpis[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
kpi_registry_defined == true
event_mappings_defined == true
definitions_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-08

ANL-08 — Experimentation Support (flags/assignments)
Header Block
template_id: ANL-08
title: Experimentation Support (flags/assignments)
type: analytics_experimentation_support
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-08_Experimentation_Support.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["FFCFG-01", "FFCFG-02", "ANL-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FFCFG-01", "FFCFG-02", "ANL-04", "ANL-07", "ANL-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical support for experimentation in analytics: how users are assigned to variants, how assignments are logged, what fields are required in events for analysis, and how experiments interact with feature flags and privacy constraints.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Feature flag registry: {{xref:FFCFG-01}} | OPTIONAL
Flag behavior spec: {{xref:FFCFG-02}} | OPTIONAL
Identity model: {{xref:ANL-04}} | OPTIONAL
Funnels/KPIs: {{xref:ANL-07}} | OPTIONAL
Event schema spec: {{xref:ANL-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Experiment identifier model (exp_id, variant_id)
Assignment unit (user/anon/tenant)
Randomization rule (hash-based)
Sticky assignment rule (how persisted)
Exposure event requirement (when log exposure)
Event enrichment rule (include exp_id/variant)
Analysis constraints (min sample size?)
Privacy constraints (no sensitive segmentation)
Kill-switch behavior (flag off)
Telemetry requirements (assignment coverage, exposure rates)

Optional Fields
Multi-armed bandit notes | OPTIONAL

A/A testing rule | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Assignments must be deterministic for a given unit during an experiment.
Exposure events should be logged once per user per experiment (or defined frequency).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Identifiers
   exp_id_rule: {{ids.exp_id_rule}}
   variant_id_rule: {{ids.variant_id_rule}}
2. Assignment
   unit: {{assign.unit}}
   randomization_rule: {{assign.randomization_rule}}
   sticky_rule: {{assign.sticky_rule}}
3. Exposure
   exposure_event_rule: {{exposure.event_rule}}
   exposure_event_name: {{exposure.event_name}} | OPTIONAL
4. Event Enrichment
   required_fields: {{enrich.fields}}
   schema_ref: {{xref:ANL-03}} | OPTIONAL
5. Analysis Constraints
   constraints: {{analysis.constraints}}
6. Privacy
   segmentation_rule: {{privacy.segmentation_rule}}
7. Kill-Switch
   kill_switch_rule: {{kill.rule}}
   flag_ref: {{kill.flag_ref}} (expected: {{xref:FFCFG-02}}) | OPTIONAL
8. Telemetry
   assignment_coverage_metric: {{telemetry.assignment_coverage_metric}}
   exposure_rate_metric: {{telemetry.exposure_rate_metric}} | OPTIONAL
9. References
   Flag registry: {{xref:FFCFG-01}} | OPTIONAL
   Flag behavior: {{xref:FFCFG-02}} | OPTIONAL
   Identity model: {{xref:ANL-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FFCFG-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-07}}, {{xref:ANL-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define assignment unit, sticky rule, and exposure rule.
intermediate: Required. Define enrichment fields and privacy segmentation rule and telemetry.
advanced: Required. Add A/A rules, bandit notes, and stricter analysis constraints and kill-switch governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, exposure event name, optional telemetry metric, bandit/A-A notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If assign.unit is UNKNOWN → block Completeness Gate.
If assign.randomization_rule is UNKNOWN → block Completeness Gate.
If exposure.event_rule is UNKNOWN → block Completeness Gate.
If privacy.segmentation_rule is UNKNOWN → block Completeness Gate.
If telemetry.assignment_coverage_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
assignment_defined == true
exposure_and_enrichment_defined == true
privacy_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-09

ANL-09 — Retention & Deletion for Analytics (ties to PRIV-05)
Header Block
template_id: ANL-09
title: Retention & Deletion for Analytics (ties to PRIV-05)
type: analytics_retention_deletion
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-09_Retention_Deletion_for_Analytics.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["PRIV-05", "ANL-03", "ANL-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRIV-05", "PRIV-06", "ANL-03", "ANL-04", "COMP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical retention and deletion behavior for analytics data: event retention periods, deletion request handling (account delete/DSAR), propagation to warehouses/third parties, and verification. This template must align with PRIV-05 and any vendor sharing rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Retention/deletion policy: {{xref:PRIV-05}} | OPTIONAL
Data sharing map (analytics vendors): {{xref:PRIV-06}} | OPTIONAL
Event schema spec: {{xref:ANL-03}} | OPTIONAL
Identity model: {{xref:ANL-04}} | OPTIONAL
Regulatory requirements: {{xref:COMP-06}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Analytics retention classes (class_id list)
Retention days by event category/type
Deletion request types supported (account delete, DSAR erase)
Deletion workflow for analytics stores (steps)
Propagation rules (warehouse, vendor tools)
Identity deletion rule (how to find user rows via hashed IDs)
Legal hold interaction (if any)
Verification rule (prove deletion completed)
Audit requirements (deletion actions recorded)
Telemetry requirements (deletion backlog, completion time)

Optional Fields
Aggregation policy (keep aggregated stats?) | OPTIONAL

Vendor limitations notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Deletion must propagate to analytics vendors where data was shared (or explicit exception).
Do not rely on raw IDs; use the identity model's hashed identifiers for deletion targeting.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Retention Classes
   classes: {{retention.classes}}
2. Retention Days
   Policy
   event_category: {{ttls[0].event_category}}
   ttl_days: {{ttls[0].ttl_days}}
   notes: {{ttls[0].notes}} | OPTIONAL

(Repeat per policy.)

3. Deletion Requests
   types: {{delete.types}}
4. Deletion Workflow
   steps:
   {{delete.steps[0]}}
   {{delete.steps[1]}}
   {{delete.steps[2]}} | OPTIONAL
5. Propagation
   warehouse_rule: {{prop.warehouse_rule}}
   vendor_rule: {{prop.vendor_rule}} | OPTIONAL
6. Identity Targeting
   id_fields: {{id.fields}}
   lookup_rule: {{id.lookup_rule}} | OPTIONAL
7. Legal Hold
   supported: {{hold.supported}}
   override_rule: {{hold.override_rule}} | OPTIONAL
8. Verification
   verification_rule: {{verify.rule}}
9. Audit
   audit_required: {{audit.required}}
   audit_ref: {{xref:AUDIT-01}} | OPTIONAL
10. Telemetry
    deletion_backlog_metric: {{telemetry.backlog_metric}}
    deletion_completion_time_metric: {{telemetry.completion_time_metric}} | OPTIONAL

11. References
    Retention policy: {{xref:PRIV-05}} | OPTIONAL
    Identity model: {{xref:ANL-04}} | OPTIONAL
    Sharing map: {{xref:PRIV-06}} | OPTIONAL

Cross-References
Upstream: {{xref:PRIV-05}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ANL-10}}, {{xref:COMP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define retention classes/days and deletion workflow and verification rule.
intermediate: Required. Define propagation rules and identity targeting and telemetry metrics.
advanced: Required. Add aggregation policy and vendor limitations notes and strict exception records.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, vendor rule, lookup rule, legal hold override, audit ref, optional metrics, aggregation/vendor notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If retention.classes is UNKNOWN → block Completeness Gate.
If delete.steps[0] is UNKNOWN → block Completeness Gate.
If verify.rule is UNKNOWN → block Completeness Gate.
If telemetry.backlog_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
retention_defined == true
deletion_workflow_defined == true
propagation_defined == true
verification_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ANL-10

ANL-10 — Analytics Access Controls (who can query/export)
Header Block
template_id: ANL-10
title: Analytics Access Controls (who can query/export)
type: analytics_access_controls
template_version: 1.0.0
output_path: 10_ops/analytics/ANL-10_Analytics_Access_Controls.md
compliance_gate_id: TMP-05.PRIMARY.ANL
upstream_dependencies: ["IAM-06", "PRIV-02", "COMP-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IAM-06", "PRIV-02", "PRIV-06", "COMP-03", "ANL-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical access control policy for analytics tools and data: who can query, who can export, what approval is needed, how access is reviewed, and how sensitive fields are protected. This template must align with privileged access policy and PII tiering.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Privileged access policy: {{xref:IAM-06}} | OPTIONAL
PII classification: {{xref:PRIV-02}} | OPTIONAL
Data sharing map (analytics vendors): {{xref:PRIV-06}} | OPTIONAL
Vendor risk management: {{xref:COMP-03}} | OPTIONAL
Analytics retention/deletion: {{xref:ANL-09}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Analytics systems in scope (tool_id list)
Access roles (viewer, analyst, admin)
Query access rules (least privilege)
Export access rules (approvals required?)
Sensitive field protection (masking/unmasking)
Audit logging requirements (who queried/exported)
Access review cadence (periodic review)
Vendor access constraints (third-party accounts)
Telemetry requirements (exports, unusual queries)
Incident rule (suspicious access → IRP)

Optional Fields
Break-glass access for investigations | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Exports of sensitive data should be rare, approved, and logged.
Access must be periodically reviewed and revoked when not needed.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Systems
   tools: {{tools.list}}
2. Roles
   roles: {{roles.list}}
3. Query Access
   query_rule: {{access.query_rule}}
   role_mapping: {{access.role_mapping}} | OPTIONAL
4. Export Access
   export_rule: {{export.rule}}
   approval_rule: {{export.approval_rule}} | OPTIONAL
5. Sensitive Fields
   masking_rule: {{sensitive.masking_rule}}
   unmask_rule: {{sensitive.unmask_rule}} | OPTIONAL
6. Audit
   audit_required: {{audit.required}}
   audit_events: {{audit.events}} | OPTIONAL
   audit_ref: {{xref:AUDIT-01}} | OPTIONAL
7. Reviews
   review_cadence: {{reviews.cadence}}
   review_ref: {{xref:IAM-09}} | OPTIONAL
8. Vendor Constraints
   vendor_rule: {{vendor.rule}}
   account_management_rule: {{vendor.account_management_rule}} | OPTIONAL
9. Telemetry
   export_count_metric: {{telemetry.export_count_metric}}
   unusual_query_metric: {{telemetry.unusual_query_metric}} | OPTIONAL
10. Incident Handling
    incident_rule: {{incident.rule}}
    irp_ref: {{xref:IRP-02}} | OPTIONAL

Cross-References
Upstream: {{xref:IAM-06}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:COMP-10}}, {{xref:AUDIT-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define tools, roles, query/export rules, and masking rule.
intermediate: Required. Define audit events, reviews cadence, and telemetry.
advanced: Required. Add break-glass and strict vendor account governance and incident triggers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, role mapping, approval rule, unmask rule, audit events, review ref, vendor account mgmt, optional metrics, break-glass, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tools.list is UNKNOWN → block Completeness Gate.
If roles.list is UNKNOWN → block Completeness Gate.
If access.query_rule is UNKNOWN → block Completeness Gate.
If export.rule is UNKNOWN → block Completeness Gate.
If telemetry.export_count_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ANL
Pass conditions:
required_fields_present == true
tools_and_roles_defined == true
query_and_export_defined == true
audit_and_reviews_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Logging & Tracing Standards (LTS)

LTS-01

LTS-01 — Logging Standard (levels, structure, fields)
Header Block
template_id: LTS-01
title: Logging Standard (levels, structure, fields)
type: logging_standard
template_version: 1.0.0
output_path: 10_ops/logging/LTS-01_Logging_Standard.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["OBS-02", "SKM-09", "CER-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-02", "SKM-09", "CER-05", "AUDIT-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical logging standard across services and clients: log levels, structured format, required fields, correlation IDs, and redaction constraints. This standard is the source of truth for how logs are produced and consumed.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Telemetry schema standard: {{xref:OBS-02}} | OPTIONAL
Secrets/log redaction: {{xref:SKM-09}} | OPTIONAL
Client logging policy: {{xref:CER-05}} | OPTIONAL
Audit schema (correlation fields): {{xref:AUDIT-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Log levels and when to use (debug/info/warn/error)
Structured logging format rule (JSON fields)
Required common fields (service, env, timestamp)
Correlation fields required (request_id, trace_id)
Message field rule (human-readable + stable code)
Error object shape standard (error_code, stack policy)
PII/secret denylist rule (no tokens)
Sampling/volume control linkage (OBS-09/LTS-08)
Retention class rule (short/standard)
Validation/enforcement rule (lint/tests)
Telemetry requirements (missing field rate, invalid logs)

Optional Fields
Local dev logging behavior | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Logs must be structured; avoid free-form-only logs in prod.
Never log secrets/tokens; redact before emission.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Levels
   levels: {{levels.list}}
   usage_rules: {{levels.usage_rules}} | OPTIONAL
2. Format
   format: {{format.type}} (json/UNKNOWN)
   field_rule: {{format.field_rule}} | OPTIONAL
3. Required Fields
   required_fields: {{fields.required}}
4. Correlation
   request_id_field: {{corr.request_id_field}}
   trace_id_field: {{corr.trace_id_field}} | OPTIONAL
5. Message & Codes
   message_rule: {{msg.rule}}
   stable_code_rule: {{msg.stable_code_rule}} | OPTIONAL
6. Errors
   error_shape: {{err.shape}}
   stack_policy_ref: {{xref:LTS-03}} | OPTIONAL
7. Redaction
   denylist_fields: {{redact.denylist_fields}}
   redaction_ref: {{xref:SKM-09}} | OPTIONAL
8. Sampling / Volume
   volume_control_ref: {{xref:LTS-08}} | OPTIONAL
   sampling_ref: {{xref:OBS-09}} | OPTIONAL
9. Retention
   retention_rule: {{retention.rule}}
10. Enforcement
    enforcement_rule: {{enforce.rule}}
    ci_gate_ref: {{enforce.ci_gate_ref}} | OPTIONAL
11. Telemetry
    missing_field_metric: {{telemetry.missing_field_metric}}
    invalid_log_metric: {{telemetry.invalid_log_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OBS-05}}, {{xref:LTS-03}}, {{xref:LTS-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define levels, required fields, denylist, retention, enforcement.
intermediate: Required. Define correlation fields, error shape, sampling linkage, telemetry metrics.
advanced: Required. Add stable codes conventions and local dev behavior and strict CI gating.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, usage rules, field rule, trace id field, stable code rule, refs, ci gate ref, optional metrics, local dev, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If levels.list is UNKNOWN → block Completeness Gate.
If fields.required is UNKNOWN → block Completeness Gate.
If redact.denylist_fields is UNKNOWN → block Completeness Gate.
If retention.rule is UNKNOWN → block Completeness Gate.
If enforce.rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
levels_and_format_defined == true
correlation_defined == true
redaction_defined == true
retention_defined == true
enforcement_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-02

LTS-02 — Tracing Standard (trace/span IDs, propagation)
Header Block
template_id: LTS-02
title: Tracing Standard (trace/span IDs, propagation)
type: tracing_standard
template_version: 1.0.0
output_path: 10_ops/logging/LTS-02_Tracing_Standard.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["OBS-04", "OBS-02", "API-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-02", "OBS-04", "API-02", "JBS-02", "WS-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical tracing standard used across all services/clients: which propagation format is used, how trace/span IDs are handled, what must be propagated across boundaries, and what is prohibited in spans. This standard is the shared rulebook that OBS-04 implements.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Telemetry schema: {{xref:OBS-02}} | OPTIONAL
Trace instrumentation standard: {{xref:OBS-04}} | OPTIONAL
Endpoint specs (auth/errors): {{xref:API-02}} | OPTIONAL
Job specs: {{xref:JBS-02}} | OPTIONAL
WebSocket spec: {{xref:API-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Propagation format (w3c tracecontext/b3/UNKNOWN)
Inbound extraction rule (read incoming trace headers)
Outbound injection rule (set outgoing headers)
Trace ID / Span ID format rule
Cross-boundary rules (HTTP, WS, jobs, events)
Correlation with logs (fields align with LTS-01/OBS-02)
Error status mapping (span status + error tags)
Privacy/redaction rule for attributes
Sampling rule reference (OBS-09)
Verification rule (propagation tests)
Telemetry requirements (missing propagation rate)

Optional Fields
Baggage usage rules | OPTIONAL
Third-party SDK constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Propagation must be consistent across services; do not mix formats without explicit gateways.
Never include raw PII or secrets in trace attributes or baggage.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Propagation Format
   format: {{prop.format}}
   headers: {{prop.headers}} | OPTIONAL
2. Extraction / Injection
   inbound_rule: {{prop.inbound_rule}}
   outbound_rule: {{prop.outbound_rule}}
3. ID Formats
   trace_id_rule: {{ids.trace_id_rule}}
   span_id_rule: {{ids.span_id_rule}} | OPTIONAL
4. Boundaries
   http_rule: {{bound.http_rule}}
   ws_rule: {{bound.ws_rule}} | OPTIONAL
   jobs_rule: {{bound.jobs_rule}} | OPTIONAL
   events_rule: {{bound.events_rule}} | OPTIONAL
5. Log Correlation
   log_fields_alignment_rule: {{corr.log_alignment_rule}}
   logging_ref: {{xref:LTS-01}} | OPTIONAL
6. Error Mapping
   span_status_rule: {{err.span_status_rule}}
   error_tag_rule: {{err.error_tag_rule}} | OPTIONAL
7. Privacy / Redaction
   attribute_denylist: {{privacy.attribute_denylist}}
   baggage_denylist: {{privacy.baggage_denylist}} | OPTIONAL
8. Sampling
   sampling_ref: {{xref:OBS-09}} | OPTIONAL
   sampling_rule: {{sampling.rule}} | OPTIONAL
9. Verification
   tests_rule: {{verify.tests_rule}}
   coverage_target: {{verify.coverage_target}} | OPTIONAL
10. Telemetry
    missing_propagation_metric: {{telemetry.missing_propagation_metric}}

Cross-References
Upstream: {{xref:OBS-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LTS-03}}, {{xref:OBS-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define format and inbound/outbound rules and boundary handling.
intermediate: Required. Define log correlation and error mapping and denylist rules and telemetry.
advanced: Required. Add baggage/third-party constraints and strict verification coverage targets.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, headers, span id rule, boundary optional rules, logging ref, error tag rule, baggage denylist, sampling rule, coverage target, baggage/sdk constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If prop.format is UNKNOWN → block Completeness Gate.
If prop.inbound_rule is UNKNOWN → block Completeness Gate.
If prop.outbound_rule is UNKNOWN → block Completeness Gate.
If bound.http_rule is UNKNOWN → block Completeness Gate.
If telemetry.missing_propagation_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
propagation_defined == true
boundary_rules_defined == true
privacy_defined == true
verification_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-03

LTS-03 — Error Logging Standard (error shapes, stack rules)
Header Block
template_id: LTS-03
title: Error Logging Standard (error shapes, stack rules)
type: error_logging_standard
template_version: 1.0.0
output_path: 10_ops/logging/LTS-03_Error_Logging_Standard.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["API-03", "LTS-01", "OBS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-03", "LTS-01", "OBS-02", "CER-01", "IRP-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical standard for logging errors and exceptions: consistent error objects, stable error codes, stack trace handling rules, and correlation requirements. This standard ensures errors are diagnosable without leaking sensitive data.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Error/status policy: {{xref:API-03}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Telemetry schema: {{xref:OBS-02}} | OPTIONAL
Client error boundary strategy: {{xref:CER-01}} | OPTIONAL
Incident severity model: {{xref:IRP-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Error object schema (fields list)
Stable error code rules (namespace, format)
Error classification (user error vs system error)
Stack trace rules (when allowed, how truncated)
PII/secret redaction rule (no raw payloads)
Correlation requirement (request_id/trace_id always)
Mapping to HTTP status codes (API-03 refs)
Client vs server error logging differences
Alerting trigger guidance (what counts as paging)
Telemetry requirements (error rate metric, unhandled exceptions)

Optional Fields
Error fingerprinting rule | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Never log request/response bodies containing PII/secrets.
Error codes must be stable across releases (avoid changing semantics).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Error Object Schema
   fields: {{err.fields}}
   required_fields: {{err.required_fields}} | OPTIONAL
2. Error Codes
   format_rule: {{codes.format_rule}}
   namespacing_rule: {{codes.namespacing_rule}} | OPTIONAL
3. Classification
   classification_rules: {{class.rules}}
4. Stack Traces
   stack_allowed_rule: {{stack.allowed_rule}}
   truncation_rule: {{stack.truncation_rule}} | OPTIONAL
5. Redaction
   denylist_fields: {{redact.denylist_fields}}
   payload_logging_rule: {{redact.payload_rule}} | OPTIONAL
6. Correlation
   correlation_required_fields: {{corr.required_fields}}
7. Status Mapping
   status_mapping_ref: {{xref:API-03}} | OPTIONAL
   mapping_rule: {{status.mapping_rule}} | OPTIONAL
8. Client vs Server
   client_rule: {{diff.client_rule}}
   server_rule: {{diff.server_rule}} | OPTIONAL
9. Alerting Guidance
   paging_rule: {{alert.paging_rule}}
   severity_ref: {{xref:IRP-01}} | OPTIONAL
10. Telemetry
    error_rate_metric: {{telemetry.error_rate_metric}}
    unhandled_exception_metric: {{telemetry.unhandled_exception_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:LTS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define error object fields, correlation fields, and redaction rules.
intermediate: Required. Define code format + status mapping + paging rule + telemetry.
advanced: Required. Add fingerprinting and strict stack rules and client/server differentiation rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, required fields list, namespacing rule, truncation rule, payload rule, status mapping rule, server rule, severity ref, optional metrics, fingerprinting, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If err.fields is UNKNOWN → block Completeness Gate.
If codes.format_rule is UNKNOWN → block Completeness Gate.
If corr.required_fields is UNKNOWN → block Completeness Gate.
If alert.paging_rule is UNKNOWN → block Completeness Gate.
If telemetry.error_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
error_schema_defined == true
correlation_defined == true
redaction_defined == true
alerting_guidance_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-04

LTS-04 — PII/Secret Redaction Standard (denylist, masking)
Header Block
template_id: LTS-04
title: PII/Secret Redaction Standard (denylist, masking)
type: pii_secret_redaction_standard
template_version: 1.0.0
output_path: 10_ops/logging/LTS-04_PII_Secret_Redaction_Standard.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["PRIV-02", "SKM-09", "OBS-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PRIV-02", "PRIV-03", "SKM-09", "OBS-06", "AUDIT-08"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical redaction standard used across logs, traces, analytics, crash reports, and audit-adjacent logs: denylist fields, masking rules, hashing rules, and enforcement. This is the global rulebook for “what must never appear” in telemetry.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
PII classification model: {{xref:PRIV-02}} | OPTIONAL
Data minimization rules: {{xref:PRIV-03}} | OPTIONAL
Secrets logging rules: {{xref:SKM-09}} | OPTIONAL
Telemetry redaction rules: {{xref:OBS-06}} | OPTIONAL
Audit PII handling: {{xref:AUDIT-08}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Global denylist fields (auth headers, cookies, tokens, secrets)
PII tier rules (what is masked/hashed/dropped)
Masking patterns (email/phone)
Hashing rule (salted hash)
Structured redaction rule (field-based)
Unstructured redaction rule (regex scanning)
Redaction application points (before export, before sink)
Allowlist policy (what can remain)
Verification rule (tests/scanners)
Incident procedure if leakage detected (SKM-08)
Telemetry requirements (redaction hits, leakage detections)

Optional Fields
Per-sink differences (SIEM vs app logs) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Redaction must occur before data leaves the process (not just downstream).
Default to drop over mask when uncertain.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Denylist
   fields: {{deny.fields}}
2. PII Tier Rules
   tier_rules: {{pii.tier_rules}}
3. Masking
   patterns: {{mask.patterns}}
   examples: {{mask.examples}} | OPTIONAL
4. Hashing
   hash_rule: {{hash.rule}}
   salt_key_ref: {{hash.salt_key_ref}} | OPTIONAL
5. Structured Redaction
   field_based_rule: {{struct.rule}}
6. Unstructured Redaction
   regex_rule: {{unstruct.regex_rule}}
   scanner_rule: {{unstruct.scanner_rule}} | OPTIONAL
7. Application Points
   apply_before_emit_rule: {{apply.before_emit_rule}}
   apply_before_export_rule: {{apply.before_export_rule}} | OPTIONAL
8. Allowlist
   allowlist_policy: {{allow.policy}}
   allow_fields: {{allow.fields}} | OPTIONAL
9. Verification
   verification_rule: {{verify.rule}}
   tooling_ref: {{verify.tooling_ref}} | OPTIONAL
10. Leakage Procedure
    leak_rule: {{leak.rule}}
    compromise_ref: {{xref:SKM-08}} | OPTIONAL
11. Telemetry
    redaction_hit_metric: {{telemetry.redaction_hit_metric}}
    leak_detect_metric: {{telemetry.leak_detect_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PRIV-02}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:LTS-09}}, {{xref:ALRT-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define denylist, tier rules, apply-before-emit rule, verification rule.
intermediate: Required. Define masking patterns and hashing rule and telemetry metrics.
advanced: Required. Add per-sink differences and strict allowlist governance and scanner coverage targets.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, examples, salt key ref, scanner rule, apply-before-export, allow fields, tooling ref, optional metric, per-sink, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If deny.fields is UNKNOWN → block Completeness Gate.
If pii.tier_rules is UNKNOWN → block Completeness Gate.
If apply.before_emit_rule is UNKNOWN → block Completeness Gate.
If verify.rule is UNKNOWN → block Completeness Gate.
If telemetry.redaction_hit_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
denylist_and_tier_rules_defined == true
masking_and_hashing_defined == true
verification_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-05

LTS-05 — Log Routing & Storage (sinks, retention classes)
Header Block
template_id: LTS-05
title: Log Routing & Storage (sinks, retention classes)
type: log_routing_storage
template_version: 1.0.0
output_path: 10_ops/logging/LTS-05_Log_Routing_Storage.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["LTS-01", "AUDIT-05", "OBS-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LTS-01", "LTS-04", "AUDIT-05", "OBS-09", "COST-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical routing and storage model for logs: which sinks exist, what routes to where, retention classes/durations, and access controls. This template must align with audit retention controls and sampling/cost budgets.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Redaction standard: {{xref:LTS-04}} | OPTIONAL
Audit retention/access controls: {{xref:AUDIT-05}} | OPTIONAL
Sampling/cardinality policy: {{xref:OBS-09}} | OPTIONAL
Cost drivers: {{xref:COST-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Log sinks inventory (sink_id list)
sink_id (stable identifier)
Sink type (stdout/agent/siem/object_store)
Routing rules (which logs go to which sink)
Retention classes (short/standard/long)
Retention days per class
Partitioning/indexing rule (by env/service)
Access control rule (who can query)
Export policy (who can export)
Encryption rule (at rest/in transit)
Telemetry requirements (ingest failures, lag)

Optional Fields
Cost controls (caps/quotas) | OPTIONAL
Archive policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Audit logs must follow AUDIT retention/access rules; do not co-mingle with app logs without clear boundaries.
Redaction must happen before routing to sinks.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Sinks
   sinks: {{sinks.list}}
2. Routing
   routes: {{routes.rules}}
3. Retention
   classes: {{retention.classes}}
   short_days: {{retention.short_days}}
   standard_days: {{retention.standard_days}}
   long_days: {{retention.long_days}}
4. Partitioning / Indexing
   partition_rule: {{storage.partition_rule}}
   index_rule: {{storage.index_rule}} | OPTIONAL
5. Access / Export
   query_access_rule: {{access.query_rule}}
   export_rule: {{access.export_rule}} | OPTIONAL
6. Encryption
   encryption_rule: {{crypto.encryption_rule}}
7. Telemetry
   ingest_fail_metric: {{telemetry.ingest_fail_metric}}
   ingest_lag_metric: {{telemetry.ingest_lag_metric}} | OPTIONAL
8. References
   Redaction standard: {{xref:LTS-04}} | OPTIONAL
   Audit retention: {{xref:AUDIT-05}} | OPTIONAL
   Sampling policy: {{xref:OBS-09}} | OPTIONAL

Cross-References
Upstream: {{xref:LTS-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LTS-08}}, {{xref:ALRT-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define sinks, routing, retention days, and access rule.
intermediate: Required. Define export policy, encryption rule, and telemetry metrics.
advanced: Required. Add cost caps/archive policy and strict partitioning/indexing guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, routes rules, index rule, export rule, optional metrics, cost/archive, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If sinks.list is UNKNOWN → block Completeness Gate.
If retention.standard_days is UNKNOWN → block Completeness Gate.
If access.query_rule is UNKNOWN → block Completeness Gate.
If crypto.encryption_rule is UNKNOWN → block Completeness Gate.
If telemetry.ingest_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
sinks_and_retention_defined == true
routing_defined == true
access_and_encryption_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-06

LTS-06 — Client Logging Standard (mobile/web constraints)
Header Block
template_id: LTS-06
title: Client Logging Standard (mobile/web constraints)
type: client_logging_standard
template_version: 1.0.0
output_path: 10_ops/logging/LTS-06_Client_Logging_Standard.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["CER-05", "CSec-02", "LTS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CER-05", "CSec-02", "LTS-01", "OBS-02", "MOB-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical logging standard for client apps (web + mobile): what is allowed to be logged on-device, what can be shipped to servers, redaction rules, offline buffering limits, and crash logging behavior. This standard must align with client security and privacy constraints.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Client logging/crash reporting: {{xref:CER-05}} | OPTIONAL
Client data protection: {{xref:CSec-02}} | OPTIONAL
Global logging standard: {{xref:LTS-01}} | OPTIONAL
Telemetry schema: {{xref:OBS-02}} | OPTIONAL
App lifecycle/state: {{xref:MOB-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Allowed client log types (ui, network, crash)
On-device storage limits (size/time)
Upload/shipping rules (when logs sent)
Redaction rules (same denylist, extra client rules)
PII/sensitive UI content rule (no screenshots text)
Network logging constraints (no headers/bodies)
Offline buffering rule (queue + flush)
Crash reporting rule (what captured)
User controls (opt-out, privacy)
Telemetry requirements (upload failures, dropped logs)

Optional Fields
Debug mode rules (dev only) | OPTIONAL

Platform differences (iOS/Android/web) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Never log auth tokens/cookies or request bodies on clients.
Client logs must be bounded and safe to ship (redacted).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Allowed Types
   types: {{types.allowed}}
2. On-Device Limits
   max_storage_mb: {{local.max_storage_mb}}
   max_age_days: {{local.max_age_days}} | OPTIONAL
3. Shipping
   upload_rule: {{ship.upload_rule}}
   flush_triggers: {{ship.flush_triggers}} | OPTIONAL
4. Redaction
   denylist_fields: {{redact.denylist_fields}}
   client_extra_rules: {{redact.client_extra_rules}} | OPTIONAL
5. Sensitive UI Content
   no_sensitive_ui_rule: {{ui.no_sensitive_ui_rule}}
6. Network Logging
   no_headers_rule: {{net.no_headers_rule}}
   no_body_rule: {{net.no_body_rule}} | OPTIONAL
7. Offline Buffering
   buffer_rule: {{offline.buffer_rule}}
   retry_rule: {{offline.retry_rule}} | OPTIONAL
8. Crash Reporting
   crash_payload_rule: {{crash.payload_rule}}
   stack_policy: {{crash.stack_policy}} | OPTIONAL
9. User Controls
   opt_out_rule: {{user.opt_out_rule}}
   consent_ref: {{xref:PRIV-04}} | OPTIONAL
10. Telemetry
    client_log_upload_fail_metric: {{telemetry.upload_fail_metric}}
    client_log_drop_metric: {{telemetry.drop_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:CER-05}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LTS-09}}, {{xref:ALRT-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define allowed types, on-device limits, redaction, and network constraints.
intermediate: Required. Define offline buffering, crash rules, user controls, telemetry metrics.
advanced: Required. Add debug/platform differences and strict shipping/flush triggers with safety checks.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, max age days, flush triggers, extra rules, retry rule, stack policy, consent ref, optional metrics, debug/platform differences, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If types.allowed is UNKNOWN → block Completeness Gate.
If local.max_storage_mb is UNKNOWN → block Completeness Gate.
If ship.upload_rule is UNKNOWN → block Completeness Gate.
If redact.denylist_fields is UNKNOWN → block Completeness Gate.
If ui.no_sensitive_ui_rule is UNKNOWN → block Completeness Gate.
If telemetry.upload_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
types_and_limits_defined == true
shipping_and_redaction_defined == true
network_and_ui_constraints_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-07

LTS-07 — Audit vs App Logs Boundary (what goes where)
Header Block
template_id: LTS-07
title: Audit vs App Logs Boundary (what goes where)
type: audit_vs_app_logs_boundary
template_version: 1.0.0
output_path: 10_ops/logging/LTS-07_Audit_vs_App_Logs_Boundary.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["AUDIT-01", "AUDIT-02", "LTS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "AUDIT-01", "AUDIT-02", "LTS-01", "ADMIN-03", "AUDIT-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical boundary between audit logs (security/compliance evidence) and application logs (debug/ops): what events must be emitted as AUDIT events, what can remain in app logs, duplication rules, and retention/access implications. This reduces compliance risk and avoids missing critical evidence.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Audit event catalog: {{xref:AUDIT-01}} | OPTIONAL
Audit schema: {{xref:AUDIT-02}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Admin audit trail: {{xref:ADMIN-03}} | OPTIONAL
Audit capture rules: {{xref:AUDIT-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Definition of audit log vs app log
Event classification rules (security, admin, payment, data access)
Must-be-audit events list (event_type refs)
Allowed-in-app-log events list (log_event refs)
Duplication rule (when both emitted)
PII constraints (audit may include minimal identifiers, app logs stricter)
Access control implications (who can read audit vs app logs)
Retention implications (audit long, app shorter)
Implementation guidance (where instrumentation happens)
Telemetry requirements (misclassified events count)

Optional Fields
Examples (audit vs app log examples) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Privileged actions and authorization decisions should be AUDIT, not app-only logs.
Do not copy full audit payloads into app logs.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Definitions
   audit_log_definition: {{defs.audit}}
   app_log_definition: {{defs.app}}
2. Classification Rules
   rules: {{class.rules}}
3. Must-Be-Audit
   event_types: {{audit.must_event_types}}
4. Allowed-in-App
   log_events: {{app.allowed_log_events}}
5. Duplication
   dup_rule: {{dup.rule}}
6. PII Constraints
   audit_pii_rule: {{pii.audit_rule}}
   app_pii_rule: {{pii.app_rule}} | OPTIONAL
7. Access & Retention
   audit_access_ref: {{xref:AUDIT-05}} | OPTIONAL
   app_log_storage_ref: {{xref:LTS-05}} | OPTIONAL
   retention_rule: {{retain.rule}}
8. Implementation Guidance
   capture_points: {{impl.capture_points}}
   anti_duplication_rule: {{impl.anti_dup_rule}} | OPTIONAL
9. Telemetry
   misclassification_metric: {{telemetry.misclassification_metric}}
10. References
    Audit catalog: {{xref:AUDIT-01}} | OPTIONAL
    Audit schema: {{xref:AUDIT-02}} | OPTIONAL
    Capture rules: {{xref:AUDIT-03}} | OPTIONAL

Cross-References
Upstream: {{xref:AUDIT-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LTS-05}}, {{xref:AUDIT-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define definitions and must-be-audit list and retention rule.
intermediate: Required. Define classification rules, duplication, access implications, telemetry.
advanced: Required. Add examples and strict guidance for preventing audit payload duplication into app logs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, app pii rule, access refs, anti-dup rule, examples, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If defs.audit is UNKNOWN → block Completeness Gate.
If audit.must_event_types is UNKNOWN → block Completeness Gate.
If class.rules is UNKNOWN → block Completeness Gate.
If retain.rule is UNKNOWN → block Completeness Gate.
If telemetry.misclassification_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
definitions_defined == true
classification_defined == true
retention_and_access_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-08

LTS-08 — Log Volume Controls (sampling, rate limits)
Header Block
template_id: LTS-08
title: Log Volume Controls (sampling, rate limits)
type: log_volume_controls
template_version: 1.0.0
output_path: 10_ops/logging/LTS-08_Log_Volume_Controls.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["OBS-09", "LTS-01", "COST-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LTS-01", "OBS-09", "LTS-05", "COST-02", "ALRT-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical controls for log volume and cost: per-level policies, sampling rules, rate limiting, and guardrails that prevent log storms. This template must align with sampling/cardinality policy and log routing/storage costs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Sampling/cardinality policy: {{xref:OBS-09}} | OPTIONAL
Log routing/storage: {{xref:LTS-05}} | OPTIONAL
Cost drivers: {{xref:COST-02}} | OPTIONAL
Noise reduction: {{xref:ALRT-05}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Volume control scope (server/client)
Per-level volume rule (debug in prod?)
Sampling rules for noisy logs
Rate limiting rule (per message key)
Burst handling (log storms)
Allowlist for non-sampled critical logs (security/audit-adjacent)
Backpressure rule (drop strategy)
Telemetry requirements (dropped logs, rate-limit hits)
Runbook references (how to respond to log storms)
Change control rule (who can change sampling)

Optional Fields
Auto-throttling notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Do not sample away security/audit-critical logs beyond defined allowlist.
Prefer dropping DEBUG/INFO before WARN/ERROR under pressure.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Scope
   scope: {{scope.value}}
2. Per-Level Rules
   rules: {{levels.rules}}
3. Sampling
   sampling_rules: {{sampling.rules}}
   critical_allowlist: {{sampling.critical_allowlist}} | OPTIONAL
4. Rate Limiting
   rate_limit_rule: {{rl.rule}}
   keying_rule: {{rl.keying_rule}} | OPTIONAL
5. Burst Handling
   storm_rule: {{storm.rule}}
   drop_priority_rule: {{storm.drop_priority_rule}} | OPTIONAL
6. Backpressure
   backpressure_rule: {{bp.rule}}
7. Telemetry
   dropped_log_metric: {{telemetry.dropped_log_metric}}
   rate_limit_hit_metric: {{telemetry.rate_limit_hit_metric}} | OPTIONAL
8. Runbooks
   log_storm_runbook_ref: {{xref:LTS-10}} | OPTIONAL
   incident_ref: {{xref:IRP-02}} | OPTIONAL
9. Change Control
   who_can_change: {{change.who_can_change}}
   approval_rule: {{change.approval_rule}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-09}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:COST-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define per-level rules, sampling rules, and telemetry metric(s).

intermediate: Required. Define rate limiting, storm handling, and change control.
advanced: Required. Add auto-throttling and strict critical allowlists + runbook mapping.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, critical allowlist, keying rule, drop priority rule, optional metrics, runbook refs, approval rule, auto-throttling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If levels.rules is UNKNOWN → block Completeness Gate.
If sampling.rules is UNKNOWN → block Completeness Gate.
If bp.rule is UNKNOWN → block Completeness Gate.
If telemetry.dropped_log_metric is UNKNOWN → block Completeness Gate.
If change.who_can_change is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
volume_controls_defined == true
storm_and_backpressure_defined == true
telemetry_defined == true
change_control_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-09

LTS-09 — Logging Verification (tests, scanners)
Header Block
template_id: LTS-09
title: Logging Verification (tests, scanners)
type: logging_verification
template_version: 1.0.0
output_path: 10_ops/logging/LTS-09_Logging_Verification.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["LTS-01", "LTS-04", "CICD-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LTS-01", "LTS-04", "OBS-02", "CICD-04", "QA-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical verification approach that ensures logging standards are followed: tests for required fields, scanners for secrets/PII, validation for structured logging, and CI gating rules. This template prevents log regressions and sensitive data leakage.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Logging standard: {{xref:LTS-01}} | OPTIONAL
Redaction standard: {{xref:LTS-04}} | OPTIONAL
Telemetry schema: {{xref:OBS-02}} | OPTIONAL
Security gates (SAST/SCA/secret scan): {{xref:CICD-04}} | OPTIONAL
Release quality gates: {{xref:QA-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Verification layers (unit/integration/e2e)
Structured logging validation rule (JSON parse + schema)
Required field checks (OBS-02/LTS-01)
Secret/PII scanning rule (LTS-04 denylist)
Snapshot/contract tests for log events (OBS-05)
CI gate rule (block merges on failures)
Allowed exceptions process (how waived)
Telemetry requirements (verification failures rate)
Ownership (who maintains scanners/tests)
Runbook for failed gates

Optional Fields
Staging canary validation | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
CI must block on secret/PII leakage detection.
Verification should be deterministic; avoid flaky scans.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Layers
   layers: {{layers.list}}
2. Structured Validation
   schema_rule: {{validate.schema_rule}}
   parser_rule: {{validate.parser_rule}} | OPTIONAL
3. Required Field Checks
   fields_rule: {{fields.rule}}
   required_fields: {{fields.required_fields}} | OPTIONAL
4. Scanning
   denylist_ref: {{xref:LTS-04}} | OPTIONAL
   scanner_rule: {{scan.scanner_rule}}
   false_positive_rule: {{scan.false_positive_rule}} | OPTIONAL
5. Contract Tests
   log_event_catalog_ref: {{xref:OBS-05}} | OPTIONAL
   contract_test_rule: {{contract.rule}}
6. CI Gate
   ci_gate_rule: {{ci.rule}}
   pipeline_ref: {{xref:CICD-04}} | OPTIONAL
7. Exceptions
   waiver_rule: {{exceptions.waiver_rule}}
   exceptions_ref: {{xref:COMP-08}} | OPTIONAL
8. Telemetry
   verification_fail_metric: {{telemetry.verification_fail_metric}}
9. Ownership
   owner: {{owner.team}}
10. Runbook
    failure_runbook: {{runbook.failure_runbook}}

Cross-References
Upstream: {{xref:LTS-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:QA-07}}, {{xref:SEC-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define layers, scanner rule, CI gate rule, runbook.
intermediate: Required. Define contract tests, exceptions, ownership, telemetry.
advanced: Required. Add staging canary validation and strict false-positive handling and waiver governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, parser rule, required fields list, denylist ref, false positive rule, pipeline ref, exceptions ref, open_questions, staging canary
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If layers.list is UNKNOWN → block Completeness Gate.
If scan.scanner_rule is UNKNOWN → block Completeness Gate.
If ci.rule is UNKNOWN → block Completeness Gate.
If telemetry.verification_fail_metric is UNKNOWN → block Completeness Gate.
If runbook.failure_runbook is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
verification_layers_defined == true
scanning_defined == true
ci_gate_defined == true
runbook_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LTS-10

LTS-10 — Logging Runbooks (common issues)
Header Block
template_id: LTS-10
title: Logging Runbooks (common issues)
type: logging_runbooks
template_version: 1.0.0
output_path: 10_ops/logging/LTS-10_Logging_Runbooks.md
compliance_gate_id: TMP-05.PRIMARY.LTS
upstream_dependencies: ["LTS-08", "LTS-09", "IRP-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LTS-08", "LTS-09", "LTS-05", "ALRT-02", "IRP-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical operational runbooks for logging incidents and common failures: log storms, missing correlation IDs, redaction leaks, sink ingest failures, and noisy services. This template provides step-by-step response procedures.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Log volume controls: {{xref:LTS-08}} | OPTIONAL
Logging verification: {{xref:LTS-09}} | OPTIONAL
Log routing/storage: {{xref:LTS-05}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Incident workflow: {{xref:IRP-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Runbook registry (runbook_id list)
runbook_id (stable identifier)
Scenario type (storm/missing_fields/leak/ingest_fail)
Trigger signals (alerts/metrics)
Triage steps (ordered)
Containment steps (ordered)
Verification steps (prove fixed)
Escalation rules (who to page)
Post-incident follow-ups (prevent recurrence)
Telemetry requirements (runbook invoked, time-to-contain)

Optional Fields
Automation hooks | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Runbooks must be executable and ordered; no vague steps.
Containment must prefer safe actions (reduce volume, preserve evidence).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_runbooks: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Runbooks (repeat)
   Runbook
   runbook_id: {{books[0].runbook_id}}
   scenario: {{books[0].scenario}}
   triggers: {{books[0].triggers}}
   triage_steps:
   {{books[0].triage_steps[0]}}
   {{books[0].triage_steps[1]}}
   containment_steps:
   {{books[0].containment_steps[0]}}
   {{books[0].containment_steps[1]}} | OPTIONAL
   verification_steps: {{books[0].verification_steps}}
   escalation_rule: {{books[0].escalation_rule}}
   followups: {{books[0].followups}} | OPTIONAL
   telemetry_metric: {{books[0].telemetry_metric}}
   automation_hooks: {{books[0].automation_hooks}} | OPTIONAL
   open_questions:
   {{books[0].open_questions[0]}} | OPTIONAL

(Repeat per runbook.)

3. References
   Log routing: {{xref:LTS-05}} | OPTIONAL
   Volume controls: {{xref:LTS-08}} | OPTIONAL
   Verification gates: {{xref:LTS-09}} | OPTIONAL
   Incident workflow: {{xref:IRP-02}} | OPTIONAL

Cross-References
Upstream: {{xref:LTS-08}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:IRP-06}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define at least core runbooks with triage/contain/verify steps.
intermediate: Required. Add escalation and follow-ups and telemetry metrics.
advanced: Required. Add automation hooks and tighter trigger mapping + evidence preservation.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, meta notes, optional containment step, followups, automation hooks, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If books[*].runbook_id is UNKNOWN → block Completeness Gate.
If books[*].triage_steps[0] is UNKNOWN → block Completeness Gate.
If books[*].containment_steps[0] is UNKNOWN → block Completeness Gate.
If books[*].verification_steps is UNKNOWN → block Completeness Gate.
If books[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LTS
Pass conditions:
required_fields_present == true
runbooks_defined == true
triage_and_containment_defined == true
verification_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Monitoring & Alerting (ALRT)

ALRT-01

ALRT-01 — Alerting Overview (principles, severity model)
Header Block
template_id: ALRT-01
title: Alerting Overview (principles, severity model)
type: alerting_overview
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-01_Alerting_Overview.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["IRP-01", "SLO-01", "OBS-08"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IRP-01", "SLO-01", "OBS-08", "ALRT-02", "ALRT-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of alerting for the system: what alerts are for, how severity is defined, what should page vs ticket, and how alerts tie into incident response and SLO burn. This document anchors the ALRT set.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Incident severity model: {{xref:IRP-01}} | OPTIONAL
SLO/SLI overview: {{xref:SLO-01}} | OPTIONAL
Alert routing/ownership: {{xref:OBS-08}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Oncall policy: {{xref:ALRT-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Alerting principles list (actionable, low noise)
Severity model (sev levels + meaning)
Paging vs non-paging criteria
SLO burn linkage rule (alerts tied to error budgets)
Ownership model (who owns alerts)
Alert lifecycle (create, tune, retire)
Change control rule (who edits alerts)
Runbook requirement (alerts must link to runbooks)
Telemetry requirements (alerts fired, false positives)

Optional Fields
Alert review cadence | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Paging alerts must be actionable and owned; otherwise convert to ticket.
Every paging alert must have a runbook and escalation path.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Principles
   {{principles[0]}}
   {{principles[1]}}
   {{principles[2]}} | OPTIONAL
2. Severity Model
   levels: {{sev.levels}}
   definitions: {{sev.definitions}} | OPTIONAL
3. Paging Criteria
   paging_rule: {{page.rule}}
   non_paging_rule: {{page.non_paging_rule}} | OPTIONAL
4. SLO / Error Budget Linkage
   slo_link_rule: {{slo.link_rule}}
   burn_alert_ref: {{xref:SLO-07}} | OPTIONAL
5. Ownership
   ownership_rule: {{own.rule}}
   routing_ref: {{xref:OBS-08}} | OPTIONAL
6. Lifecycle
   lifecycle_steps: {{life.steps}}
   retire_rule: {{life.retire_rule}} | OPTIONAL
7. Change Control
   who_can_edit: {{change.who_can_edit}}
   approval_rule: {{change.approval_rule}} | OPTIONAL
8. Runbooks
   runbook_required_rule: {{runbooks.required_rule}}
   runbook_index_ref: {{xref:IRP-10}} | OPTIONAL
9. Telemetry
   alerts_fired_metric: {{telemetry.alerts_fired_metric}}
   false_positive_metric: {{telemetry.false_positive_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:IRP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:ALRT-03}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define severity model, paging rule, ownership, runbook requirement.

intermediate: Required. Define SLO linkage, lifecycle steps, change control, telemetry.
advanced: Required. Add review cadence and explicit tuning/retirement thresholds.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional principles, definitions, non-paging rule, burn ref, retire rule, approval rule, runbook index ref, optional metric, review cadence, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If sev.levels is UNKNOWN → block Completeness Gate.
If page.rule is UNKNOWN → block Completeness Gate.
If own.rule is UNKNOWN → block Completeness Gate.
If runbooks.required_rule is UNKNOWN → block Completeness Gate.
If telemetry.alerts_fired_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
severity_and_paging_defined == true
ownership_defined == true
runbooks_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-02

ALRT-02 — Alert Catalog (by alert_id)
Header Block
template_id: ALRT-02
title: Alert Catalog (by alert_id)
type: alert_catalog
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-02_Alert_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["OBS-03", "SLO-02", "OBS-08"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-03", "SLO-02", "SLO-03", "OBS-08", "IRP-10"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of alerts, indexed by alert_id, including what each alert monitors, severity, routing, runbook links, and SLO relationships. This catalog is the source of truth for operational alert coverage.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Endpoint SLO catalog: {{xref:SLO-03}} | OPTIONAL
Alert routing/ownership: {{xref:OBS-08}} | OPTIONAL
Incident runbooks index: {{xref:IRP-10}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Alert registry (alert_id list)
alert_id (stable identifier)
Name/summary
Metric(s) used (metric_id refs)
Condition/threshold summary
Severity (sev1/sev2/sev3/UNKNOWN)
Routing target (team/oncall)
Runbook reference
SLO binding (slo_id) | OPTIONAL
Customer impact flag (yes/no/UNKNOWN)
Noise controls (dedupe/suppression)
Owner (team)
Telemetry requirements (alerts fired, ack time)

Optional Fields
Dashboard reference | OPTIONAL
Example alert payload | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Paging alerts must have owners, routing, and runbooks.
Alerts must reference existing metric_ids; do not invent metrics.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_alerts: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Alerts (repeat per alert_id)
   Alert
   alert_id: {{alerts[0].alert_id}}
   name: {{alerts[0].name}}
   metrics: {{alerts[0].metrics}}
   condition_summary: {{alerts[0].condition_summary}}
   severity: {{alerts[0].severity}}
   routing: {{alerts[0].routing}}
   owner: {{alerts[0].owner}}
   runbook_ref: {{alerts[0].runbook_ref}}
   slo_binding: {{alerts[0].slo_binding}} | OPTIONAL
   customer_impact: {{alerts[0].customer_impact}}
   noise_controls: {{alerts[0].noise_controls}}
   dashboard_ref: {{alerts[0].dashboard_ref}} | OPTIONAL
   example_payload: {{alerts[0].example_payload}} | OPTIONAL
   telemetry_metric: {{alerts[0].telemetry_metric}}
   ack_time_metric: {{alerts[0].ack_time_metric}} | OPTIONAL
   open_questions:
   {{alerts[0].open_questions[0]}} | OPTIONAL

(Repeat per alert.)

3. References
   Alert rule spec: {{xref:ALRT-03}} | OPTIONAL
   Oncall policy: {{xref:ALRT-04}} | OPTIONAL
   Noise reduction: {{xref:ALRT-05}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-03}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:ALRT-03}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define alerts with metrics, severity, routing, runbook ref, owner.
intermediate: Required. Define noise controls, customer impact, telemetry metrics.
advanced: Required. Add SLO bindings, dashboard refs, and example payloads + stricter thresholds docs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, slo binding, dashboard ref, example payload, ack time metric, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If alerts[*].alert_id is UNKNOWN → block Completeness Gate.*
*If alerts[*].metrics is UNKNOWN → block Completeness Gate.
If alerts[*].severity is UNKNOWN → block Completeness Gate.*
*If alerts[*].routing is UNKNOWN → block Completeness Gate.
If alerts[*].runbook_ref is UNKNOWN → block Completeness Gate.*
*If alerts[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
alert_registry_defined == true
metrics_and_conditions_defined == true
ownership_and_runbooks_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-03

ALRT-03 — Alert Rule Spec (condition, threshold, window)
Header Block
template_id: ALRT-03
title: Alert Rule Spec (condition, threshold, window)
type: alert_rule_spec
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-03_Alert_Rule_Spec.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["ALRT-02", "OBS-03", "SLO-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ALRT-02", "OBS-03", "SLO-06", "ALRT-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical specification format for alert rules: how conditions are written, threshold and window conventions, severity mapping, and required metadata. This template standardizes alert rules so they are consistent, reviewable, and tunable.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
SLO measurement rules: {{xref:SLO-06}} | OPTIONAL
Noise reduction rules: {{xref:ALRT-05}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Rule schema (fields list)
Condition syntax rule (query language)
Threshold conventions (absolute vs rate)
Window conventions (5m/15m/1h)
Severity mapping rules (condition → severity)
Dedup keys rule (how alerts grouped)
Suppression rules (maintenance windows)
Required metadata (owner, runbook, dashboard)
Testing/validation rule (dry runs)
Telemetry requirements (rule evaluation errors)

Optional Fields
Auto-tuning notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Rules must be explainable and stable; avoid overly complex conditions.
Every rule must declare dedupe keys and owner/runbook metadata.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Rule Schema
   fields: {{schema.fields}}
2. Condition Syntax
   syntax_rule: {{cond.syntax_rule}}
   examples: {{cond.examples}} | OPTIONAL
3. Thresholds
   threshold_rule: {{thresh.rule}}
   rate_rule: {{thresh.rate_rule}} | OPTIONAL
4. Windows
   window_rule: {{window.rule}}
   default_windows: {{window.defaults}} | OPTIONAL
5. Severity Mapping
   mapping_rule: {{sev.mapping_rule}}
6. Dedupe
   dedupe_key_rule: {{dedupe.key_rule}}
7. Suppression
   suppression_rule: {{suppress.rule}}
   maintenance_ref: {{xref:REL-07}} | OPTIONAL
8. Metadata
   required_metadata: {{meta.required}}
   runbook_rule: {{meta.runbook_rule}} | OPTIONAL
9. Validation
   dry_run_rule: {{validate.dry_run_rule}}
   review_rule: {{validate.review_rule}} | OPTIONAL
10. Telemetry
    eval_error_metric: {{telemetry.eval_error_metric}}

Cross-References
Upstream: {{xref:ALRT-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-05}}, {{xref:ALRT-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define schema, condition syntax, windows, and required metadata.
intermediate: Required. Define thresholds, dedupe, suppression, validation, telemetry.
advanced: Required. Add auto-tuning notes and strict review/approval requirements.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, examples, rate rule, defaults, maintenance ref, runbook rule, review rule, auto-tuning, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If schema.fields is UNKNOWN → block Completeness Gate.
If cond.syntax_rule is UNKNOWN → block Completeness Gate.
If window.rule is UNKNOWN → block Completeness Gate.
If meta.required is UNKNOWN → block Completeness Gate.
If telemetry.eval_error_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
rule_schema_defined == true
metadata_defined == true
validation_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-04

ALRT-04 — Oncall & Escalation Policy (rota, paging)
Header Block
template_id: ALRT-04
title: Oncall & Escalation Policy (rota, paging)
type: oncall_escalation_policy
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-04_Oncall_Escalation_Policy.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["IRP-03", "OBS-08", "ALRT-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IRP-03", "OBS-08", "ALRT-01", "ALRT-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical oncall and escalation policy: how paging works, escalation ladders, coverage expectations, and handoff rules. This template standardizes “who gets paged” and ties alert routing to incident roles.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Incident roles/responsibilities: {{xref:IRP-03}} | OPTIONAL
Alert routing/ownership: {{xref:OBS-08}} | OPTIONAL
Alerting overview: {{xref:ALRT-01}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Coverage model (24/7 vs business hours)
Primary/secondary escalation ladder
Paging channels (phone, sms, app)
Ack expectations (time to acknowledge)
Escalation timing rules (minutes)
Handoff rules (shift change)
Backup roles (manager/escalation)
Incident commander assignment rule
Telemetry requirements (ack time, missed pages)
Change control rule (who edits schedules)

Optional Fields
Follow-the-sun policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Paging must have a primary owner and a backup path; no dead-ends.
Ack expectations should map to severity (sev1 faster).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Coverage
   coverage: {{coverage.model}}
   hours: {{coverage.hours}} | OPTIONAL
2. Ladders
   primary_ladder: {{ladder.primary}}
   secondary_ladder: {{ladder.secondary}} | OPTIONAL
3. Paging Channels
   channels: {{channels.list}}
4. Ack / Escalation Timing
   ack_sla_minutes: {{timing.ack_sla_minutes}}
   escalation_steps: {{timing.escalation_steps}}
5. Handoff
   handoff_rule: {{handoff.rule}}
   shift_change_rule: {{handoff.shift_change_rule}} | OPTIONAL
6. IC Assignment
   ic_rule: {{ic.rule}}
7. Telemetry
   ack_time_metric: {{telemetry.ack_time_metric}}
   missed_page_metric: {{telemetry.missed_page_metric}} | OPTIONAL
8. Change Control
   who_can_edit_schedules: {{change.who_can_edit_schedules}}
   approval_rule: {{change.approval_rule}} | OPTIONAL

Cross-References
Upstream: {{xref:IRP-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OBS-08}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define coverage model, ladders, ack SLA, and IC rule.
intermediate: Required. Define escalation steps, handoff rules, telemetry metrics.
advanced: Required. Add follow-the-sun and strict schedule change governance and severity-based ack mapping.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, hours, secondary ladder, shift change rule, approval rule, optional metrics, follow-the-sun, open_questions

If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If coverage.model is UNKNOWN → block Completeness Gate.
If ladder.primary is UNKNOWN → block Completeness Gate.
If timing.ack_sla_minutes is UNKNOWN → block Completeness Gate.
If timing.escalation_steps is UNKNOWN → block Completeness Gate.
If ic.rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
coverage_and_ladders_defined == true
timing_defined == true
handoff_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-05

ALRT-05 — Noise Reduction Rules (dedupe, suppression)
Header Block
template_id: ALRT-05
title: Noise Reduction Rules (dedupe, suppression)
type: alert_noise_reduction_rules
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-05_Noise_Reduction_Rules.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["ALRT-03", "OBS-09", "REL-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ALRT-03", "ALRT-02", "OBS-09", "REL-07", "IRP-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules that reduce alert noise: deduplication keys, grouping, suppression during maintenance, escalation dampening, and anti-flap policies. This template ensures alerting stays actionable.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Alert rule spec: {{xref:ALRT-03}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Sampling/cardinality policy: {{xref:OBS-09}} | OPTIONAL
Maintenance windows policy: {{xref:REL-07}} | OPTIONAL
Incident workflow: {{xref:IRP-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Noise control principles list
Deduplication rule (dedupe keys)
Grouping rule (by service/env)
Suppression rules (maintenance windows)
Anti-flap rule (cooldowns, hysteresis)
Escalation dampening (avoid repeated pages)
Ticketing conversion rule (non-paging)
Telemetry requirements (pages per day, suppressed count)
Review/tuning cadence

Optional Fields
Auto-silence rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Noise controls must not suppress true sev1 outages; suppression must be scoped and time-bounded.
Every suppression should have an owner and expiry.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Principles
   {{principles[0]}}
   {{principles[1]}} | OPTIONAL
2. Deduplication
   dedupe_rule: {{dedupe.rule}}
3. Grouping
   group_rule: {{group.rule}}
4. Suppression
   maintenance_ref: {{xref:REL-07}} | OPTIONAL
   suppression_rule: {{suppress.rule}}
   expiry_rule: {{suppress.expiry_rule}} | OPTIONAL
5. Anti-Flap
   anti_flap_rule: {{flap.rule}}
   cooldown_minutes: {{flap.cooldown_minutes}} | OPTIONAL
6. Escalation Dampening
   dampening_rule: {{dampen.rule}}
7. Ticketing Conversion
   ticket_rule: {{ticket.rule}}
8. Telemetry
   pages_per_day_metric: {{telemetry.pages_per_day_metric}}
   suppressed_alert_metric: {{telemetry.suppressed_alert_metric}} | OPTIONAL
9. Review Cadence
   cadence: {{review.cadence}}

Cross-References
Upstream: {{xref:ALRT-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-10}}, {{xref:IRP-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dedupe/group/suppression and telemetry.
intermediate: Required. Define anti-flap and dampening and ticketing rule and cadence.
advanced: Required. Add auto-silence and strict sev1 protection constraints and expiry governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional principles, expiry rule, cooldown minutes, optional metric, auto-silence, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dedupe.rule is UNKNOWN → block Completeness Gate.
If suppress.rule is UNKNOWN → block Completeness Gate.
If flap.rule is UNKNOWN → block Completeness Gate.
If telemetry.pages_per_day_metric is UNKNOWN → block Completeness Gate.
If review.cadence is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
noise_controls_defined == true
suppression_and_anti_flap_defined == true
telemetry_defined == true
cadence_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-06

ALRT-06 — Incident Triggers (alert → IRP runbook)
Header Block
template_id: ALRT-06
title: Incident Triggers (alert → IRP runbook)
type: incident_triggers
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-06_Incident_Triggers.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["ALRT-02", "IRP-02", "IRP-10"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ALRT-02", "IRP-02", "IRP-01", "IRP-10", "SLO-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mapping between alerts and incident response: which alerts automatically trigger incident declaration, which runbooks to follow, and what metadata is required to start incident management quickly and consistently.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Incident workflow: {{xref:IRP-02}} | OPTIONAL
Incident severity model: {{xref:IRP-01}} | OPTIONAL
Incident runbooks index: {{xref:IRP-10}} | OPTIONAL
SLO burn alerts: {{xref:SLO-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Trigger registry (trigger_id list)
trigger_id (stable identifier)
Alert binding (alert_id list)
Default incident severity mapping (sev)
Runbook reference (IRP runbook_id)
Auto-declare rule (yes/no/UNKNOWN)
Escalation rule (who to page)
Customer impact comms rule (when to update status)
Verification/ack rule (when trigger considered resolved)
Telemetry requirements (incidents created, false triggers)

Optional Fields
Automation hook (create incident ticket) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Sev1/Sev2 incident triggers must have explicit runbooks and comms rules.
Do not invent alert_ids; reference ALRT-02.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_triggers: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Triggers (repeat)
   Trigger
   trigger_id: {{triggers[0].trigger_id}}
   alert_ids: {{triggers[0].alert_ids}}
   severity: {{triggers[0].severity}}
   runbook_ref: {{triggers[0].runbook_ref}}
   auto_declare: {{triggers[0].auto_declare}}
   escalation_rule: {{triggers[0].escalation_rule}}
   customer_comms_rule: {{triggers[0].customer_comms_rule}}
   resolve_rule: {{triggers[0].resolve_rule}}
   telemetry_metric: {{triggers[0].telemetry_metric}}
   false_trigger_metric: {{triggers[0].false_trigger_metric}} | OPTIONAL
   automation_hook: {{triggers[0].automation_hook}} | OPTIONAL
   open_questions:
   {{triggers[0].open_questions[0]}} | OPTIONAL

(Repeat per trigger.)

3. References
   Alert catalog: {{xref:ALRT-02}} | OPTIONAL
   Incident workflow: {{xref:IRP-02}} | OPTIONAL
   Status page rules: {{xref:IRP-05}} | OPTIONAL

Cross-References
Upstream: {{xref:ALRT-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IRP-02}}, {{xref:IRP-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define triggers with alert bindings, severity, runbook ref, resolve rule.

intermediate: Required. Define comms rules, escalation rule, telemetry metrics.
advanced: Required. Add automation hooks and strict false-trigger tracking + review process.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, false trigger metric, automation hook, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If triggers[*].trigger_id is UNKNOWN → block Completeness Gate.
If triggers[*].alert_ids is UNKNOWN → block Completeness Gate.
If triggers[*].runbook_ref is UNKNOWN → block Completeness Gate.
If triggers[*].resolve_rule is UNKNOWN → block Completeness Gate.
If triggers[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
trigger_registry_defined == true
alert_bindings_defined == true
runbooks_defined == true
comms_and_resolution_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-07

ALRT-07 — Customer Impact Rules (status page, comms)
Header Block
template_id: ALRT-07
title: Customer Impact Rules (status page, comms)
type: customer_impact_rules
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-07_Customer_Impact_Rules.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["IRP-04", "IRP-05", "ALRT-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IRP-04", "IRP-05", "ALRT-06", "SLO-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for determining customer impact and triggering customer-facing communications: when to update a status page, what channels are used, what approvals are required, and how communications map to incident severity and SLAs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Incident communications plan: {{xref:IRP-04}} | OPTIONAL
Status page/customer updates: {{xref:IRP-05}} | OPTIONAL
Incident triggers mapping: {{xref:ALRT-06}} | OPTIONAL
SLA commitments: {{xref:SLO-05}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Customer impact definition (what counts as impact)
Severity-to-comms mapping (sev → actions)
Status page update trigger rule
Comms channels (email/in-app/status)
Approval roles (who can publish)
Update cadence during incident (every X minutes)
Content constraints (no sensitive/internal details)
Post-incident follow-up rule (RCA summary?)
Telemetry requirements (time-to-first-update, update count)

Optional Fields
Regional impact rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Comms must be timely and consistent with severity and SLA commitments.
Do not include sensitive security details or internal identifiers in public updates.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Impact Definition
   definition: {{impact.definition}}
2. Severity Mapping
   mapping: {{sev.mapping}}
3. Status Page Triggers
   trigger_rule: {{status.trigger_rule}}
   status_page_ref: {{xref:IRP-05}} | OPTIONAL
4. Channels
   channels: {{channels.list}}
5. Approvals
   approver_roles: {{approve.roles}}
   publish_rule: {{approve.publish_rule}} | OPTIONAL
6. Update Cadence
   cadence_minutes: {{cadence.minutes}}
   cadence_rule: {{cadence.rule}} | OPTIONAL
7. Content Constraints
   constraints: {{content.constraints}}
8. Post-Incident
   followup_rule: {{post.followup_rule}}
   postmortem_ref: {{xref:IRP-06}} | OPTIONAL
9. Telemetry
   time_to_first_update_metric: {{telemetry.ttfu_metric}}
   updates_sent_metric: {{telemetry.updates_sent_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:ALRT-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IRP-04}}, {{xref:IRP-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define impact definition, trigger rule, channels, cadence.
intermediate: Required. Define severity mapping, approvals, content constraints, telemetry.
advanced: Required. Add regional rules and stricter SLA alignment + templated messaging constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, status page ref, publish rule, cadence

rule, postmortem ref, optional metrics, regional rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If impact.definition is UNKNOWN → block Completeness Gate.
If status.trigger_rule is UNKNOWN → block Completeness Gate.
If channels.list is UNKNOWN → block Completeness Gate.
If content.constraints is UNKNOWN → block Completeness Gate.
If telemetry.ttfu_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
impact_and_severity_mapping_defined == true
triggers_and_channels_defined == true
content_constraints_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-08

ALRT-08 — Alert Testing & Drills (synthetics, fire drills)
Header Block
template_id: ALRT-08
title: Alert Testing & Drills (synthetics, fire drills)
type: alert_testing_drills
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-08_Alert_Testing_Drills.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["ALRT-02", "IRP-09", "QA-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ALRT-02", "ALRT-03", "IRP-09", "QA-07", "SLO-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical approach for validating alerts and running drills: synthetics where applicable, periodic fire drills, and verification that alerts page the right people and link to usable runbooks. This template ensures alerting actually works before incidents.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Alert rule spec: {{xref:ALRT-03}} | OPTIONAL
Incident drills: {{xref:IRP-09}} | OPTIONAL
Release quality gates: {{xref:QA-07}} | OPTIONAL
SLO burn alerting: {{xref:SLO-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Testing types (synthetic, unit, dry-run)
Drill cadence (monthly/quarterly/UNKNOWN)
Drill scenarios list (scenario_id)
Success criteria (page delivered, ack, runbook followed)
Synthetics coverage rule (what gets synthetic tests)
Dry-run validation rule (rule fires in test env)
Ownership (who runs drills)
Telemetry requirements (drill pass rate, time-to-ack)
Post-drill remediation workflow (fix + retest)

Optional Fields
Game day notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Drills must be tracked and remediation assigned when failures occur.
Paging drills should be scheduled to avoid undue disruption.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Testing Types
   types: {{tests.types}}
2. Cadence
   cadence: {{cadence.value}}
3. Scenarios
   scenarios: {{scenarios.list}}
4. Success Criteria
   criteria: {{success.criteria}}
5. Synthetics
   coverage_rule: {{synthetics.coverage_rule}}
   tooling: {{synthetics.tooling}} | OPTIONAL
6. Dry Runs
   dry_run_rule: {{dry.rule}}
7. Ownership
   owner: {{owner.team}}
8. Telemetry
   drill_pass_rate_metric: {{telemetry.pass_rate_metric}}
   drill_ack_time_metric: {{telemetry.ack_time_metric}} | OPTIONAL
9. Remediation
   workflow: {{fix.workflow}}
   retest_rule: {{fix.retest_rule}} | OPTIONAL

Cross-References
Upstream: {{xref:ALRT-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-10}}, {{xref:IRP-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define cadence, scenarios, success criteria, remediation workflow.
intermediate: Required. Define synthetics coverage, dry-run rule, telemetry.
advanced: Required. Add game day planning and stricter success criteria and tooling integrations.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, synthetics tooling, ack metric, retest rule, game day notes, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tests.types is UNKNOWN → block Completeness Gate.
If cadence.value is UNKNOWN → block Completeness Gate.
If scenarios.list is UNKNOWN → block Completeness Gate.
If success.criteria is UNKNOWN → block Completeness Gate.
If telemetry.pass_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
cadence_and_scenarios_defined == true
success_criteria_defined == true
remediation_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-09

ALRT-09 — Alert Dashboards (what to watch)
Header Block
template_id: ALRT-09
title: Alert Dashboards (what to watch)
type: alert_dashboards
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-09_Alert_Dashboards.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["OBS-07", "ALRT-02", "SLO-10"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-07", "ALRT-02", "SLO-10", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical dashboards used specifically for alert monitoring and oncall: what panels are required, how they map to alert coverage and SLO burn, and how oncall uses them during incidents.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
SLO reporting: {{xref:SLO-10}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Oncall dashboard registry (dash_id list)
dash_id (stable identifier)
Purpose (alert overview, burn, noise)
Required panels list (alerts firing, ack times, burn rates)
Alert coverage mapping (which alerts appear)
SLO burn mapping (which SLOs appear)
Ownership (team)
Staleness/review cadence
Telemetry requirements (dashboard usage, staleness)

Optional Fields
Links/drilldowns | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Oncall dashboards must cover critical alerts and SLO burn indicators.
Dashboards must reference existing alerts/SLOs; do not invent IDs.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_oncall_dashboards: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Dashboards (repeat)
   Dashboard
   dash_id: {{dash[0].dash_id}}
   purpose: {{dash[0].purpose}}
   required_panels: {{dash[0].required_panels}}
   alert_mapping: {{dash[0].alert_mapping}}
   slo_mapping: {{dash[0].slo_mapping}}
   owner: {{dash[0].owner}}
   review_cadence: {{dash[0].review_cadence}}
   telemetry_metric: {{dash[0].telemetry_metric}}
   links: {{dash[0].links}} | OPTIONAL
   open_questions:
   {{dash[0].open_questions[0]}} | OPTIONAL

(Repeat per dashboard.)

3. References
   Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
   Alert catalog: {{xref:ALRT-02}} | OPTIONAL
   SLO reporting: {{xref:SLO-10}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-07}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IRP-02}}, {{xref:ALRT-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dashboards with required panels and mappings and owners.
intermediate: Required. Define review cadence and telemetry and coverage mapping.
advanced: Required. Add drilldowns and stricter staleness thresholds + enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, links, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dash[*].dash_id is UNKNOWN → block Completeness Gate.*
*If dash[*].required_panels is UNKNOWN → block Completeness Gate.
If dash[*].alert_mapping is UNKNOWN → block Completeness Gate.*
*If dash[*].owner is UNKNOWN → block Completeness Gate.
If dash[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
dashboard_registry_defined == true
panels_and_mappings_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

ALRT-10

ALRT-09 — Alert Dashboards (what to watch)
Header Block
template_id: ALRT-09
title: Alert Dashboards (what to watch)
type: alert_dashboards
template_version: 1.0.0
output_path: 10_ops/alerting/ALRT-09_Alert_Dashboards.md
compliance_gate_id: TMP-05.PRIMARY.ALRT
upstream_dependencies: ["OBS-07", "ALRT-02", "SLO-10"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-07", "ALRT-02", "SLO-10", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical dashboards used specifically for alert monitoring and oncall: what panels are required, how they map to alert coverage and SLO burn, and how oncall uses them during incidents.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
SLO reporting: {{xref:SLO-10}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Oncall dashboard registry (dash_id list)
dash_id (stable identifier)
Purpose (alert overview, burn, noise)
Required panels list (alerts firing, ack times, burn rates)
Alert coverage mapping (which alerts appear)
SLO burn mapping (which SLOs appear)
Ownership (team)
Staleness/review cadence
Telemetry requirements (dashboard usage, staleness)

Optional Fields
Links/drilldowns | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Oncall dashboards must cover critical alerts and SLO burn indicators.
Dashboards must reference existing alerts/SLOs; do not invent IDs.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_oncall_dashboards: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Dashboards (repeat)
   Dashboard
   dash_id: {{dash[0].dash_id}}
   purpose: {{dash[0].purpose}}
   required_panels: {{dash[0].required_panels}}
   alert_mapping: {{dash[0].alert_mapping}}
   slo_mapping: {{dash[0].slo_mapping}}
   owner: {{dash[0].owner}}
   review_cadence: {{dash[0].review_cadence}}
   telemetry_metric: {{dash[0].telemetry_metric}}
   links: {{dash[0].links}} | OPTIONAL
   open_questions:
   {{dash[0].open_questions[0]}} | OPTIONAL

(Repeat per dashboard.)

3. References
   Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
   Alert catalog: {{xref:ALRT-02}} | OPTIONAL
   SLO reporting: {{xref:SLO-10}} | OPTIONAL

Cross-References
Upstream: {{xref:OBS-07}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IRP-02}}, {{xref:ALRT-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dashboards with required panels and mappings and owners.
intermediate: Required. Define review cadence and telemetry and coverage mapping.
advanced: Required. Add drilldowns and stricter staleness thresholds + enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, links, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dash[*].dash_id is UNKNOWN → block Completeness Gate.*
*If dash[*].required_panels is UNKNOWN → block Completeness Gate.
If dash[*].alert_mapping is UNKNOWN → block Completeness Gate.*
*If dash[*].owner is UNKNOWN → block Completeness Gate.
If dash[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.ALRT
Pass conditions:
required_fields_present == true
dashboard_registry_defined == true
panels_and_mappings_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLOs/SLAs & Error Budgets (SLO)

SLO-01

SLO-01 — SLO/SLI Overview (definitions, scope)
Header Block
template_id: SLO-01
title: SLO/SLI Overview (definitions, scope)
type: slo_sli_overview
template_version: 1.0.0
output_path: 10_ops/slo/SLO-01_SLO_SLI_Overview.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["OBS-03", "ALRT-01", "IRP-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-03", "ALRT-01", "IRP-01", "SLO-02", "SLO-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical overview of reliability targets: what SLIs/SLOs/SLAs mean for this system, what is in-scope, and how error budgets drive engineering decisions. This document anchors the SLO set and links observability to reliability governance.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Alerting overview: {{xref:ALRT-01}} | OPTIONAL
Incident severity model: {{xref:IRP-01}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Error budget policy: {{xref:SLO-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Definitions (SLI, SLO, SLA)
In-scope services/surfaces
Out-of-scope statement (explicit)
SLI measurement principles (user-centric)
SLO target selection principles
Error budget concept and use
Ownership model (who owns SLOs)
Review cadence (monthly/quarterly)
Tooling references (dashboards/alerts)
Telemetry requirements (SLO calculation failures)

Optional Fields
Initial MVP targets notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
SLOs must be measurable with defined SLIs; no vague targets.
Do not claim external SLAs unless explicitly defined in contracts.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Definitions
   sli: {{defs.sli}}
   slo: {{defs.slo}}
   sla: {{defs.sla}}
2. Scope
   in_scope: {{scope.in_scope}}
   out_of_scope: {{scope.out_of_scope}} | OPTIONAL
3. Measurement Principles
   principles: {{measure.principles}}
4. Target Selection
   selection_rule: {{targets.selection_rule}}
5. Error Budgets
   budget_definition: {{budget.definition}}
   budget_usage_rule: {{budget.usage_rule}} | OPTIONAL
6. Ownership & Cadence
   owner_rule: {{owners.owner_rule}}
   review_cadence: {{owners.review_cadence}}
7. Tooling
   dashboards_ref: {{xref:OBS-07}} | OPTIONAL
   alerts_ref: {{xref:ALRT-02}} | OPTIONAL
8. Telemetry
   slo_calc_fail_metric: {{telemetry.slo_calc_fail_metric}}

Cross-References
Upstream: {{xref:OBS-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-02}}, {{xref:SLO-04}}, {{xref:SLO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define definitions, scope, and ownership/cadence.
intermediate: Required. Define measurement principles and error budget usage and telemetry.
advanced: Required. Add MVP targets notes and strict selection rationale + traceability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, out of scope, budget usage, tooling refs, MVP notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If defs.sli or defs.slo is UNKNOWN → block Completeness Gate.
If scope.in_scope is UNKNOWN → block Completeness Gate.
If owners.owner_rule is UNKNOWN → block Completeness Gate.
If owners.review_cadence is UNKNOWN → block Completeness Gate.
If telemetry.slo_calc_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
definitions_defined == true
scope_defined == true
ownership_and_cadence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-02

SLO-02 — Service SLO Catalog (by service_id)
Header Block
template_id: SLO-02
title: Service SLO Catalog (by service_id)
type: service_slo_catalog
template_version: 1.0.0
output_path: 10_ops/slo/SLO-02_Service_SLO_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["OBS-03", "OBS-07", "SLO-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-01", "OBS-03", "OBS-07", "API-01", "RELIA-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of service-level SLOs, indexed by service_id, including SLIs, targets, measurement windows, error budget policy references, and ownership. This catalog is the source of truth for reliability targets at the service boundary.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
API endpoint catalog: {{xref:API-01}} | OPTIONAL
Failure modes catalog: {{xref:RELIA-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Service registry (service_id list)
service_id (stable identifier)
Service name/description
SLIs (availability, latency, error rate)
Metric references (metric_id list)
SLO target (e.g., 99.9%)
Window (30d/7d/UNKNOWN)
Error budget policy reference (SLO-04)
Owner (team)
Alert linkage (alert_id list)
Dashboard linkage (dashboard_id list)
Telemetry requirements (slo evaluation errors)

Optional Fields
Dependencies considered (critical deps list) | OPTIONAL
Notes on exclusions (planned maintenance) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
SLOs must reference existing metric_ids from OBS-03; do not invent metrics.
Each service must have at least one SLI and one target (or explicit exception).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_services: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Services (repeat per service_id)
   Service SLO
   service_id: {{services[0].service_id}}
   name: {{services[0].name}}
   slis: {{services[0].slis}}
   metric_ids: {{services[0].metric_ids}}
   target: {{services[0].target}}
   window: {{services[0].window}}
   error_budget_ref: {{xref:SLO-04}} | OPTIONAL
   owner: {{services[0].owner}}
   alert_ids: {{services[0].alert_ids}}
   dashboard_ids: {{services[0].dashboard_ids}}
   deps: {{services[0].deps}} | OPTIONAL
   maintenance_exclusions: {{services[0].maintenance_exclusions}} | OPTIONAL
   telemetry_metric: {{services[0].telemetry_metric}}
   open_questions:
   {{services[0].open_questions[0]}} | OPTIONAL

(Repeat per service.)

3. References
   Error budget policy: {{xref:SLO-04}} | OPTIONAL
   Measurement rules: {{xref:SLO-06}} | OPTIONAL
   Reporting: {{xref:SLO-10}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-04}}, {{xref:SLO-07}}, {{xref:ALRT-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define service_id, SLIs, target, owner, and metric references.
intermediate: Required. Define windows, alert/dashboard linkage, and telemetry metric.
advanced: Required. Add dependency considerations and maintenance exclusions and strict exceptions governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, deps, maintenance exclusions, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If services[*].service_id is UNKNOWN → block Completeness Gate.
If services[*].metric_ids is UNKNOWN → block Completeness Gate.
If services[*].target is UNKNOWN → block Completeness Gate.
If services[*].owner is UNKNOWN → block Completeness Gate.
If services[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
service_registry_defined == true
sli_and_targets_defined == true
metric_refs_defined == true
ownership_and_linkage_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-03

SLO-03 — Endpoint SLO Catalog (by endpoint_id)
Header Block
template_id: SLO-03
title: Endpoint SLO Catalog (by endpoint_id)
type: endpoint_slo_catalog
template_version: 1.0.0
output_path: 10_ops/slo/SLO-03_Endpoint_SLO_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["API-01", "OBS-03", "SLO-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-01", "API-02", "OBS-03", "SLO-01", "SLO-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of endpoint-level SLOs, indexed by endpoint_id, for critical API routes: latency and error-rate targets, SLI definitions, measurement windows, and ownership. This template is used when endpoint-level guarantees matter beyond service-level SLOs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Endpoint specs: {{xref:API-02}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Measurement rules: {{xref:SLO-06}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Endpoint registry (endpoint_id list)
endpoint_id (stable identifier)
Route/method summary
SLI definition (latency/error)
Metric references (metric_id list)
Target (p95 latency, error rate)
Window (7d/30d/UNKNOWN)
Exclusions (what doesn’t count)
Owner (team/service)
Alert linkage (alert_id list)
Telemetry requirements (endpoint SLO calc failures)

Optional Fields
Tiering (critical/standard) | OPTIONAL
Client impact notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Endpoint SLOs must reference existing metric_ids; do not invent metrics.
Endpoint SLOs should be used only for critical endpoints; otherwise rely on service SLOs.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_endpoints: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Endpoints (repeat per endpoint_id)
   Endpoint SLO
   endpoint_id: {{endpoints[0].endpoint_id}}
   route: {{endpoints[0].route}}
   method: {{endpoints[0].method}}
   sli: {{endpoints[0].sli}}
   metric_ids: {{endpoints[0].metric_ids}}
   target: {{endpoints[0].target}}
   window: {{endpoints[0].window}}
   exclusions: {{endpoints[0].exclusions}}
   owner: {{endpoints[0].owner}}
   alert_ids: {{endpoints[0].alert_ids}}
   tier: {{endpoints[0].tier}} | OPTIONAL
   client_impact: {{endpoints[0].client_impact}} | OPTIONAL
   telemetry_metric: {{endpoints[0].telemetry_metric}}
   open_questions:
   {{endpoints[0].open_questions[0]}} | OPTIONAL

(Repeat per endpoint.)

3. References
   Service SLOs: {{xref:SLO-02}} | OPTIONAL
   Burn alerts: {{xref:SLO-07}} | OPTIONAL

Cross-References
Upstream: {{xref:API-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:SLO-07}}, {{xref:PERF-03}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define endpoint_id, SLI, target, owner, metric references.
intermediate: Required. Define exclusions, alert linkage, telemetry metric.
advanced: Required. Add tiering and client impact notes and strict endpoint selection rationale.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, tiering, client impact, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If endpoints[*].endpoint_id is UNKNOWN → block Completeness Gate.
If endpoints[*].metric_ids is UNKNOWN → block Completeness Gate.
If endpoints[*].target is UNKNOWN → block Completeness Gate.
If endpoints[*].owner is UNKNOWN → block Completeness Gate.
If endpoints[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
endpoint_registry_defined == true
sli_targets_defined == true
metric_refs_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-04

SLO-04 — Error Budget Policy (burn rates, actions)
Header Block
template_id: SLO-04
title: Error Budget Policy (burn rates, actions)
type: error_budget_policy
template_version: 1.0.0
output_path: 10_ops/slo/SLO-04_Error_Budget_Policy.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["SLO-01", "ALRT-01", "REL-08"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-01", "SLO-02", "ALRT-01", "REL-08", "IRP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for error budgets: how budgets are calculated, what burn rates mean, and what actions are required when budgets are consumed. This template links reliability targets to release/change decisions and incident response.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Alerting overview: {{xref:ALRT-01}} | OPTIONAL
Change risk classification: {{xref:REL-08}} | OPTIONAL
Postmortem template: {{xref:IRP-06}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Budget calculation rule (per SLO/window)
Budget states (healthy/warn/breached)
Burn rate definition (fast/slow burn)
Burn alert triggers (SLO-07 linkage)
Actions per state (freeze, reduce risk, focus on reliability)
Release policy interaction (REL refs)
Exception rule (when can ship despite breach)
Ownership (who enforces)
Review cadence (weekly/monthly)
Telemetry requirements (budget remaining, burn rate)

Optional Fields
Service tiering (critical vs standard) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Actions must be concrete and enforceable (not aspirational).
Exceptions must be explicit and time-bound.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Calculation
   calc_rule: {{calc.rule}}
   window_rule: {{calc.window_rule}} | OPTIONAL
2. States
   states: {{states.list}}
   thresholds: {{states.thresholds}} | OPTIONAL
3. Burn Rates
   definition: {{burn.definition}}
   fast_burn_rule: {{burn.fast_rule}} | OPTIONAL
   slow_burn_rule: {{burn.slow_rule}} | OPTIONAL
4. Burn Alerts
   burn_alert_ref: {{xref:SLO-07}} | OPTIONAL
   trigger_rule: {{alerts.trigger_rule}}
5. Actions
   healthy_actions: {{actions.healthy}}
   warn_actions: {{actions.warn}}
   breach_actions: {{actions.breach}}
6. Release Interaction
   release_rule: {{release.rule}}
   change_risk_ref: {{xref:REL-08}} | OPTIONAL
7. Exceptions
   exception_rule: {{exceptions.rule}}
   exceptions_ref: {{xref:COMP-08}} | OPTIONAL
8. Ownership & Cadence
   owner: {{owner.team}}
   review_cadence: {{review.cadence}}
9. Telemetry
   budget_remaining_metric: {{telemetry.budget_remaining_metric}}
   burn_rate_metric: {{telemetry.burn_rate_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:SLO-07}}, {{xref:REL-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define states and actions and review cadence and owner.
intermediate: Required. Define burn rate triggers, release interaction, telemetry metrics.
advanced: Required. Add tiering and strict exceptions governance with auditability and postmortem linkage.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, window rule, thresholds, fast/slow rules, burn alert ref, change risk ref, exceptions ref, optional metric, tiering, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If calc.rule is UNKNOWN → block Completeness Gate.
If states.list is UNKNOWN → block Completeness Gate.
If actions.breach is UNKNOWN → block Completeness Gate.
If review.cadence is UNKNOWN → block Completeness Gate.
If telemetry.budget_remaining_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
calculation_defined == true
states_and_actions_defined == true
release_and_exceptions_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-05

SLO-05 — SLA Commitments (external, contracts)
Header Block
template_id: SLO-05
title: SLA Commitments (external, contracts)
type: sla_commitments
template_version: 1.0.0
output_path: 10_ops/slo/SLO-05_SLA_Commitments.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["SLO-01", "COMP-01", "REL-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-01", "COMP-01", "REL-01", "SLO-02", "ALRT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical external SLA commitments (if any): what is promised to customers, how availability is measured, what credits/remedies exist, and how SLA reporting is performed. This template must not invent SLAs; it only documents explicit commitments.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Compliance scope/contracts context: {{xref:COMP-01}} | OPTIONAL
Release policy (versioning): {{xref:REL-01}} | OPTIONAL
Service SLO catalog (internal targets): {{xref:SLO-02}} | OPTIONAL
Customer impact rules: {{xref:ALRT-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
SLA registry (sla_id list)
sla_id (stable identifier)
Customer segment/applicability (who covered)
Availability definition (what counts as downtime)
Measurement window (monthly/UNKNOWN)
Exclusions (planned maintenance)
Credit/remedy terms (or NONE/UNKNOWN)
Reporting process (how calculated, published)
Owner (legal/ops)
Telemetry requirements (sla calc failures, credits issued)

Optional Fields
Service mapping (service_id list) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Do not claim an SLA exists unless provided; use UNKNOWN or NONE.
SLA definitions must reference measurable signals (SLO/metrics).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_slas: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. SLAs (repeat)
   SLA
   sla_id: {{slas[0].sla_id}}
   applicability: {{slas[0].applicability}}
   availability_definition: {{slas[0].availability_definition}}
   window: {{slas[0].window}}
   exclusions: {{slas[0].exclusions}}
   credits: {{slas[0].credits}}
   reporting_process: {{slas[0].reporting_process}}
   owner: {{slas[0].owner}}
   service_mapping: {{slas[0].service_mapping}} | OPTIONAL
   telemetry_metric: {{slas[0].telemetry_metric}}
   credits_issued_metric: {{slas[0].credits_issued_metric}} | OPTIONAL
   open_questions:
   {{slas[0].open_questions[0]}} | OPTIONAL

(Repeat per SLA.)

3. References
   Internal SLOs: {{xref:SLO-02}} | OPTIONAL
   Customer comms: {{xref:ALRT-07}} | OPTIONAL
   Reporting: {{xref:SLO-10}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-10}}, {{xref:COMP-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define sla registry and availability definition and exclusions and reporting

process.
intermediate: Required. Define credits terms, owner, telemetry metrics.
advanced: Required. Add service mapping and strict evidence pointers to contracts and measurement queries.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, credits terms, service mapping, optional metrics, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If slas[*].sla_id is UNKNOWN → block Completeness Gate.
If slas[*].availability_definition is UNKNOWN → block Completeness Gate.
If slas[*].reporting_process is UNKNOWN → block Completeness Gate.
If slas[*].owner is UNKNOWN → block Completeness Gate.
If slas[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
sla_registry_defined == true
definitions_defined == true
reporting_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-06

SLO-06 — Measurement Rules (windows, aggregation)
Header Block
template_id: SLO-06
title: Measurement Rules (windows, aggregation)
type: slo_measurement_rules
template_version: 1.0.0
output_path: 10_ops/slo/SLO-06_Measurement_Rules.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["OBS-03", "OBS-02", "SLO-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-01", "OBS-03", "OBS-02", "SLO-02", "SLO-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for calculating SLIs/SLOs: measurement windows, aggregation methods, latency percentile rules, availability definitions, and how missing data is handled. This standard ensures SLOs are consistent and comparable across services.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SLO overview: {{xref:SLO-01}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Telemetry schema: {{xref:OBS-02}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Endpoint SLO catalog: {{xref:SLO-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Supported windows (7d/30d/rolling)
Aggregation rules (sum/avg/ratio)
Availability calculation rule (good/total)
Latency percentile rules (p50/p95/p99)
Error rate rules (5xx, timeouts)
SLO evaluation frequency (how often computed)
Handling missing/partial telemetry (UNKNOWN vs assume bad)
Exclusion rules (maintenance windows)
Metric label selection rules (env/service)
Telemetry requirements (calc job failures)

Optional Fields
Multi-region aggregation rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
SLO calculations must be reproducible and based on defined metric sources.
Exclude planned maintenance only when explicitly defined.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Windows
   supported: {{win.supported}}
   default_window: {{win.default}} | OPTIONAL
2. Aggregation
   aggregation_rules: {{agg.rules}}
3. Availability
   good_total_rule: {{avail.good_total_rule}}
   status_code_rule: {{avail.status_code_rule}} | OPTIONAL
4. Latency
   percentiles: {{lat.percentiles}}
   histogram_rule: {{lat.histogram_rule}} | OPTIONAL
5. Error Rate
   error_classes: {{err.classes}}
   timeout_inclusion_rule: {{err.timeout_rule}} | OPTIONAL
6. Frequency
   eval_frequency_minutes: {{freq.minutes}}
7. Missing Data
   missing_data_rule: {{missing.rule}}
8. Exclusions
   maintenance_exclusion_rule: {{exclude.maintenance_rule}}
   maintenance_ref: {{xref:REL-07}} | OPTIONAL
9. Label Selection
   label_rule: {{labels.rule}}
10. Telemetry
    slo_calc_job_fail_metric: {{telemetry.calc_fail_metric}}

Cross-References
Upstream: {{xref:OBS-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-07}}, {{xref:SLO-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define windows, availability rule, latency percentiles, missing data rule.

intermediate: Required. Define aggregation, exclusions, label selection, telemetry.
advanced: Required. Add multi-region rules and stricter reproducibility and auditability of calculations.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, default window, status code rule, histogram rule, timeout rule, maintenance ref, multi-region rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If win.supported is UNKNOWN → block Completeness Gate.
If avail.good_total_rule is UNKNOWN → block Completeness Gate.
If lat.percentiles is UNKNOWN → block Completeness Gate.
If missing.rule is UNKNOWN → block Completeness Gate.
If telemetry.calc_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
windows_and_rules_defined == true
missing_data_defined == true
exclusions_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-07

SLO-07 — Alerting on Burn (ties to ALRT)
Header Block
template_id: SLO-07
title: Alerting on Burn (ties to ALRT)
type: slo_burn_alerting
template_version: 1.0.0
output_path: 10_ops/slo/SLO-07_Alerting_on_Burn.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["SLO-04", "ALRT-03", "SLO-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-04", "SLO-06", "ALRT-03", "ALRT-02", "SLO-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for alerting on error budget burn: burn-rate alert tiers, window pairs, severity mapping, routing, and runbook linkage. This template standardizes SLO burn alerting so it is consistent and low-noise.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Error budget policy: {{xref:SLO-04}} | OPTIONAL
Measurement rules: {{xref:SLO-06}} | OPTIONAL
Alert rule spec: {{xref:ALRT-03}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Burn alert tiers (tier_id list)
Window pairs (short/long windows)
Burn rate threshold rules (e.g., 2x, 10x)
Severity mapping (tier → sev)
Routing rule (team/oncall)
Dedupe/suppression rules (ALRT-05 tie-in)
Runbook linkage rule (what to do)
SLO coverage rule (which SLOs require burn alerts)
Telemetry requirements (burn alerts fired, budget remaining)

Optional Fields
Per-tier actions (freeze triggers) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Burn alerts must be actionable and map to clear actions in SLO-04.
Burn alert rules must reference actual SLO metrics; do not invent sources.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Tiers
   tiers: {{tiers.list}}
2. Windows
   window_pairs: {{windows.pairs}}
3. Thresholds
   threshold_rule: {{thresh.rule}}
   example_thresholds: {{thresh.examples}} | OPTIONAL
4. Severity
   mapping: {{sev.mapping}}
5. Routing & Noise Controls
   routing_rule: {{route.rule}}
   noise_ref: {{xref:ALRT-05}} | OPTIONAL
6. Runbooks
   runbook_rule: {{runbooks.rule}}
   error_budget_ref: {{xref:SLO-04}} | OPTIONAL
7. Coverage
   coverage_rule: {{cover.rule}}
8. Telemetry
   burn_alert_metric: {{telemetry.burn_alert_metric}}
   budget_remaining_metric: {{telemetry.budget_remaining_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ALRT-02}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define tiers, windows, threshold rule, severity mapping, coverage rule.
intermediate: Required. Define routing/noise controls and telemetry metrics.
advanced: Required. Add per-tier actions and tighter examples and strict alignment to freeze triggers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, examples, noise ref, optional metric, per-tier actions, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If tiers.list is UNKNOWN → block Completeness Gate.
If windows.pairs is UNKNOWN → block Completeness Gate.
If thresh.rule is UNKNOWN → block Completeness Gate.
If cover.rule is UNKNOWN → block Completeness Gate.
If telemetry.burn_alert_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
tiers_and_windows_defined == true
thresholds_and_severity_defined == true
coverage_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-08

SLO-08 — Review Cadence & Ownership (monthly reviews)
Header Block
template_id: SLO-08
title: Review Cadence & Ownership (monthly reviews)
type: slo_review_cadence_ownership
template_version: 1.0.0
output_path: 10_ops/slo/SLO-08_Review_Cadence_Ownership.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["SLO-02", "SLO-04", "IRP-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-02", "SLO-04", "SLO-10", "IRP-06", "REL-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical process for SLO reviews: cadence, participants, required inputs/outputs, and how actions are tracked. This template ensures reliability governance is routine and tied to error budgets, incidents, and change freezes.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Error budget policy: {{xref:SLO-04}} | OPTIONAL
SLO reporting: {{xref:SLO-10}} | OPTIONAL
Postmortem template: {{xref:IRP-06}} | OPTIONAL
Change calendar/freeze: {{xref:REL-09}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Cadence (monthly/quarterly/UNKNOWN)
Participants/roles (SRE, service owners)
Required inputs (SLO reports, incidents)
Agenda template (what reviewed)
Decision rules (when to adjust SLOs)
Action items tracking rule (owners, due dates)
Escalation rule (budget breach → leadership)
Documentation storage location
Telemetry requirements (reviews held, actions closed)

Optional Fields
Executive summary format | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Reviews must produce action items when budgets are unhealthy.
Changes to SLO targets must be documented and approved.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Cadence
   cadence: {{cadence.value}}
2. Participants
   roles: {{roles.list}}
3. Inputs
   inputs: {{inputs.list}}
4. Agenda
   agenda: {{agenda.items}}
5. Decisions
   adjustment_rule: {{decisions.adjustment_rule}}
   approval_rule: {{decisions.approval_rule}} | OPTIONAL
6. Action Items
   tracking_rule: {{actions.tracking_rule}}
   storage_location: {{actions.storage_location}} | OPTIONAL
7. Escalation
   escalation_rule: {{escalate.rule}}
8. Storage
   docs_location: {{docs.location}}
9. Telemetry
   reviews_held_metric: {{telemetry.reviews_held_metric}}
   actions_closed_metric: {{telemetry.actions_closed_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-09}}, {{xref:REL-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define cadence, roles, agenda, and tracking rule.
intermediate: Required. Define decision/approval rules and escalation and telemetry metrics.
advanced: Required. Add exec summary format and stricter governance for SLO target changes and freezes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, approval rule, storage location, actions closed metric, exec summary, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If cadence.value is UNKNOWN → block Completeness Gate.
If roles.list is UNKNOWN → block Completeness Gate.
If agenda.items is UNKNOWN → block Completeness Gate.
If actions.tracking_rule is UNKNOWN → block Completeness Gate.
If telemetry.reviews_held_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
cadence_and_roles_defined == true
agenda_defined == true
tracking_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-09

SLO-09 — Exceptions & Adjustments (temporary changes)
Header Block
template_id: SLO-09
title: Exceptions & Adjustments (temporary changes)
type: slo_exceptions_adjustments
template_version: 1.0.0
output_path: 10_ops/slo/SLO-09_Exceptions_Adjustments.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["SLO-04", "COMP-08", "REL-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-04", "COMP-08", "REL-09", "SLO-08", "AUDIT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical process for temporary SLO exceptions and adjustments: when an SLO target/window can be temporarily changed, how it is approved, how it expires, and how it is documented/audited. This template prevents “silent” reliability downgrades.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Error budget policy: {{xref:SLO-04}} | OPTIONAL
Exception management: {{xref:COMP-08}} | OPTIONAL
Change calendar/freeze rules: {{xref:REL-09}} | OPTIONAL
SLO review process: {{xref:SLO-08}} | OPTIONAL
Privileged change audit: {{xref:AUDIT-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Allowed exception types (measurement gap, migration)
Request fields (why, duration, compensating actions)
Approver roles (service owner, SRE)
Max duration rule
Expiry/renewal rule
Communication rule (who notified)
Audit logging rule (record changes)
Reversion rule (auto revert to baseline)
Telemetry requirements (active exceptions, overdue)

Optional Fields
Emergency exception rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
No exceptions without expiry and compensating controls/actions.
Temporary changes must be reversible and documented.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Exception Types
   types: {{types.list}}
2. Request
   required_fields: {{request.fields}}
3. Approvals
   approver_roles: {{approve.roles}}
   workflow_ref: {{xref:COMP-08}} | OPTIONAL
4. Duration
   max_days: {{duration.max_days}}
   renewal_rule: {{duration.renewal_rule}} | OPTIONAL
5. Communications
   notify_rule: {{comms.notify_rule}}
6. Audit
   audit_required: {{audit.required}}
   audit_ref: {{xref:AUDIT-07}} | OPTIONAL
7. Reversion
   revert_rule: {{revert.rule}}
8. Telemetry
   active_exceptions_metric: {{telemetry.active_metric}}
   overdue_exceptions_metric: {{telemetry.overdue_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-10}}, {{xref:REL-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define types, required fields, max days, revert rule.
intermediate: Required. Define approvals, comms, audit rule, telemetry.
advanced: Required. Add emergency rules and stricter renewal/rollback governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, workflow ref, renewal rule, audit ref, optional metric, emergency rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If types.list is UNKNOWN → block Completeness Gate.

If request.fields is UNKNOWN → block Completeness Gate.
If duration.max_days is UNKNOWN → block Completeness Gate.
If revert.rule is UNKNOWN → block Completeness Gate.
If telemetry.active_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
exception_process_defined == true
expiry_and_reversion_defined == true
audit_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SLO-10

SLO-10 — Reporting (dashboards, exec summaries)
Header Block
template_id: SLO-10
title: Reporting (dashboards, exec summaries)
type: slo_reporting
template_version: 1.0.0
output_path: 10_ops/slo/SLO-10_SLO_Reporting.md
compliance_gate_id: TMP-05.PRIMARY.SLO
upstream_dependencies: ["SLO-02", "OBS-07", "ALRT-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-02", "SLO-03", "OBS-07", "ALRT-09", "SLO-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical reporting outputs for SLOs/error budgets: dashboards, periodic reports, and executive summaries, including what is included, cadence, and distribution. This template makes reliability visible and actionable.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Endpoint SLO catalog: {{xref:SLO-03}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Alert dashboards: {{xref:ALRT-09}} | OPTIONAL
Error budget policy: {{xref:SLO-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Report types (dashboard, weekly report, exec summary)
Cadence per report type
Audience/distribution list
Required contents (SLO status, burn rates, incidents)
Dashboard references (dashboard_id list)
Exec summary format rules
Action items section rule (what actions required)
Ownership (who publishes)
Telemetry requirements (reports published, missed cadence)

Optional Fields
Customer-facing reporting rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Reports must be based on measured data and link to dashboards.
When budgets are unhealthy, reports must include action items and owners.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Report Types
   types: {{report.types}}
2. Cadence
   cadence: {{report.cadence}}
3. Audience
   audience: {{audience.list}}
4. Contents
   required_sections: {{content.sections}}
5. Dashboards
   dashboard_ids: {{dashboards.ids}}
6. Exec Summary
   format_rule: {{exec.format_rule}}
   template: {{exec.template}} | OPTIONAL
7. Actions
   action_section_rule: {{actions.rule}}
8. Ownership
   owner: {{owner.team}}
9. Telemetry
   reports_published_metric: {{telemetry.published_metric}}
   missed_cadence_metric: {{telemetry.missed_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SLO-08}}, {{xref:REL-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define report types, cadence, required contents, owner.
intermediate: Required. Define exec summary format, dashboard refs, telemetry.
advanced: Required. Add customer-facing rules and strict distribution + accountability practices.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, exec template, missed metric, customer-facing rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If report.types is UNKNOWN → block Completeness Gate.

If report.cadence is UNKNOWN → block Completeness Gate.
If content.sections is UNKNOWN → block Completeness Gate.
If dashboards.ids is UNKNOWN → block Completeness Gate.
If telemetry.published_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SLO
Pass conditions:
required_fields_present == true
reporting_defined == true
cadence_and_audience_defined == true
dashboard_linkage_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Performance & Scalability (PERF)

PERF-01

PERF-01 — Performance Overview (targets, constraints)
Header Block
template_id: PERF-01
title: Performance Overview (targets, constraints)
type: performance_overview
template_version: 1.0.0
output_path: 10_ops/performance/PERF-01_Performance_Overview.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["PBP-01", "SLO-01", "CPR-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-01", "SLO-01", "CPR-01", "PERF-02", "PERF-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of performance for the system: what “fast” means, what targets exist for web/mobile/backend, and what constraints shape performance work (cost, reliability, feature scope). This document anchors the PERF set.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Performance budgets overview: {{xref:PBP-01}} | OPTIONAL
SLO overview (latency expectations): {{xref:SLO-01}} | OPTIONAL
Client performance budget: {{xref:CPR-01}} | OPTIONAL
Critical path inventory: {{xref:PERF-02}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Performance targets summary (web/mobile/api)
In-scope surfaces (screens/endpoints)
Out-of-scope statement (explicit)
Primary constraints (cost, device, network)
Measurement approach (tools/metrics)
Ownership (who owns perf)
Perf regression policy reference (PERF-09)
Release gate linkage (block conditions)
Telemetry requirements (perf regressions detected)

Optional Fields
MVP vs later targets | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Targets must be measurable (define metrics and windows).
Do not invent budgets; reference PBP/CPR docs where defined.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Targets
   web_targets: {{targets.web}}
   mobile_targets: {{targets.mobile}}
   api_targets: {{targets.api}}
2. Scope
   in_scope: {{scope.in}}
   out_of_scope: {{scope.out}} | OPTIONAL
3. Constraints
   constraints: {{constraints.list}}
4. Measurement
   metrics: {{measure.metrics}}
   tooling: {{measure.tooling}} | OPTIONAL
5. Ownership
   owner: {{owner.team}}
6. Regression Policy
   regression_ref: {{xref:PERF-09}} | OPTIONAL
   block_rule: {{regress.block_rule}}
7. Release Linkage
   release_gate_rule: {{release.gate_rule}}
8. Telemetry
   perf_regression_metric: {{telemetry.regression_metric}}

Cross-References
Upstream: {{xref:PBP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PERF-02}}, {{xref:PERF-09}}, {{xref:LOAD-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define targets, constraints, measurement, regression block rule.
intermediate: Required. Define scope and ownership and release gate linkage and telemetry.
advanced: Required. Add MVP vs later targets and strict tie-ins to budgets and SLOs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, tooling, regression ref, optional scope, MVP targets, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If targets.web or targets.api is UNKNOWN → block Completeness Gate.
If constraints.list is UNKNOWN → block Completeness Gate.
If regress.block_rule is UNKNOWN → block Completeness Gate.
If telemetry.regression_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
targets_and_constraints_defined == true
measurement_defined == true
regression_and_release_linkage_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-02

PERF-02 — Critical Path Inventory (user journeys)
Header Block
template_id: PERF-02
title: Critical Path Inventory (user journeys)
type: performance_critical_path_inventory
template_version: 1.0.0
output_path: 10_ops/performance/PERF-02_Critical_Path_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["FE-01", "RJT-01", "PERF-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "RJT-01", "PERF-01", "PERF-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical inventory of performance-critical user journeys (“critical paths”), including the screens/endpoints involved and why they matter. This inventory is used to prioritize profiling, regression gates, and load testing.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Route map/layout: {{xref:FE-01}} | OPTIONAL
Journey inventory: {{xref:RJT-01}} | OPTIONAL
Performance overview: {{xref:PERF-01}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Critical path registry (path_id list)
path_id (stable identifier)
Journey name/summary
User segment (who uses it)
Screens/routes involved (screen_id/route refs)
Endpoints involved (endpoint_id refs)
Success timing target (p95)
Priority (P0/P1/P2/UNKNOWN)
Owner (team)
Telemetry requirements (journey timing metric)

Optional Fields
Dependencies (third-party) | OPTIONAL

Notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Critical paths should map to real routes/screens and endpoint IDs.
Timing targets should reference budgets where defined; otherwise use UNKNOWN.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_paths: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Paths (repeat)
   Critical Path
   path_id: {{paths[0].path_id}}
   name: {{paths[0].name}}
   user_segment: {{paths[0].user_segment}}
   screens: {{paths[0].screens}}
   endpoints: {{paths[0].endpoints}}
   target_p95_ms: {{paths[0].target_p95_ms}}
   priority: {{paths[0].priority}}
   owner: {{paths[0].owner}}
   dependencies: {{paths[0].dependencies}} | OPTIONAL
   journey_metric: {{paths[0].journey_metric}}
   open_questions:
   {{paths[0].open_questions[0]}} | OPTIONAL

(Repeat per path.)

3. References
   Latency budgets: {{xref:PERF-03}} | OPTIONAL
   Perf gates: {{xref:PERF-09}} | OPTIONAL
   Load testing: {{xref:LOAD-03}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-05}}, {{xref:LOAD-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define path registry, screens/endpoints, and owner and priority.
intermediate: Required. Define timing targets and telemetry metric per path.
advanced: Required. Add dependency notes and strict mapping to budgets and journey tests.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, dependencies, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If paths[*].path_id is UNKNOWN → block Completeness Gate.
If paths[*].screens is UNKNOWN → block Completeness Gate.
If paths[*].endpoints is UNKNOWN → block Completeness Gate.
If paths[*].owner is UNKNOWN → block Completeness Gate.
If paths[*].journey_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
critical_paths_defined == true
mappings_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-03

PERF-03 — Latency Budgets (per surface)
Header Block
template_id: PERF-03
title: Latency Budgets (per surface)
type: latency_budgets
template_version: 1.0.0
output_path: 10_ops/performance/PERF-03_Latency_Budgets.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["PBP-04", "SLO-06", "PERF-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-04", "SLO-06", "PERF-02", "API-01", "FE-01", "MOB-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical latency budgets for key surfaces (web screens, mobile screens, API endpoints, background jobs): target p50/p95 values and the budget breakdown (client vs network vs server) where applicable. This template is used to set performance expectations and gates.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API budget spec: {{xref:PBP-04}} | OPTIONAL
Measurement rules: {{xref:SLO-06}} | OPTIONAL
Critical path inventory: {{xref:PERF-02}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Route map/layout: {{xref:FE-01}} | OPTIONAL
Mobile navigation map: {{xref:MOB-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Budget registry (budget_id list)
budget_id (stable identifier)
Surface type (api/screen/job/ws)
Surface selector (endpoint_id/screen_id/job_id)
p50 target (ms)
p95 target (ms)
Budget breakdown (client/network/server)
Measurement location (client/server)
Owner (team)
Telemetry requirements (p95 metric per budget)

Optional Fields
Notes (device/network assumptions) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must map to real surface IDs (endpoint_id/screen_id/job_id).
Targets must be measurable with referenced metrics.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_budgets: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Budgets (repeat)
   Budget
   budget_id: {{budgets[0].budget_id}}
   surface_type: {{budgets[0].surface_type}}
   surface_selector: {{budgets[0].surface_selector}}
   p50_ms: {{budgets[0].p50_ms}}
   p95_ms: {{budgets[0].p95_ms}}
   breakdown: {{budgets[0].breakdown}}
   measurement: {{budgets[0].measurement}}
   owner: {{budgets[0].owner}}
   metric_ref: {{budgets[0].metric_ref}} | OPTIONAL
   notes: {{budgets[0].notes}} | OPTIONAL
   telemetry_metric: {{budgets[0].telemetry_metric}}
   open_questions:
   {{budgets[0].open_questions[0]}} | OPTIONAL

(Repeat per budget.)

3. References
   Critical paths: {{xref:PERF-02}} | OPTIONAL
   Perf gates: {{xref:PERF-09}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PERF-09}}, {{xref:LOAD-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define budgets with selectors and p50/p95 targets.
intermediate: Required. Define breakdown + measurement location + telemetry metric.

advanced: Required. Add device/network notes and strict linkage to critical paths and SLO windows.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, metric ref, notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If budgets[*].budget_id is UNKNOWN → block Completeness Gate.
If budgets[*].surface_selector is UNKNOWN → block Completeness Gate.
If budgets[*].p95_ms is UNKNOWN → block Completeness Gate.
If budgets[*].owner is UNKNOWN → block Completeness Gate.
If budgets[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
budget_registry_defined == true
targets_defined == true
measurement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-04

PERF-04 — Throughput Targets (RPS, concurrency)
Header Block
template_id: PERF-04
title: Throughput Targets (RPS, concurrency)
type: throughput_targets
template_version: 1.0.0
output_path: 10_ops/performance/PERF-04_Throughput_Targets.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["LOAD-02", "COST-03", "PERF-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-01", "API-01", "LOAD-02", "COST-03", "SLO-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical throughput and concurrency targets for the system: expected and peak RPS, concurrent users/sessions, background job throughput, and platform limits. This template is used to size systems and to define load test pass/fail criteria.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Performance overview: {{xref:PERF-01}} | OPTIONAL
API endpoint catalog: {{xref:API-01}} | OPTIONAL
Workload models: {{xref:LOAD-02}} | OPTIONAL
Cost forecast model: {{xref:COST-03}} | OPTIONAL
Service SLO catalog: {{xref:SLO-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Target registry (target_id list)
target_id (stable identifier)
Surface (api/job/ws/client)
Surface selector (service_id/endpoint_id/job_id)
Expected throughput (RPS or jobs/min)
Peak throughput (burst)
Concurrency target (users/sessions/workers)
Assumptions (traffic mix, regions)
Owner (team)
Telemetry requirements (current throughput vs target)

Optional Fields
Headroom policy (e.g., 2x) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Targets must be tied to workload models and used in load tests; avoid arbitrary numbers.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_targets: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Targets (repeat)
   Target
   target_id: {{targets[0].target_id}}
   surface: {{targets[0].surface}}
   selector: {{targets[0].selector}}
   expected_rate: {{targets[0].expected_rate}}
   peak_rate: {{targets[0].peak_rate}}
   concurrency: {{targets[0].concurrency}}
   assumptions: {{targets[0].assumptions}}
   owner: {{targets[0].owner}}
   headroom_policy: {{targets[0].headroom_policy}} | OPTIONAL
   telemetry_metric: {{targets[0].telemetry_metric}}
   open_questions:
   {{targets[0].open_questions[0]}} | OPTIONAL

(Repeat per target.)

3. References
   Load models: {{xref:LOAD-02}} | OPTIONAL
   Load criteria: {{xref:LOAD-06}} | OPTIONAL
   Capacity findings: {{xref:LOAD-07}} | OPTIONAL

Cross-References
Upstream: {{xref:LOAD-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-06}}, {{xref:COST-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define targets with expected/peak and concurrency and owner.
intermediate: Required. Define assumptions and telemetry metric per target.

advanced: Required. Add headroom policy and stricter linkage to cost forecasts and scaling strategy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, headroom policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If targets[*].target_id is UNKNOWN → block Completeness Gate.
If targets[].expected_rate is UNKNOWN → block Completeness Gate.
If targets[*].peak_rate is UNKNOWN → block Completeness Gate.
If targets[].owner is UNKNOWN → block Completeness Gate.
If targets[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
targets_defined == true
assumptions_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-05

PERF-05 — Scaling Strategy (horizontal/vertical/caching)
Header Block
template_id: PERF-05
title: Scaling Strategy (horizontal/vertical/caching)
type: scaling_strategy
template_version: 1.0.0
output_path: 10_ops/performance/PERF-05_Scaling_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["PERF-04", "IAC-06", "PERF-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-04", "IAC-06", "PERF-06", "PERF-07", "RELIA-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical scaling strategy for the system: how services scale under load (horizontal/vertical), where caching is used, how autoscaling is configured, and what backpressure protections exist. This template ties throughput targets to concrete scaling mechanics.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Compute baseline/autoscaling: {{xref:IAC-06}} | OPTIONAL
DB performance policy: {{xref:PERF-06}} | OPTIONAL
Cache strategy: {{xref:PERF-07}} | OPTIONAL
Backpressure policy: {{xref:RELIA-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Scaling units (service/workers/db)
Horizontal scaling rule (replicas/autoscale)
Vertical scaling rule (instance sizing)
Autoscaling triggers (CPU/RPS/queue depth)
Caching usage rule (what can be cached)
Capacity headroom rule (target utilization)
Backpressure safeguards (limits, shedding)
Multi-region considerations (if any)
Telemetry requirements (scaling events, saturation)

Optional Fields
Cost tradeoffs notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Scaling strategy must include protection against thundering herds and overload.
If multi-region is not supported, state it explicitly in rules or UNKNOWN.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Units
   units: {{units.list}}
2. Horizontal
   rule: {{horiz.rule}}
   autoscale: {{horiz.autoscale}} | OPTIONAL
3. Vertical
   rule: {{vert.rule}}
4. Triggers
   triggers: {{scale.triggers}}
5. Caching
   cache_rule: {{cache.rule}}
   cache_ref: {{xref:PERF-07}} | OPTIONAL
6. Headroom
   utilization_target: {{headroom.utilization_target}}
   headroom_rule: {{headroom.rule}} | OPTIONAL
7. Backpressure
   backpressure_rule: {{bp.rule}}
   shed_rule: {{bp.shed_rule}} | OPTIONAL
8. Multi-Region
   supported: {{region.supported}}
   rule: {{region.rule}} | OPTIONAL
9. Telemetry
   scaling_event_metric: {{telemetry.scaling_event_metric}}
   saturation_metric: {{telemetry.saturation_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-01}}, {{xref:COST-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define horizontal/vertical rules and triggers and headroom target.
intermediate: Required. Define caching usage and backpressure safeguards and telemetry.

advanced: Required. Add multi-region rules and cost tradeoffs notes and strict capacity planning tie-ins.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, autoscale details, cache ref, headroom rule, shed rule, region rule, optional metrics, cost tradeoffs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If horiz.rule is UNKNOWN → block Completeness Gate.
If scale.triggers is UNKNOWN → block Completeness Gate.
If headroom.utilization_target is UNKNOWN → block Completeness Gate.
If bp.rule is UNKNOWN → block Completeness Gate.
If telemetry.scaling_event_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
scaling_rules_defined == true
triggers_defined == true
backpressure_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-06

PERF-06 — Database Performance Policy (indexes, query limits)
Header Block
template_id: PERF-06
title: Database Performance Policy (indexes, query limits)
type: database_performance_policy
template_version: 1.0.0
output_path: 10_ops/performance/PERF-06_Database_Performance_Policy.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["DATA-06", "PFS-05", "PERF-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "DATA-06", "PFS-05", "PERF-01", "PERF-09", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical database performance policy: query complexity limits, indexing expectations, pagination constraints, and observability requirements for DB performance. This template keeps DB performance predictable as the system scales.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Canonical data schemas: {{xref:DATA-06}} | OPTIONAL
Performance constraints (API): {{xref:PFS-05}} | OPTIONAL
Performance overview: {{xref:PERF-01}} | OPTIONAL
Perf regression gates: {{xref:PERF-09}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Query performance principles (avoid N+1)
Indexing policy (what must be indexed)
Query timeout rule (max ms)
Pagination requirements (cursor/limits)
Join/aggregation limits (max joins)
Read/write separation rules (if any)
Migration performance rules (backfills)
Observability requirements (slow query logs, metrics)
Change control rule (review for new queries)
Telemetry requirements (p95 query time metric)

Optional Fields
DB tiering rules (hot/cold data) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Any endpoint that can return many rows must use pagination and bounded queries.
Slow queries must be observable and attributable to endpoints.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Principles
   principles: {{principles.list}}
2. Indexing
   index_rule: {{index.rule}}
   required_indexes: {{index.required}} | OPTIONAL
3. Timeouts
   query_timeout_ms: {{timeouts.query_timeout_ms}}
   statement_timeout_rule: {{timeouts.statement_timeout_rule}} | OPTIONAL
4. Pagination
   pagination_ref: {{xref:PFS-02}} | OPTIONAL
   pagination_rule: {{paging.rule}}
5. Limits
   max_joins: {{limits.max_joins}}
   aggregation_rule: {{limits.aggregation_rule}} | OPTIONAL
6. Migrations / Backfills
   backfill_rule: {{migrations.backfill_rule}}
   safe_migration_ref: {{xref:ADMIN-04}} | OPTIONAL
7. Observability
   slow_query_log_rule: {{obs.slow_query_log_rule}}
   metrics: {{obs.metrics}} | OPTIONAL
8. Change Control
   review_rule: {{change.review_rule}}
9. Telemetry
   p95_query_time_metric: {{telemetry.p95_query_time_metric}}
   slow_query_count_metric: {{telemetry.slow_query_count_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:DATA-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-03}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define timeouts, pagination rule, max joins, and review rule.

intermediate: Required. Define indexing policy and backfill rule and telemetry metrics.
advanced: Required. Add DB tiering and strict observability metrics and query review enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, required indexes, statement timeout rule, pagination ref, aggregation rule, safe migration ref, obs metrics, optional metrics, DB tiering, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If timeouts.query_timeout_ms is UNKNOWN → block Completeness Gate.
If paging.rule is UNKNOWN → block Completeness Gate.
If migrations.backfill_rule is UNKNOWN → block Completeness Gate.
If change.review_rule is UNKNOWN → block Completeness Gate.
If telemetry.p95_query_time_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
timeouts_and_pagination_defined == true
indexing_and_limits_defined == true
observability_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-07

PERF-07 — Cache Strategy (tiers, invalidation)
Header Block
template_id: PERF-07
title: Cache Strategy (tiers, invalidation)
type: cache_strategy
template_version: 1.0.0
output_path: 10_ops/performance/PERF-07_Cache_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["PERF-06", "RLIM-01", "PRIV-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-06", "API-01", "PRIV-03", "RLIM-01", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical caching strategy for the system: cache tiers, what is cacheable, TTL rules, invalidation strategy, and privacy constraints. This template ensures caching improves performance without breaking correctness or privacy.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
DB performance policy: {{xref:PERF-06}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Data minimization rules: {{xref:PRIV-03}} | OPTIONAL
Rate limit policy: {{xref:RLIM-01}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Cache tiers (client, edge, server)
Cacheable data rules (what can be cached)
Non-cacheable rules (user-specific/PII)
TTL rules (default TTLs)
Invalidation strategy (write-through, event-driven)
Cache key rules (avoid raw PII in keys)
Staleness tolerance rules (stale-while-revalidate)
Observability requirements (hit rate, evictions)
Telemetry requirements (hit rate metric, stale serves)

Optional Fields
Cache warming strategy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL
Do not cache or key by raw PII; hashed identifiers only when allowed.
Cache invalidation must be deterministic for data that must be consistent.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Tiers
   tiers: {{tiers.list}}
2. Cacheability
   cacheable_rule: {{cache.rule}}
   non_cacheable_rule: {{cache.non_rule}}
3. TTL
   default_ttl_seconds: {{ttl.default_seconds}}
   ttl_overrides: {{ttl.overrides}} | OPTIONAL
4. Invalidation
   strategy: {{inv.strategy}}
   events_ref: {{xref:EVT-01}} | OPTIONAL
5. Keys
   key_rule: {{keys.rule}}
   pii_key_rule: {{keys.pii_key_rule}} | OPTIONAL
6. Staleness
   stale_rule: {{stale.rule}}
   swr_rule: {{stale.swr_rule}} | OPTIONAL
7. Observability
   hit_rate_metric_ref: {{obs.hit_rate_metric_ref}} | OPTIONAL
   eviction_metric_ref: {{obs.eviction_metric_ref}} | OPTIONAL
8. Telemetry
   cache_hit_rate_metric: {{telemetry.hit_rate_metric}}
   stale_serve_metric: {{telemetry.stale_serve_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PERF-05}}, {{xref:LOAD-03}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define tiers, cacheable/non-cacheable, TTL, invalidation strategy, key rule.
intermediate: Required. Define staleness rules and observability metrics and telemetry.
advanced: Required. Add warming strategy and strict event-driven invalidation + correctness constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ttl overrides, events ref, pii key rule, swr

rule, obs metric refs, optional metrics, warming, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tiers.list is UNKNOWN → block Completeness Gate.
If cache.rule is UNKNOWN → block Completeness Gate.
If ttl.default_seconds is UNKNOWN → block Completeness Gate.
If inv.strategy is UNKNOWN → block Completeness Gate.
If keys.rule is UNKNOWN → block Completeness Gate.
If telemetry.hit_rate_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
cache_policy_defined == true
ttl_and_keys_defined == true
invalidation_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-08

PERF-08 — Queue/Async Performance (jobs, backpressure)
Header Block
template_id: PERF-08
title: Queue/Async Performance (jobs, backpressure)
type: async_queue_performance
template_version: 1.0.0
output_path: 10_ops/performance/PERF-08_Queue_Async_Performance.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["JBS-02", "RELIA-04", "PERF-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "JBS-02", "JBS-05", "RELIA-04", "PERF-04", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical performance requirements for asynchronous work: job throughput targets, queue depth limits, worker concurrency, backpressure behavior, and observability. This template ensures background processing remains reliable under spikes.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Job specs: {{xref:JBS-02}} | OPTIONAL
Idempotency/concurrency: {{xref:JBS-05}} | OPTIONAL
Backpressure policy: {{xref:RELIA-04}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Queue/work types covered (job_id list or queue_id list)
Throughput targets (jobs/min)
Worker concurrency rules (max workers)
Queue depth thresholds (warn/breach)
Latency targets for async completion (p95)
Backpressure rule (shedding, pause intake)
Retry interaction (DLQ policy refs)
Observability requirements (queue depth metric, lag)
Telemetry requirements (queue lag metric, DLQ count)

Optional Fields
Priority queue rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Async performance targets must be measurable (queue metrics, job durations).
Backpressure must activate before overload cascades.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Scope
   queues_or_jobs: {{scope.items}}
2. Targets
   throughput: {{targets.throughput}}
   completion_p95_ms: {{targets.completion_p95_ms}}
3. Concurrency
   worker_rule: {{workers.rule}}
   max_workers: {{workers.max}} | OPTIONAL
4. Thresholds
   warn_depth: {{depth.warn}}
   breach_depth: {{depth.breach}} | OPTIONAL
5. Backpressure
   backpressure_rule: {{bp.rule}}
6. Retry/DLQ Interaction
   retry_ref: {{xref:JBS-04}} | OPTIONAL
   idempotency_ref: {{xref:JBS-05}} | OPTIONAL
   dlq_rule: {{dlq.rule}} | OPTIONAL
7. Observability
   queue_depth_metric: {{obs.queue_depth_metric}}
   queue_lag_metric: {{obs.queue_lag_metric}} | OPTIONAL
8. Telemetry
   queue_lag_telemetry_metric: {{telemetry.queue_lag_metric}}
   dlq_count_metric: {{telemetry.dlq_count_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:JBS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-03}}, {{xref:IRP-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define scope, throughput targets, backpressure rule, queue depth metric.
intermediate: Required. Define concurrency, thresholds, DLQ interaction, telemetry.
advanced: Required. Add priority queue rules and stricter multi-tenant fairness and capacity planning linkages.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, max workers, breach depth,

retry/idempotency refs, dlq rule, optional obs/telemetry metrics, priority queue, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scope.items is UNKNOWN → block Completeness Gate.
If targets.throughput is UNKNOWN → block Completeness Gate.
If bp.rule is UNKNOWN → block Completeness Gate.
If obs.queue_depth_metric is UNKNOWN → block Completeness Gate.
If telemetry.queue_lag_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
targets_defined == true
backpressure_defined == true
observability_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-09

PERF-09 — Performance Regression Gates (thresholds, CI checks)
Header Block
template_id: PERF-09
title: Performance Regression Gates (thresholds, CI checks)
type: performance_regression_gates
template_version: 1.0.0
output_path: 10_ops/performance/PERF-09_Performance_Regression_Gates.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["PBP-06", "CICD-03", "QA-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-06", "CICD-03", "QA-07", "PERF-03", "RJT-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical performance gating rules that block regressions: which tests run in CI, what thresholds apply, what is allowed to fail, and how waivers work. This template turns performance budgets into enforceable release gates.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Perf CI gates: {{xref:PBP-06}} | OPTIONAL
Test pipeline spec: {{xref:CICD-03}} | OPTIONAL
Release quality gates: {{xref:QA-07}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Journey test cadence: {{xref:RJT-05}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Gate registry (gate_id list)
gate_id (stable identifier)
Surface type (web/mobile/api)
Test suite binding (unit/e2e/load-lite)
Metrics measured (LCP, startup time, p95 latency)
Thresholds per metric (budget values)
Baseline comparison rule (vs main branch)
Fail behavior (block/warn)
Waiver/exception process (COMP-08)
Evidence artifact rule (store reports)
Telemetry requirements (gate failures, trend)

Optional Fields
Canary/perf lab notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Gates must be deterministic and repeatable; flaky perf tests must be quarantined.
Waivers must be time-bound and tracked.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_gates: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Gates (repeat)
   Gate
   gate_id: {{gates[0].gate_id}}
   surface: {{gates[0].surface}}
   test_suite: {{gates[0].test_suite}}
   metrics: {{gates[0].metrics}}
   thresholds: {{gates[0].thresholds}}
   baseline_rule: {{gates[0].baseline_rule}}
   fail_behavior: {{gates[0].fail_behavior}}
   waiver_rule: {{gates[0].waiver_rule}}
   evidence_rule: {{gates[0].evidence_rule}}
   telemetry_metric: {{gates[0].telemetry_metric}}
   canary_notes: {{gates[0].canary_notes}} | OPTIONAL
   open_questions:
   {{gates[0].open_questions[0]}} | OPTIONAL

(Repeat per gate.)

3. References
   Budget spec: {{xref:PBP-06}} | OPTIONAL
   Exceptions: {{xref:COMP-08}} | OPTIONAL
   Reporting: {{xref:PBP-07}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:REL-06}}, {{xref:LOAD-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define gates with metrics/thresholds and fail behavior and evidence rule.

intermediate: Required. Define baseline rule and waiver process and telemetry.
advanced: Required. Add canary/lab notes and strict flake handling and trend governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, canary/lab notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If gates[*].gate_id is UNKNOWN → block Completeness Gate.
If gates[*].thresholds is UNKNOWN → block Completeness Gate.
If gates[*].fail_behavior is UNKNOWN → block Completeness Gate.
If gates[*].evidence_rule is UNKNOWN → block Completeness Gate.
If gates[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
gate_registry_defined == true
thresholds_defined == true
waiver_defined == true
evidence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PERF-10

PERF-10 — Performance Runbooks (triage playbooks)
Header Block
template_id: PERF-10
title: Performance Runbooks (triage playbooks)
type: performance_runbooks
template_version: 1.0.0
output_path: 10_ops/performance/PERF-10_Performance_Runbooks.md
compliance_gate_id: TMP-05.PRIMARY.PERF
upstream_dependencies: ["PERF-09", "OBS-07", "IRP-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-09", "OBS-07", "OBS-03", "IRP-02", "LOAD-08"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical runbooks for performance issues: high latency, low throughput, cache misses, DB slowdowns, and client performance regressions. Runbooks must be step-by-step and reference the dashboards/metrics needed for triage.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Perf regression gates: {{xref:PERF-09}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Incident workflow: {{xref:IRP-02}} | OPTIONAL
Bottleneck analysis template: {{xref:LOAD-08}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Runbook registry (runbook_id list)
runbook_id (stable identifier)
Scenario type (api_latency/client_perf/db_slow/cache_miss)
Trigger signals (alerts/metrics thresholds)
Triage steps (ordered)
Isolation steps (identify component)
Mitigation steps (rollback/cache flush/scale)
Verification steps (prove recovery)
Escalation rule (who to involve)
Telemetry requirements (runbook invoked, time-to-mitigate)

Optional Fields
Automation hooks | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Runbooks must reference metric_ids/dashboard_ids; do not rely on tribal knowledge.
Mitigation steps must prefer safe actions first (reduce load before risky changes).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_runbooks: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Runbooks (repeat)
   Runbook
   runbook_id: {{books[0].runbook_id}}
   scenario: {{books[0].scenario}}
   triggers: {{books[0].triggers}}
   triage_steps:
   {{books[0].triage_steps[0]}}
   {{books[0].triage_steps[1]}}
   isolation_steps:
   {{books[0].isolation_steps[0]}}
   {{books[0].isolation_steps[1]}} | OPTIONAL
   mitigation_steps:
   {{books[0].mitigation_steps[0]}}
   {{books[0].mitigation_steps[1]}} | OPTIONAL
   verification_steps: {{books[0].verification_steps}}
   escalation_rule: {{books[0].escalation_rule}}
   telemetry_metric: {{books[0].telemetry_metric}}
   automation_hooks: {{books[0].automation_hooks}} | OPTIONAL
   open_questions:
   {{books[0].open_questions[0]}} | OPTIONAL

(Repeat per runbook.)

3. References
   Dashboards: {{xref:OBS-07}} | OPTIONAL
   Metrics: {{xref:OBS-03}} | OPTIONAL
   Incident workflow: {{xref:IRP-02}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-09}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:IRP-06}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define core runbooks with triage/mitigate/verify steps.
intermediate: Required. Define escalation rules and telemetry metrics.
advanced: Required. Add automation hooks and strict trigger mapping to alerts and budgets.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, meta notes, optional steps, automation hooks, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If books[*].runbook_id is UNKNOWN → block Completeness Gate.
If books[*].triage_steps[0] is UNKNOWN → block Completeness Gate.
If books[*].mitigation_steps[0] is UNKNOWN → block Completeness Gate.
If books[*].verification_steps is UNKNOWN → block Completeness Gate.
If books[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PERF
Pass conditions:
required_fields_present == true
runbooks_defined == true
triage_and_mitigation_defined == true
verification_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Load/Stress Planning (LOAD)

LOAD-01

LOAD-01 — Load Testing Overview (goals, scope)
Header Block
template_id: LOAD-01
title: Load Testing Overview (goals, scope)
type: load_testing_overview
template_version: 1.0.0
output_path: 10_ops/load/LOAD-01_Load_Testing_Overview.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["PERF-04", "SLO-06", "LOAD-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-04", "PERF-03", "SLO-06", "LOAD-02", "LOAD-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of load and stress testing: why we run it, what's in-scope, what success means, and how it connects to throughput targets and SLOs. This document anchors the LOAD set.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
SLO measurement rules: {{xref:SLO-06}} | OPTIONAL
Workload models: {{xref:LOAD-02}} | OPTIONAL
Pass/fail criteria: {{xref:LOAD-06}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Goals (capacity confidence, bottleneck discovery)
In-scope systems/surfaces
Out-of-scope statement (explicit)
Test types (load, stress, soak, spike)
Environment requirements (prod-like)
Data requirements (seed, privacy constraints)
Ownership (who runs tests)
Cadence (per release/quarterly)
Evidence artifact rule (store reports)
Telemetry requirements (test failures, trends)

Optional Fields
Tooling notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Load tests must be repeatable and have clear pass/fail criteria.
Never use real PII in load test data unless explicitly allowed by PRIV rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Goals
   {{goals[0]}}
   {{goals[1]}}
   {{goals[2]}} | OPTIONAL
2. Scope
   in_scope: {{scope.in}}
   out_of_scope: {{scope.out}} | OPTIONAL
3. Test Types
   types: {{types.list}}
4. Environments
   env_rule: {{env.rule}}
   envs: {{env.envs}} | OPTIONAL
5. Data
   data_rule: {{data.rule}}
   privacy_ref: {{xref:PRIV-02}} | OPTIONAL
6. Cadence & Ownership
   cadence: {{cadence.value}}
   owner: {{owner.team}}
7. Evidence
   report_artifacts: {{evidence.report_artifacts}}
   storage_location: {{evidence.storage_location}} | OPTIONAL
8. Telemetry
   load_test_fail_metric: {{telemetry.fail_metric}}
   trend_metric: {{telemetry.trend_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-02}}, {{xref:LOAD-03}}, {{xref:LOAD-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define scope, test types, env rule, cadence, evidence artifacts.

intermediate: Required. Define data rule and ownership and telemetry metrics.
advanced: Required. Add tooling notes and strict repeatability + change-trigger guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional goal, out of scope, envs list, privacy ref, storage location, optional metrics, tooling notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scope.in is UNKNOWN → block Completeness Gate.
If types.list is UNKNOWN → block Completeness Gate.
If env.rule is UNKNOWN → block Completeness Gate.
If evidence.report_artifacts is UNKNOWN → block Completeness Gate.
If telemetry.fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
scope_defined == true
test_types_defined == true
env_and_data_defined == true
evidence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-02

LOAD-02 — Workload Models (traffic shapes, personas)
Header Block
template_id: LOAD-02
title: Workload Models (traffic shapes, personas)
type: workload_models
template_version: 1.0.0
output_path: 10_ops/load/LOAD-02_Workload_Models.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["PERF-04", "ANL-07", "PERF-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-04", "PERF-02", "ANL-07", "API-01", "RJT-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical workload models used for load tests: traffic shapes, personas, endpoint mixes, and ramp patterns. This template ensures load tests simulate realistic usage and map to throughput targets and critical paths.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Critical paths: {{xref:PERF-02}} | OPTIONAL
Funnels/KPIs (usage distribution): {{xref:ANL-07}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Journey inventory: {{xref:RJT-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Model registry (model_id list)
model_id (stable identifier)
Persona definition (user type)
Traffic shape (steady/ramp/spike/soak)
Duration (minutes)
Target load (RPS/concurrency)
Endpoint mix (endpoint_id → %)
Think time rules (user pacing)
Data requirements (accounts, fixtures)
Owner (team)
Telemetry requirements (actual vs intended load)

Optional Fields
Regional distribution | OPTIONAL
Cache warmup phase | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Workload models must reference real endpoint IDs; do not invent endpoint_id values.
Target load must map to PERF-04 targets or explicitly state UNKNOWN.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_models: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Models (repeat)
   Model
   model_id: {{models[0].model_id}}
   persona: {{models[0].persona}}
   traffic_shape: {{models[0].traffic_shape}}
   duration_minutes: {{models[0].duration_minutes}}
   target_load: {{models[0].target_load}}
   endpoint_mix: {{models[0].endpoint_mix}}
   think_time_rule: {{models[0].think_time_rule}}
   data_requirements: {{models[0].data_requirements}}
   owner: {{models[0].owner}}
   regional_distribution: {{models[0].regional_distribution}} | OPTIONAL
   cache_warmup: {{models[0].cache_warmup}} | OPTIONAL
   telemetry_metric: {{models[0].telemetry_metric}}
   open_questions:
   {{models[0].open_questions[0]}} | OPTIONAL

(Repeat per model.)

3. References
   Scenarios catalog: {{xref:LOAD-03}} | OPTIONAL
   Setup rules: {{xref:LOAD-04}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-03}}, {{xref:LOAD-06}}, {{xref:LOAD-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define models with traffic shape, duration, target load, endpoint mix.
intermediate: Required. Define think time and data requirements and telemetry.
advanced: Required. Add regional distribution and cache warmup and strict mapping to KPIs/critical paths.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, regional distribution, cache warmup, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If models[*].model_id is UNKNOWN → block Completeness Gate.
If models[*].target_load is UNKNOWN → block Completeness Gate.
If models[*].endpoint_mix is UNKNOWN → block Completeness Gate.
If models[*].owner is UNKNOWN → block Completeness Gate.
If models[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
model_registry_defined == true
target_load_and_mix_defined == true
data_and_pacing_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-03

LOAD-03 — Test Scenarios Catalog (by scenario_id)
Header Block
template_id: LOAD-03
title: Test Scenarios Catalog (by scenario_id)
type: load_test_scenarios_catalog
template_version: 1.0.0
output_path: 10_ops/load/LOAD-03_Test_Scenarios_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["LOAD-02", "LOAD-06", "API-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LOAD-02", "LOAD-06", "API-01", "PERF-03", "PERF-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of load test scenarios, indexed by scenario_id, including the workload model used, the systems exercised, the duration, and the pass/fail rules. This catalog is used to standardize load testing and track coverage.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Workload models: {{xref:LOAD-02}} | OPTIONAL
Pass/fail criteria: {{xref:LOAD-06}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Scenario registry (scenario_id list)
scenario_id (stable identifier)
Scenario name/summary
Workload model binding (model_id)
Systems/surfaces exercised (service_id/endpoint_id list)
Duration (minutes)
Ramp pattern (if separate from model)
Pass/fail criteria binding (criteria_id or rules)
Evidence artifacts produced (reports, graphs)
Owner (team)
Telemetry requirements (scenario pass rate, run duration)

Optional Fields
Cost estimate notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Scenarios must reference existing model_ids and real endpoint/service IDs.
Pass/fail must reference measurable metrics (latency/error/throughput).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_scenarios: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Scenarios (repeat)
   Scenario
   scenario_id: {{scenarios[0].scenario_id}}
   name: {{scenarios[0].name}}
   model_id: {{scenarios[0].model_id}}
   surfaces: {{scenarios[0].surfaces}}
   duration_minutes: {{scenarios[0].duration_minutes}}
   ramp_pattern: {{scenarios[0].ramp_pattern}} | OPTIONAL
   pass_fail: {{scenarios[0].pass_fail}}
   evidence_artifacts: {{scenarios[0].evidence_artifacts}}
   owner: {{scenarios[0].owner}}
   telemetry_metric: {{scenarios[0].telemetry_metric}}
   cost_notes: {{scenarios[0].cost_notes}} | OPTIONAL
   open_questions:
   {{scenarios[0].open_questions[0]}} | OPTIONAL

(Repeat per scenario.)

3. References
   Harness/tooling: {{xref:LOAD-05}} | OPTIONAL
   Findings report: {{xref:LOAD-07}} | OPTIONAL

Cross-References
Upstream: {{xref:LOAD-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-07}}, {{xref:LOAD-08}}, {{xref:COST-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define scenario_id/name/model binding/duration and evidence artifacts.
intermediate: Required. Define pass/fail bindings and telemetry metric.

advanced: Required. Add cost estimate notes and strict mapping to budgets and targets + change triggers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, ramp pattern, cost notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scenarios[*].scenario_id is UNKNOWN → block Completeness Gate.
If scenarios[*].model_id is UNKNOWN → block Completeness Gate.
If scenarios[*].pass_fail is UNKNOWN → block Completeness Gate.
If scenarios[*].evidence_artifacts is UNKNOWN → block Completeness Gate.
If scenarios[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
scenario_registry_defined == true
bindings_defined == true
evidence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-04

LOAD-04 — Data & Environment Setup (fixtures, isolation)
Header Block
template_id: LOAD-04
title: Data & Environment Setup (fixtures, isolation)
type: load_data_environment_setup
template_version: 1.0.0
output_path: 10_ops/load/LOAD-04_Data_Environment_Setup.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["TDE-01", "TDE-04", "LOAD-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "TDE-01", "TDE-04", "ENV-01", "LOAD-05", "PRIV-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical setup rules for load tests: environment selection, isolation constraints, fixture/seed data requirements, and how to reset between runs. This template prevents load tests from polluting shared environments and ensures repeatability.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Environment overview: {{xref:TDE-01}} | OPTIONAL
Data reset/isolation rules: {{xref:TDE-04}} | OPTIONAL
Environment matrix: {{xref:ENV-01}} | OPTIONAL
Load tooling/harness: {{xref:LOAD-05}} | OPTIONAL
PII classification: {{xref:PRIV-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Environment selection rule (stage/prod-like)
Isolation rule (dedicated tenant/accounts)
Seed data requirements (dataset versions)
Account provisioning rule (how to create test users)
Reset rule (how to clean up)
External dependency rule (mock or real?)
Rate limit/abuse control overrides (if any)
Safety guardrails (no prod impact)
Telemetry requirements (setup failures, data drift)

Optional Fields
Cost controls (limit duration) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never use real customer PII in load fixtures unless explicitly allowed.
Load tests must not disable security controls in a way that invalidates results without being documented.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Environment
   selection_rule: {{env.selection_rule}}
   envs: {{env.envs}} | OPTIONAL
2. Isolation
   isolation_rule: {{isolation.rule}}
   tenant_strategy: {{isolation.tenant_strategy}} | OPTIONAL
3. Seed Data
   datasets: {{data.datasets}}
   versioning_rule: {{data.versioning_rule}} | OPTIONAL
4. Accounts
   provision_rule: {{acct.provision_rule}}
   account_pool_rule: {{acct.pool_rule}} | OPTIONAL
5. Reset / Cleanup
   reset_rule: {{reset.rule}}
6. External Dependencies
   dependency_rule: {{deps.rule}}
7. Overrides
   rate_limit_override_rule: {{overrides.rate_limit_rule}}
8. Guardrails
   safety_rule: {{guard.safety_rule}}
9. Telemetry
   setup_fail_metric: {{telemetry.setup_fail_metric}}
   data_drift_metric: {{telemetry.data_drift_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:TDE-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-07}}, {{xref:LOAD-08}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define env selection, isolation, seed data, reset rule, guardrails.
intermediate: Required. Define dependency rule, overrides, telemetry metrics.
advanced: Required. Add cost controls and strict dataset versioning and drift detection processes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, env list, tenant strategy, versioning rule, account pool, optional metrics, cost controls, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If env.selection_rule is UNKNOWN → block Completeness Gate.
If isolation.rule is UNKNOWN → block Completeness Gate.
If data.datasets is UNKNOWN → block Completeness Gate.
If reset.rule is UNKNOWN → block Completeness Gate.
If guard.safety_rule is UNKNOWN → block Completeness Gate.
If telemetry.setup_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
env_and_isolation_defined == true
data_and_reset_defined == true
guardrails_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-05

LOAD-05 — Tooling & Harness (runner, metrics capture)
Header Block
template_id: LOAD-05
title: Tooling & Harness (runner, metrics capture)
type: load_tooling_harness
template_version: 1.0.0
output_path: 10_ops/load/LOAD-05_Tooling_Harness.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["CICD-03", "OBS-03", "QAH-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CICD-03", "OBS-03", "OBS-07", "QAH-01", "LOAD-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical tooling and harness used to run load tests: the runner, scenario wiring, metrics capture, result storage, and CI integration. This template standardizes how load is generated and how results are made comparable.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Test pipeline spec: {{xref:CICD-03}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Harness overview: {{xref:QAH-01}} | OPTIONAL
Load testing overview: {{xref:LOAD-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Tooling inventory (tool_id list)
tool_id (stable identifier)
Runner selection (k6/locust/jmeter/UNKNOWN)
Scenario wiring rule (scenario_id → runner scripts)
Metrics capture rule (what captured, where)
Result storage location (artifacts repo/bucket)
CI integration rule (when tests run)
Secrets handling rule (test creds)
Environment config rule (endpoints, tokens)
Telemetry requirements (runner failures, missing metrics)

Optional Fields
Distributed load generation notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Test credentials must be handled via secrets management; never hardcode.
Load results must include enough metadata to reproduce (commit SHA, env).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Tools
   tools: {{tools.list}}
2. Runner
   runner: {{runner.name}}
   runner_rule: {{runner.rule}} | OPTIONAL
3. Scenario Wiring
   wiring_rule: {{wiring.rule}}
   script_location: {{wiring.script_location}} | OPTIONAL
4. Metrics Capture
   metrics: {{metrics.captured}}
   capture_pipeline: {{metrics.pipeline}} | OPTIONAL
5. Results Storage
   storage_location: {{results.storage_location}}
   retention_rule: {{results.retention_rule}} | OPTIONAL
6. CI Integration
   ci_rule: {{ci.rule}}
   pipeline_ref: {{xref:CICD-03}} | OPTIONAL
7. Secrets
   secrets_rule: {{secrets.rule}}
   secrets_ref: {{xref:SDR-02}} | OPTIONAL
8. Environment Config
   config_rule: {{env.config_rule}}
9. Telemetry
   runner_fail_metric: {{telemetry.runner_fail_metric}}
   missing_metrics_metric: {{telemetry.missing_metrics_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:LOAD-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-06}}, {{xref:LOAD-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define runner, scenario wiring, metrics capture, results storage, CI rule.

intermediate: Required. Define secrets handling, env config, telemetry metrics.
advanced: Required. Add distributed generation notes and strict reproducibility metadata requirements.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, runner rule, script location, pipeline, retention rule, pipeline ref, secrets ref, optional metrics, distributed notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If runner.name is UNKNOWN → block Completeness Gate.
If wiring.rule is UNKNOWN → block Completeness Gate.
If metrics.captured is UNKNOWN → block Completeness Gate.
If results.storage_location is UNKNOWN → block Completeness Gate.
If ci.rule is UNKNOWN → block Completeness Gate.
If telemetry.runner_fail_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
tooling_and_wiring_defined == true
metrics_and_storage_defined == true
ci_and_secrets_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-06

LOAD-06 — Pass/Fail Criteria (SLO ties)
Header Block
template_id: LOAD-06
title: Pass/Fail Criteria (SLO ties)
type: load_pass_fail_criteria
template_version: 1.0.0
output_path: 10_ops/load/LOAD-06_Pass_Fail_Criteria.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["SLO-06", "PERF-03", "PERF-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SLO-06", "PERF-03", "PERF-04", "OBS-03", "ALRT-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical pass/fail criteria for load tests: which metrics are evaluated, what thresholds apply, and what constitutes a failure. This template ties load results to latency budgets, throughput targets, and SLO measurement rules.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SLO measurement rules: {{xref:SLO-06}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Alert catalog (for overload signals): {{xref:ALRT-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Criteria registry (criteria_id list)
criteria_id (stable identifier)
Scenario binding (scenario_id)
Metrics evaluated (latency/error/throughput)
Thresholds per metric (p95 <= X, error <= Y)
Evaluation window (steady-state window)
Warmup exclusion rule (ignore first N minutes)
Failure classification (hard fail vs warn)
Evidence rule (include charts/reports)
Telemetry requirements (criteria failures count)

Optional Fields
Saturation signals (CPU, queue depth) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Criteria must be measurable with captured metrics; do not use subjective pass conditions.
If budgets/targets are unknown, criteria must flag it and default to conservative fail behavior or explicit waiver.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_criteria: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Criteria (repeat)
   Criteria
   criteria_id: {{criteria[0].criteria_id}}
   scenario_id: {{criteria[0].scenario_id}}
   metrics: {{criteria[0].metrics}}
   thresholds: {{criteria[0].thresholds}}
   eval_window: {{criteria[0].eval_window}}
   warmup_rule: {{criteria[0].warmup_rule}}
   fail_class: {{criteria[0].fail_class}}
   evidence_rule: {{criteria[0].evidence_rule}}
   saturation_signals: {{criteria[0].saturation_signals}} | OPTIONAL
   telemetry_metric: {{criteria[0].telemetry_metric}}
   open_questions:
   {{criteria[0].open_questions[0]}} | OPTIONAL

(Repeat per criteria.)

3. References
   Latency budgets: {{xref:PERF-03}} | OPTIONAL
   Throughput targets: {{xref:PERF-04}} | OPTIONAL
   Findings report: {{xref:LOAD-07}} | OPTIONAL

Cross-References
Upstream: {{xref:SLO-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-07}}, {{xref:LOAD-08}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define metrics/thresholds, warmup rule, fail class, evidence rule.

intermediate: Required. Define steady-state window and saturation signals and telemetry.
advanced: Required. Add stricter waiver behavior and multi-metric gating with trend comparisons.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, saturation signals, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If criteria[*].criteria_id is UNKNOWN → block Completeness Gate.
If criteria[].thresholds is UNKNOWN → block Completeness Gate.
If criteria[*].fail_class is UNKNOWN → block Completeness Gate.
If criteria[].evidence_rule is UNKNOWN → block Completeness Gate.
If criteria[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
criteria_registry_defined == true
thresholds_defined == true
evaluation_defined == true
evidence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-07

LOAD-07 — Capacity Findings Report (results template)
Header Block
template_id: LOAD-07
title: Capacity Findings Report (results template)
type: capacity_findings_report
template_version: 1.0.0
output_path: 10_ops/load/LOAD-07_Capacity_Findings_Report.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["LOAD-03", "LOAD-06", "LOAD-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "LOAD-03", "LOAD-06", "LOAD-05", "OBS-07", "COST-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical reporting template for load test findings: what ran, what passed/failed, bottlenecks found, capacity limits, and recommended next actions. This template standardizes result communication and creates an evidence artifact for performance readiness.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Scenarios catalog: {{xref:LOAD-03}} | OPTIONAL
Pass/fail criteria: {{xref:LOAD-06}} | OPTIONAL
Tooling/harness: {{xref:LOAD-05}} | OPTIONAL
Dashboards: {{xref:OBS-07}} | OPTIONAL
Cost reporting: {{xref:COST-09}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Report metadata (date, env, commit)
Scenario(s) executed (scenario_id list)
Workload model summary (model_id)
Key results (pass/fail)
Capacity limit statement (what max sustainable load)
Primary bottlenecks found (top 1–5)
SLO/latency impact summary
Mitigations recommended (ranked)
Follow-up tests required
Evidence artifacts links (reports/charts)
Telemetry requirements (report published, follow-ups closed)

Optional Fields
Cost impact summary | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Findings must be backed by evidence artifacts (charts, logs).
Avoid “recommendations” phrasing that introduces new scope; treat as mitigation options.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Report Metadata
   date: {{meta.date}}
   environment: {{meta.environment}}
   commit_sha: {{meta.commit_sha}}
2. What Ran
   scenario_ids: {{run.scenario_ids}}
   model_ids: {{run.model_ids}} | OPTIONAL
3. Results
   pass_fail_summary: {{results.pass_fail_summary}}
   key_metrics: {{results.key_metrics}} | OPTIONAL
4. Capacity Limit
   max_sustainable_load: {{capacity.max_sustainable_load}}
   constraints: {{capacity.constraints}} | OPTIONAL
5. Bottlenecks
   {{bottlenecks[0]}}
   {{bottlenecks[1]}} | OPTIONAL
6. Impact
   slo_impact: {{impact.slo_impact}}
   latency_impact: {{impact.latency_impact}} | OPTIONAL
7. Mitigations
   {{mitigations[0]}}
   {{mitigations[1]}} | OPTIONAL
8. Follow-Ups
   tests_required: {{followups.tests_required}}
   owners: {{followups.owners}} | OPTIONAL
9. Evidence
   artifacts: {{evidence.artifacts}}
10. Telemetry
    report_published_metric: {{telemetry.report_published_metric}}
    followups_closed_metric: {{telemetry.followups_closed_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:LOAD-03}}, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:LOAD-08}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill metadata, scenarios, pass/fail, capacity limit, evidence artifacts.
intermediate: Required. Add bottlenecks, impact summary, mitigations, telemetry.
advanced: Required. Add cost impact summary and strict follow-up tracking with owners/dates.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, model ids, key metrics, capacity constraints, optional bottlenecks/mitigations, cost impact, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If meta.date is UNKNOWN → block Completeness Gate.
If run.scenario_ids is UNKNOWN → block Completeness Gate.
If results.pass_fail_summary is UNKNOWN → block Completeness Gate.
If capacity.max_sustainable_load is UNKNOWN → block Completeness Gate.
If evidence.artifacts is UNKNOWN → block Completeness Gate.
If telemetry.report_published_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
metadata_defined == true
results_defined == true
capacity_and_evidence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-08

LOAD-08 — Bottleneck Analysis Template (what failed)
Header Block
template_id: LOAD-08
title: Bottleneck Analysis Template (what failed)
type: bottleneck_analysis_template
template_version: 1.0.0
output_path: 10_ops/load/LOAD-08_Bottleneck_Analysis_Template.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["OBS-07", "OBS-03", "PERF-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OBS-07", "OBS-03", "LTS-02", "PERF-06", "RELIA-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical analysis template used when load tests fail or degrade: how to identify the limiting component (CPU, DB, cache, queue, network), what evidence to collect, and how to propose mitigations. This template standardizes bottleneck writeups.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Tracing standard: {{xref:LTS-02}} | OPTIONAL
DB performance policy: {{xref:PERF-06}} | OPTIONAL
Failure modes catalog: {{xref:RELIA-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Incident/scenario reference (scenario_id)
Symptom summary (what degraded)
Limiting resource hypothesis (CPU/DB/IO)
Evidence checklist (metrics/logs/traces)
Primary signals (which metric_ids)
Isolation steps (how narrowed down)
Root cause statement (if known)
Mitigation options (ranked)
Retest plan (what scenario rerun)
Telemetry requirements (time-to-root-cause)

Optional Fields
Cost tradeoff notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Every claim must be backed by evidence artifacts or marked UNKNOWN.
Mitigations must be tied to the bottleneck mechanism (not generic).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Reference
   scenario_id: {{ref.scenario_id}}
   test_run_id: {{ref.test_run_id}} | OPTIONAL
2. Symptoms
   summary: {{symptoms.summary}}
   metrics_affected: {{symptoms.metrics_affected}} | OPTIONAL
3. Hypothesis
   limiting_resource: {{hyp.resource}}
   hypothesis_notes: {{hyp.notes}} | OPTIONAL
4. Evidence
   checklist: {{evidence.checklist}}
   primary_signals: {{evidence.primary_signals}}
5. Isolation
   steps: {{isolation.steps}}
6. Root Cause
   root_cause: {{rc.root_cause}}
7. Mitigations
   {{mitigations[0]}}
   {{mitigations[1]}} | OPTIONAL
8. Retest Plan
   retest_scenario_id: {{retest.scenario_id}}
   criteria_ref: {{xref:LOAD-06}} | OPTIONAL
9. Telemetry
   time_to_rca_metric: {{telemetry.time_to_rca_metric}}

Cross-References
Upstream: {{xref:LOAD-07}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PERF-10}}, {{xref:COST-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill reference, symptoms, evidence checklist, root cause (or UNKNOWN), retest plan.

intermediate: Required. Add hypothesis + isolation steps + mitigations + telemetry.
advanced: Required. Add cost tradeoffs and strict evidence linking (dashboards/trace IDs).

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, test run id, metrics affected, hypothesis notes, cost tradeoffs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ref.scenario_id is UNKNOWN → block Completeness Gate.
If evidence.checklist is UNKNOWN → block Completeness Gate.
If evidence.primary_signals is UNKNOWN → block Completeness Gate.
If retest.scenario_id is UNKNOWN → block Completeness Gate.
If telemetry.time_to_rca_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
reference_defined == true
evidence_defined == true
mitigation_and_retest_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-09

LOAD-09 — Chaos/Failure Injection Plan (optional)
Header Block
template_id: LOAD-09
title: Chaos/Failure Injection Plan (optional)
type: chaos_failure_injection_plan
template_version: 1.0.0
output_path: 10_ops/load/LOAD-09_Chaos_Failure_Injection_Plan.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["RELIA-08", "IRP-09", "RELIA-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "RELIA-08", "RELIA-02", "IRP-09", "ALRT-02", "OBS-07"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical plan for chaos/failure injection (if used): what failure modes are injected, safety guardrails, success criteria, and how learnings translate into resilience improvements. This is optional but recommended for mature reliability practices.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Chaos engineering plan: {{xref:RELIA-08}} | OPTIONAL
Failure modes catalog: {{xref:RELIA-02}} | OPTIONAL
Incident drills: {{xref:IRP-09}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Dashboards: {{xref:OBS-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Scope (services/environments)
Failure modes to test (fm_id list)
Injection methods (latency, kill pod, drop deps)
Safety guardrails (abort conditions)
Success criteria (detection + recovery)
Runbook linkage (what to follow)
Ownership (who runs experiments)
Cadence (quarterly/UNKNOWN)
Telemetry requirements (experiment pass rate, aborts)

Optional Fields
Blast radius rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Chaos tests must be safe, time-boxed, and have abort conditions.
Do not run chaos experiments in production without explicit approvals.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Scope
   services: {{scope.services}}
   environments: {{scope.envs}} | OPTIONAL
2. Failure Modes
   fm_ids: {{modes.fm_ids}}
3. Injection
   methods: {{inject.methods}}
   tooling: {{inject.tooling}} | OPTIONAL
4. Guardrails
   abort_conditions: {{guard.abort_conditions}}
   safety_rule: {{guard.safety_rule}} | OPTIONAL
5. Success Criteria
   criteria: {{success.criteria}}
6. Runbooks
   runbook_refs: {{runbooks.refs}}
   incident_ref: {{xref:IRP-02}} | OPTIONAL
7. Ownership & Cadence
   owner: {{owner.team}}
   cadence: {{cadence.value}}
8. Telemetry
   chaos_pass_rate_metric: {{telemetry.pass_rate_metric}}
   abort_metric: {{telemetry.abort_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:RELIA-08}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:IRP-06}}, {{xref:RELIA-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Optional.
intermediate: Required if chaos is used. Define scope, modes, guardrails, criteria, owner.
advanced: Required if chaos is used. Add blast radius and approvals and strict telemetry + remediation linkage.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, envs, tooling, safety rule, abort metric, blast radius, open_questions

If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If modes.fm_ids is UNKNOWN → block Completeness Gate (when used).
If guard.abort_conditions is UNKNOWN → block Completeness Gate (when used).
If success.criteria is UNKNOWN → block Completeness Gate (when used).
If telemetry.pass_rate_metric is UNKNOWN → block Completeness Gate (when used).

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions (when used):
required_fields_present == true
scope_defined == true
guardrails_defined == true
success_criteria_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

LOAD-10

LOAD-10 — Re-run Cadence & Change Triggers
Header Block
template_id: LOAD-10
title: Re-run Cadence & Change Triggers
type: load_rerun_cadence_triggers
template_version: 1.0.0
output_path: 10_ops/load/LOAD-10_Rerun_Cadence_Change_Triggers.md
compliance_gate_id: TMP-05.PRIMARY.LOAD
upstream_dependencies: ["REL-02", "LOAD-06", "PERF-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "REL-02", "LOAD-06", "PERF-04", "PERF-05", "COST-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for when load tests must be re-run: scheduled cadence, change triggers (code/config/infra), and what to do when results regress. This template prevents stale capacity assumptions.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Change management workflow: {{xref:REL-02}} | OPTIONAL
Pass/fail criteria: {{xref:LOAD-06}} | OPTIONAL
Throughput targets: {{xref:PERF-04}} | OPTIONAL
Scaling strategy: {{xref:PERF-05}} | OPTIONAL
Forecast model: {{xref:COST-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Baseline cadence (weekly/monthly/quarterly)
Trigger changes list (new endpoint, DB schema change, infra change)
Risk classification mapping (high-risk requires rerun)
Rerun scope rule (which scenarios to rerun)
Evidence retention rule (keep last N runs)
Escalation rule (regression → block release)
Ownership (who schedules reruns)
Telemetry requirements (reruns completed, overdue)

Optional Fields
Automated trigger detection notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
High-risk changes must trigger reruns before release.
Reruns must reference scenario_ids and pass/fail criteria; do not rerun “ad hoc.”
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Cadence
   baseline_cadence: {{cadence.baseline}}
2. Triggers
   triggers: {{triggers.list}}
3. Risk Mapping
   risk_rule: {{risk.rule}}
   risk_levels: {{risk.levels}} | OPTIONAL
4. Rerun Scope
   scope_rule: {{scope.rule}}
   scenario_selection: {{scope.scenario_selection}} | OPTIONAL
5. Evidence Retention
   retention_rule: {{evidence.retention_rule}}
   storage_ref: {{xref:LOAD-05}} | OPTIONAL
6. Escalation
   block_release_rule: {{esc.block_release_rule}}
   release_ref: {{xref:REL-02}} | OPTIONAL
7. Ownership
   owner: {{owner.team}}
8. Telemetry
   reruns_completed_metric: {{telemetry.completed_metric}}
   reruns_overdue_metric: {{telemetry.overdue_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:REL-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:LOAD-07}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define cadence, triggers list, scope rule, block release rule.
intermediate: Required. Define risk mapping and evidence retention and telemetry.
advanced: Required. Add automated trigger detection and strict governance for bypassing reruns.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, risk levels, scenario selection, storage ref, optional metrics, automation notes, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If cadence.baseline is UNKNOWN → block Completeness Gate.
If triggers.list is UNKNOWN → block Completeness Gate.
If scope.rule is UNKNOWN → block Completeness Gate.
If esc.block_release_rule is UNKNOWN → block Completeness Gate.
If telemetry.completed_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.LOAD
Pass conditions:
required_fields_present == true
cadence_and_triggers_defined == true
scope_defined == true
escalation_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Capacity & Cost Modeling (COST)

COST-01

COST-01 — Cost Model Overview (dimensions, owners)
Header Block
template_id: COST-01
title: Cost Model Overview (dimensions, owners)
type: cost_model_overview
template_version: 1.0.0
output_path: 10_ops/cost/COST-01_Cost_Model_Overview.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-02", "COST-08", "COMP-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-02", "COST-08", "COMP-01", "COST-03", "COST-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of cost modeling for the system: what dimensions are tracked, who owns cost governance, and how cost is measured and acted upon. This document anchors the COST set and enables cost-aware operations.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost drivers catalog: {{xref:COST-02}} | OPTIONAL
Cost attribution tags standard: {{xref:COST-08}} | OPTIONAL
Compliance scope (cost evidence/ownership): {{xref:COMP-01}} | OPTIONAL
Forecast model: {{xref:COST-03}} | OPTIONAL
Budget/alerts: {{xref:COST-04}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Cost dimensions (compute/storage/egress/3rd-party)
Attribution model (who pays: team/tenant)
Owners (finops, eng lead)
Measurement sources (billing exports, tags)
Cadence (weekly/monthly review)
Decision rules (what triggers optimization)
Cost vs performance tradeoff principle
Telemetry requirements (cost anomalies detected)

Optional Fields
Chargeback/showback notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Cost model must be consistent with tagging and reporting; no “unattributed” costs without a plan.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Dimensions
   dimensions: {{dims.list}}
2. Attribution
   attribution_model: {{attrib.model}}
   keys: {{attrib.keys}} | OPTIONAL
3. Owners
   owners: {{owners.list}}
4. Measurement Sources
   sources: {{sources.list}}
5. Cadence
   cadence: {{cadence.value}}
6. Decision Rules
   rules: {{decisions.rules}}
7. Tradeoffs
   tradeoff_principle: {{tradeoff.principle}}
8. Telemetry
   cost_anomaly_metric: {{telemetry.cost_anomaly_metric}}

Cross-References
Upstream: {{xref:COST-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-03}}, {{xref:COST-09}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define dimensions, attribution model, owners, cadence.
intermediate: Required. Define measurement sources, decision rules, telemetry.
advanced: Required. Add showback/chargeback notes and strict tradeoff guidance with performance ties.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, keys, showback notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If dims.list is UNKNOWN → block Completeness Gate.
If attrib.model is UNKNOWN → block Completeness Gate.
If owners.list is UNKNOWN → block Completeness Gate.

If cadence.value is UNKNOWN → block Completeness Gate.
If telemetry.cost_anomaly_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
dimensions_defined == true
attribution_defined == true
owners_and_cadence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-02

COST-02 — Cost Drivers Catalog (compute, storage, egress)
Header Block
template_id: COST-02
title: Cost Drivers Catalog (compute, storage, egress)
type: cost_drivers_catalog
template_version: 1.0.0
output_path: 10_ops/cost/COST-02_Cost_Drivers_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-01", "IAC-06", "LTS-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-01", "IAC-06", "IAC-07", "LTS-05", "OBS-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of cost drivers, indexed by driver_id, including what the driver is, what controls it, and which telemetry tracks it. This catalog supports budgeting, forecasting, and optimization work.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost model overview: {{xref:COST-01}} | OPTIONAL
Compute baseline: {{xref:IAC-06}} | OPTIONAL
Data baseline (storage): {{xref:IAC-07}} | OPTIONAL
Log routing/storage: {{xref:LTS-05}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Driver registry (driver_id list)
driver_id (stable identifier)
Driver name/category (compute/storage/egress/vendor)
What creates the cost (usage description)
Primary levers (scale down, cache, compress)
Owner (team/finops)
Telemetry metric refs (metric_id list)
Budget linkage (COST-04)
Optimization candidates (common fixes)
Telemetry requirements (driver cost anomalies)

Optional Fields
Provider SKU references | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Drivers should be attributable (tags/accounts) and connected to measurable telemetry.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_drivers: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Drivers (repeat)
   Driver
   driver_id: {{drivers[0].driver_id}}
   name: {{drivers[0].name}}
   category: {{drivers[0].category}}
   usage_description: {{drivers[0].usage_description}}
   levers: {{drivers[0].levers}}
   owner: {{drivers[0].owner}}
   metric_ids: {{drivers[0].metric_ids}}
   budget_ref: {{xref:COST-04}} | OPTIONAL
   optimization_candidates: {{drivers[0].optimization_candidates}}
   anomaly_metric: {{drivers[0].anomaly_metric}}
   sku_refs: {{drivers[0].sku_refs}} | OPTIONAL
   open_questions:
   {{drivers[0].open_questions[0]}} | OPTIONAL

(Repeat per driver.)

3. References
   Forecast: {{xref:COST-03}} | OPTIONAL
   Reporting: {{xref:COST-09}} | OPTIONAL

Cross-References
Upstream: {{xref:COST-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-04}}, {{xref:COST-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define driver registry, category, usage description, levers, owner.
intermediate: Required. Define telemetry metrics and optimization candidates and anomaly

metric.
advanced: Required. Add SKU refs and stricter budget linkage + attribution requirements.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, budget ref, sku refs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If drivers[*].driver_id is UNKNOWN → block Completeness Gate.
If drivers[].category is UNKNOWN → block Completeness Gate.
If drivers[*].levers is UNKNOWN → block Completeness Gate.
If drivers[].owner is UNKNOWN → block Completeness Gate.
If drivers[*].anomaly_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
driver_registry_defined == true
telemetry_defined == true
ownership_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-03

COST-03 — Forecast Model (monthly/annual)
Header Block
template_id: COST-03
title: Forecast Model (monthly/annual)
type: cost_forecast_model
template_version: 1.0.0
output_path: 10_ops/cost/COST-03_Forecast_Model.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-01", "COST-02", "PERF-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-01", "COST-02", "PERF-04", "COST-09", "ANL-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical cost forecasting model: how monthly/annual cost is projected using growth assumptions, workload projections, and unit cost drivers. This template supports budget planning and scenario analysis.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost model overview: {{xref:COST-01}} | OPTIONAL
Cost drivers catalog: {{xref:COST-02}} | OPTIONAL
Throughput targets (growth proxy): {{xref:PERF-04}} | OPTIONAL
Cost reporting baselines: {{xref:COST-09}} | OPTIONAL
Funnels/KPIs (growth): {{xref:ANL-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Forecast horizon (months)
Scenarios (base/high/low)
Input assumptions (users, RPS, storage growth)
Unit costs (per driver/category)
Computation rule (how totals derived)
Update cadence (monthly)
Owner (finops)
Telemetry requirements (forecast error metric)

Optional Fields
Sensitivity analysis notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Forecast must be explainable (inputs → outputs).
Use the cost drivers catalog for unit costs; do not invent drivers.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Horizon
   months: {{horizon.months}}
2. Scenarios
   scenarios: {{scenarios.list}}
3. Assumptions
   assumptions: {{assumptions.list}}
4. Unit Costs
   unit_costs: {{unit_costs.list}}
5. Computation
   calculation_rule: {{calc.rule}}
6. Cadence & Ownership
   cadence: {{cadence.value}}
   owner: {{owner.team}}
7. Telemetry
   forecast_error_metric: {{telemetry.forecast_error_metric}}

Cross-References
Upstream: {{xref:COST-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-04}}, {{xref:COST-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define horizon, scenarios, assumptions, and calc rule.
intermediate: Required. Define unit costs and cadence and telemetry.
advanced: Required. Add sensitivity analysis and strict linkage to growth KPIs + throughput targets.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, sensitivity notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If horizon.months is UNKNOWN → block Completeness Gate.
If scenarios.list is UNKNOWN → block Completeness Gate.
If assumptions.list is UNKNOWN → block Completeness Gate.
If calc.rule is UNKNOWN → block Completeness Gate.
If telemetry.forecast_error_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
forecast_inputs_defined == true
calculation_defined == true
ownership_and_cadence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-04

COST-04 — Budget & Alerts (thresholds, routing)
Header Block
template_id: COST-04
title: Budget & Alerts (thresholds, routing)
type: cost_budget_alerts
template_version: 1.0.0
output_path: 10_ops/cost/COST-04_Budget_Alerts.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-01", "COST-03", "ALRT-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-01", "COST-03", "COST-02", "ALRT-02", "OBS-08"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical cost budgets and alerting: what thresholds exist, how they are evaluated, who gets notified, and what actions are required when budgets are exceeded or trending poorly.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost model overview: {{xref:COST-01}} | OPTIONAL
Forecast model: {{xref:COST-03}} | OPTIONAL
Cost drivers catalog: {{xref:COST-02}} | OPTIONAL
Alert catalog: {{xref:ALRT-02}} | OPTIONAL
Routing/ownership: {{xref:OBS-08}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Budget registry (budget_id list)
budget_id (stable identifier)
Scope (global/team/service/tenant)
Budget amount (monthly)
Thresholds (warn/breach)
Evaluation cadence (daily/weekly)
Alert routing (team/oncall/email)
Owner (finops/team)
Action rules (what to do on warn/breach)
Telemetry requirements (budget breach count)

Optional Fields
Anomaly detection rule | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must be scoped and attributable; avoid “misc” budgets.
Alerts must route to owners and include actionable guidance.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_budgets: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Budgets (repeat)
   Budget
   budget_id: {{budgets[0].budget_id}}
   scope: {{budgets[0].scope}}
   amount_monthly: {{budgets[0].amount_monthly}}
   thresholds: {{budgets[0].thresholds}}
   cadence: {{budgets[0].cadence}}
   routing: {{budgets[0].routing}}
   owner: {{budgets[0].owner}}
   actions: {{budgets[0].actions}}
   anomaly_rule: {{budgets[0].anomaly_rule}} | OPTIONAL
   telemetry_metric: {{budgets[0].telemetry_metric}}
   open_questions:
   {{budgets[0].open_questions[0]}} | OPTIONAL

(Repeat per budget.)

3. References
   Cost reporting: {{xref:COST-09}} | OPTIONAL
   Optimization backlog: {{xref:COST-06}} | OPTIONAL

Cross-References
Upstream: {{xref:COST-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-06}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define budgets with scope/amount/thresholds and routing and actions.
intermediate: Required. Define cadence and telemetry metric per budget.
advanced: Required. Add anomaly detection rule and strict escalation + post-breach review practices.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, anomaly rule, open_questions

If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If budgets[*].budget_id is UNKNOWN → block Completeness Gate.
If budgets[*].amount_monthly is UNKNOWN → block Completeness Gate.
If budgets[*].thresholds is UNKNOWN → block Completeness Gate.
If budgets[*].routing is UNKNOWN → block Completeness Gate.
If budgets[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
budget_registry_defined == true
thresholds_defined == true
routing_defined == true
actions_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-05

COST-05 — Unit Economics (cost per user/action)
Header Block
template_id: COST-05
title: Unit Economics (cost per user/action)
type: cost_unit_economics
template_version: 1.0.0
output_path: 10_ops/cost/COST-05_Unit_Economics.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-02", "ANL-07", "PERF-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-02", "COST-09", "ANL-07", "PERF-04", "COST-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical unit economics model: how to compute cost per user, cost per key action, and cost per tenant using cost drivers and usage metrics. This template supports pricing decisions and cost optimization prioritization.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost drivers catalog: {{xref:COST-02}} | OPTIONAL
Cost reporting baselines: {{xref:COST-09}} | OPTIONAL
Funnels/KPIs (usage volume): {{xref:ANL-07}} | OPTIONAL
Throughput targets (action volume proxy): {{xref:PERF-04}} | OPTIONAL
Forecast model: {{xref:COST-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Unit definitions (user/action/tenant)
Action catalog for unit cost (action_id list)
Cost allocation rule (drivers → units)
Usage measurement rule (events/metrics used)
Computation formulas (cost/unit)
Cadence (monthly)
Owner (finops/product)
Telemetry requirements (unit cost trend metric)

Optional Fields
Pricing linkage notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Unit economics must be based on measurable usage and driver costs; avoid guesses.
Allocation rules must be consistent and documented.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Units
   units: {{units.list}}
2. Actions
   actions: {{actions.list}}
3. Allocation
   allocation_rule: {{alloc.rule}}
4. Usage Measurement
   usage_rule: {{usage.rule}}
   usage_sources: {{usage.sources}} | OPTIONAL
5. Formulas
   formulas: {{formulas.list}}
6. Cadence & Ownership
   cadence: {{cadence.value}}
   owner: {{owner.team}}
7. Telemetry
   unit_cost_trend_metric: {{telemetry.unit_cost_trend_metric}}

Cross-References
Upstream: {{xref:COST-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-06}}, {{xref:COST-03}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define unit definitions, allocation rule, usage rule, formulas.
intermediate: Required. Define actions list, cadence, telemetry metric.
advanced: Required. Add pricing linkage notes and stricter allocation audits + variance analysis.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, usage sources, pricing linkage, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If units.list is UNKNOWN → block Completeness Gate.
If alloc.rule is UNKNOWN → block Completeness Gate.
If usage.rule is UNKNOWN → block Completeness Gate.
If formulas.list is UNKNOWN → block Completeness Gate.
If telemetry.unit_cost_trend_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
units_and_actions_defined == true
allocation_defined == true
formulas_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-06

COST-06 — Optimization Backlog (ranked opportunities)
Header Block
template_id: COST-06
title: Optimization Backlog (ranked opportunities)
type: cost_optimization_backlog
template_version: 1.0.0
output_path: 10_ops/cost/COST-06_Optimization_Backlog.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-02", "COST-04", "COST-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-02", "COST-04", "COST-09", "PERF-05", "REL-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical backlog of cost optimization opportunities, ranked by impact and effort, with clear owners and evidence. This template turns cost insights into actionable work rather than ad hoc ideas.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost drivers: {{xref:COST-02}} | OPTIONAL
Budgets/alerts: {{xref:COST-04}} | OPTIONAL
Cost reporting: {{xref:COST-09}} | OPTIONAL
Scaling strategy: {{xref:PERF-05}} | OPTIONAL
Change management: {{xref:REL-02}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Opportunity registry (opt_id list)
opt_id (stable identifier)
Driver linkage (driver_id)
Opportunity description
Estimated monthly savings
Effort estimate (S/M/L)
Risk level (low/med/high)
Owner (team)
Status (backlog/in-progress/done)
Evidence link (cost report/chart)
Telemetry requirements (savings realized metric)

Optional Fields
Dependencies | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Each opportunity must link to a measurable driver and evidence.
Savings estimates must be grounded or marked UNKNOWN; avoid invented numbers.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_opportunities: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Opportunities (repeat)
   Opportunity
   opt_id: {{opts[0].opt_id}}
   driver_id: {{opts[0].driver_id}}
   description: {{opts[0].description}}
   savings_monthly: {{opts[0].savings_monthly}}
   effort: {{opts[0].effort}}
   risk: {{opts[0].risk}}
   owner: {{opts[0].owner}}
   status: {{opts[0].status}}
   evidence: {{opts[0].evidence}}
   deps: {{opts[0].deps}} | OPTIONAL
   telemetry_metric: {{opts[0].telemetry_metric}}
   open_questions:
   {{opts[0].open_questions[0]}} | OPTIONAL

(Repeat per opportunity.)

3. References
   Cost drivers: {{xref:COST-02}} | OPTIONAL
   Change workflow: {{xref:REL-02}} | OPTIONAL

Cross-References
Upstream: {{xref:COST-09}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-10}}, {{xref:REL-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define opportunity list, driver linkage, owner, and evidence.
intermediate: Required. Define savings estimate and effort/risk and telemetry.

advanced: Required. Add dependency mapping and strict “realized savings” verification and closure criteria.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, deps, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If opts[*].opt_id is UNKNOWN → block Completeness Gate.*
*If opts[*].driver_id is UNKNOWN → block Completeness Gate.
If opts[*].owner is UNKNOWN → block Completeness Gate.*
*If opts[*].evidence is UNKNOWN → block Completeness Gate.
If opts[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
opportunity_registry_defined == true
driver_and_evidence_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-07

COST-07 — Showback/Chargeback Policy (by team/tenant)
Header Block
template_id: COST-07
title: Showback/Chargeback Policy (by team/tenant)
type: showback_chargeback_policy
template_version: 1.0.0
output_path: 10_ops/cost/COST-07_Showback_Chargeback_Policy.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-01", "COST-08", "COST-09"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-01", "COST-08", "COST-09", "COST-05", "COMP-01"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for showback/chargeback: how costs are attributed to teams/tenants, how costs are reported, and how disputes and adjustments are handled. This template is optional for early-stage systems but required if chargeback is used.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost model overview: {{xref:COST-01}} | OPTIONAL
Attribution tags standard: {{xref:COST-08}} | OPTIONAL
Cost reporting: {{xref:COST-09}} | OPTIONAL
Unit economics: {{xref:COST-05}} | OPTIONAL
Compliance scope/contracts: {{xref:COMP-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Policy mode (showback vs chargeback)
Attribution keys (team_id, tenant_id)
Allocation rules (shared infra allocation)
Reporting cadence (monthly)
Dispute/adjustment workflow
Approval roles (finops, leadership)
Data sources (billing exports)
Telemetry requirements (unattributed cost %, disputes count)

Optional Fields
Minimum bill thresholds | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Chargeback must be explainable and auditable; allocations must be documented.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Mode
   mode: {{mode.value}}
2. Keys
   keys: {{keys.list}}
3. Allocation
   shared_allocation_rule: {{alloc.shared_rule}}
   allocation_examples: {{alloc.examples}} | OPTIONAL
4. Reporting
   cadence: {{report.cadence}}
   audience: {{report.audience}} | OPTIONAL
5. Disputes
   workflow: {{disputes.workflow}}
   sla_days: {{disputes.sla_days}} | OPTIONAL
6. Approvals
   approver_roles: {{approve.roles}}
7. Sources
   sources: {{sources.list}}
8. Telemetry
   unattributed_cost_metric: {{telemetry.unattributed_cost_metric}}
   disputes_count_metric: {{telemetry.disputes_count_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:COST-08}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-09}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Optional.
intermediate: Required if showback/chargeback is used. Define keys, allocation, disputes, telemetry.
advanced: Required if showback/chargeback is used. Add examples and strict auditability and approval SLAs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, allocation examples, audience, sla_days, optional metric, thresholds, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If mode.value is UNKNOWN → block Completeness Gate (when used).
If keys.list is UNKNOWN → block Completeness Gate (when used).
If alloc.shared_rule is UNKNOWN → block Completeness Gate (when used).
If telemetry.unattributed_cost_metric is UNKNOWN → block Completeness Gate (when used).

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions (when used):
required_fields_present == true
mode_and_keys_defined == true
allocation_defined == true
reporting_and_disputes_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-08

COST-08 — Cost Attribution Tags Standard (required tags)
Header Block
template_id: COST-08
title: Cost Attribution Tags Standard (required tags)
type: cost_attribution_tags_standard
template_version: 1.0.0
output_path: 10_ops/cost/COST-08_Cost_Attribution_Tags_Standard.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["IAC-03", "COST-01", "COMP-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IAC-03", "COST-01", "COMP-02", "COST-09", "CICD-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical tagging/labeling standard required for cost attribution across infrastructure and services: required tags, naming conventions, enforcement, and validation. This template enables showback, budgeting, and driver analysis.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
IaC module/stack standards: {{xref:IAC-03}} | OPTIONAL
Cost model overview: {{xref:COST-01}} | OPTIONAL
Control matrix (governance): {{xref:COMP-02}} | OPTIONAL
Cost reporting: {{xref:COST-09}} | OPTIONAL
Pipeline overview: {{xref:CICD-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Required tags list (key names)
Tag value conventions (allowed patterns)
Minimum tag set per resource type (compute/storage)
Ownership tags (team_id/service_id)
Environment tag rule (dev/stage/prod)
Tenant/customer tag rule (if multi-tenant)
Enforcement rule (IaC validation, CI gate)
Exception process (temporary missing tags)
Telemetry requirements (untagged cost %)

Optional Fields
Provider-specific notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Resources without required tags should fail CI or be flagged for remediation.
Tag keys must be stable; do not rename without migration plan.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Required Tags
   tags: {{tags.required}}
2. Value Conventions
   patterns: {{tags.patterns}}
   examples: {{tags.examples}} | OPTIONAL
3. Per-Resource Minimums
   resource_minimums: {{tags.resource_minimums}}
4. Ownership
   ownership_tags: {{tags.ownership_tags}}
5. Environment
   env_tag_rule: {{env.rule}}
6. Tenant/Customer
   tenant_tag_rule: {{tenant.rule}}
7. Enforcement
   enforcement_rule: {{enforce.rule}}
   ci_gate_ref: {{enforce.ci_gate_ref}} | OPTIONAL
8. Exceptions
   exception_rule: {{exceptions.rule}}
   exceptions_ref: {{xref:COMP-08}} | OPTIONAL
9. Telemetry
   untagged_cost_metric: {{telemetry.untagged_cost_metric}}

Cross-References
Upstream: {{xref:IAC-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-09}}, {{xref:COST-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define required tags, env tag rule, enforcement rule, telemetry metric.
intermediate: Required. Define resource minimums, conventions, exceptions.
advanced: Required. Add provider notes and strict migration guidance for tag schema changes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, examples, ci gate ref, provider notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If tags.required is UNKNOWN → block Completeness Gate.
If tags.resource_minimums is UNKNOWN → block Completeness Gate.
If enforce.rule is UNKNOWN → block Completeness Gate.
If telemetry.untagged_cost_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
required_tags_defined == true
resource_minimums_defined == true
enforcement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-09

COST-09 — Cost Reporting (dashboards, cadence)
Header Block
template_id: COST-09
title: Cost Reporting (dashboards, cadence)
type: cost_reporting
template_version: 1.0.0
output_path: 10_ops/cost/COST-09_Cost_Reporting.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["COST-01", "COST-08", "COST-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COST-01", "COST-08", "COST-04", "COST-02", "OBS-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical cost reporting outputs: dashboards, periodic reports, and what dimensions are required (team/service/tenant). This template makes cost visible and actionable and supports budget governance.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Cost model overview: {{xref:COST-01}} | OPTIONAL
Attribution tags standard: {{xref:COST-08}} | OPTIONAL
Budgets/alerts: {{xref:COST-04}} | OPTIONAL
Cost drivers catalog: {{xref:COST-02}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Report types (dashboard, weekly summary, monthly report)
Cadence per report type
Audience/distribution
Required dimensions (team/service/tenant/env)
Dashboard references (dashboard_id list)
Alert linkage (budget alerts)
Action items rule (what happens when costs spike)
Ownership (who publishes)
Telemetry requirements (reports published, late reports)

Optional Fields
Executive summary format | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Reporting must align to tag schema; if tags missing, report must show coverage gaps.
Reports must be tied to action rules (opt backlog, budget actions).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Report Types
   types: {{report.types}}
2. Cadence
   cadence: {{report.cadence}}
3. Audience
   audience: {{audience.list}}
4. Dimensions
   dimensions: {{dims.list}}
5. Dashboards
   dashboard_ids: {{dashboards.ids}}
6. Alerts & Actions
   budget_alert_ref: {{xref:COST-04}} | OPTIONAL
   action_rule: {{actions.rule}}
7. Ownership
   owner: {{owner.team}}
8. Telemetry
   reports_published_metric: {{telemetry.published_metric}}
   late_report_metric: {{telemetry.late_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:COST-08}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-06}}, {{xref:COST-10}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define types, cadence, dims, dashboards, owner.
intermediate: Required. Define action rule and telemetry metrics.
advanced: Required. Add exec summary format and strict coverage gap reporting + accountability.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, exec summary, optional metric, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If report.types is UNKNOWN → block Completeness Gate.
If report.cadence is UNKNOWN → block Completeness Gate.

If dims.list is UNKNOWN → block Completeness Gate.
If dashboards.ids is UNKNOWN → block Completeness Gate.
If telemetry.published_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
reporting_defined == true
cadence_and_dimensions_defined == true
dashboard_linkage_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

COST-10

COST-10 — Incident Cost Review (post-incident spend)
Header Block
template_id: COST-10
title: Incident Cost Review (post-incident spend)
type: incident_cost_review
template_version: 1.0.0
output_path: 10_ops/cost/COST-10_Incident_Cost_Review.md
compliance_gate_id: TMP-05.PRIMARY.COST
upstream_dependencies: ["IRP-06", "COST-09", "COST-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "IRP-06", "COST-09", "COST-02", "COST-04", "AUDIT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical post-incident cost review process: how to quantify incremental spend during incidents (scale-ups, egress spikes, vendor overages), how to record it, and how to feed it into optimization/backlog and governance.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Postmortem template: {{xref:IRP-06}} | OPTIONAL
Cost reporting: {{xref:COST-09}} | OPTIONAL
Cost drivers catalog: {{xref:COST-02}} | OPTIONAL
Budgets/alerts: {{xref:COST-04}} | OPTIONAL
Privileged change audit: {{xref:AUDIT-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Incident reference (incident_id)
Time window for cost analysis (start/end)
Cost deltas by driver (driver_id list)
Primary causes (what changed)
Actions taken (scale up, failover)
Avoidable vs unavoidable classification
Follow-up actions (optimization backlog items)
Owner (finops/incident lead)
Telemetry requirements (incident cost deltas tracked)

Optional Fields
Customer credits linkage | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Cost review must be evidence-based (billing + tags) and linked to incident timeline.
Follow-ups must be created when avoidable deltas are found.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Reference
   incident_id: {{ref.incident_id}}
   postmortem_ref: {{xref:IRP-06}} | OPTIONAL
2. Window
   start: {{window.start}}
   end: {{window.end}}
3. Deltas
   drivers: {{deltas.drivers}}
   total_delta: {{deltas.total}} | OPTIONAL
4. Causes
   causes: {{causes.list}}
5. Actions Taken
   actions: {{actions.list}}
6. Classification
   avoidable_rule: {{class.avoidable_rule}}
   classification: {{class.classification}} | OPTIONAL
7. Follow-Ups
   backlog_items: {{followups.backlog_items}}
   owner: {{followups.owner}} | OPTIONAL
8. Telemetry
   incident_cost_review_metric: {{telemetry.review_metric}}

Cross-References
Upstream: {{xref:IRP-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:COST-06}}, {{xref:COST-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Fill incident ref, window, driver deltas, causes, telemetry.
intermediate: Required. Add avoidable classification, follow-ups, ownership.
advanced: Required. Add customer credits linkage and stricter evidence linkage + prevention actions.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, total delta, classification optional, followups owner, customer credits, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ref.incident_id is UNKNOWN → block Completeness Gate.
If window.start or window.end is UNKNOWN → block Completeness Gate.
If deltas.drivers is UNKNOWN → block Completeness Gate.
If causes.list is UNKNOWN → block Completeness Gate.
If telemetry.review_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.COST
Pass conditions:
required_fields_present == true
reference_and_window_defined == true
deltas_defined == true
followups_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Performance Budgets & Profiling (PBP)

PBP-01

PBP-01 — Performance Budgets Overview (web/mobile/api)
Header Block
template_id: PBP-01
title: Performance Budgets Overview (web/mobile/api)
type: performance_budgets_overview
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-01_Performance_Budgets_Overview.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PERF-01", "PERF-03", "CPR-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-01", "PERF-03", "CPR-01", "PBP-02", "PBP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the single, canonical overview of performance budgets across web, mobile, and API surfaces: what budgets exist, how they are measured, and how budgets are enforced via gates. This document anchors the PBP set.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Performance overview: {{xref:PERF-01}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Client performance budget: {{xref:CPR-01}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Perf CI gates: {{xref:PBP-06}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Budget types in scope (web, mobile, api, bundle)
What budgets control (LCP, TTI, startup time, p95 latency)
Measurement sources (RUM, synthetic, CI)
Ownership model (who owns budgets)
Budget enforcement model (warn vs block)
Change control rule (who can adjust budgets)
Evidence artifact rule (store reports)
Telemetry requirements (budget violations, drift)

Optional Fields
MVP budgets vs later | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must be measurable and mapped to real surfaces (screens/endpoints).
Budget changes must be reviewed and time-bound when used as waivers.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Budget Types
   types: {{types.list}}
2. Controls
   metrics_controlled: {{controls.metrics}}
3. Measurement
   sources: {{measure.sources}}
   collection_rule: {{measure.collection_rule}} | OPTIONAL
4. Ownership
   owner_rule: {{owners.rule}}
   owners: {{owners.list}} | OPTIONAL
5. Enforcement
   enforcement_rule: {{enforce.rule}}
   block_vs_warn_rule: {{enforce.block_vs_warn_rule}} | OPTIONAL
6. Change Control
   who_can_change: {{change.who_can_change}}
   approval_rule: {{change.approval_rule}} | OPTIONAL
7. Evidence
   artifacts: {{evidence.artifacts}}
   storage_location: {{evidence.storage_location}} | OPTIONAL
8. Telemetry
   budget_violation_metric: {{telemetry.violation_metric}}
   budget_drift_metric: {{telemetry.drift_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-02}}, {{xref:PBP-06}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define types, metrics controlled, enforcement rule, evidence artifacts, violation metric.
intermediate: Required. Define ownership and change control and measurement sources.
advanced: Required. Add MVP vs later and strict governance for budget changes + drift monitoring.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, collection rule, owners list, block vs warn

rule, approval rule, storage location, optional metrics, MVP budgets, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If types.list is UNKNOWN → block Completeness Gate.
If controls.metrics is UNKNOWN → block Completeness Gate.
If enforce.rule is UNKNOWN → block Completeness Gate.
If evidence.artifacts is UNKNOWN → block Completeness Gate.
If telemetry.violation_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
budget_types_defined == true
measurement_defined == true
enforcement_defined == true
evidence_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-02

PBP-02 — Budget Registry (by budget_id)
Header Block
template_id: PBP-02
title: Budget Registry (by budget_id)
type: performance_budget_registry
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-02_Budget_Registry.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PERF-03", "PERF-02", "PBP-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-01", "PERF-03", "PERF-02", "API-01", "FE-01", "MOB-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical registry of performance budgets, indexed by budget_id, including what surface each budget applies to, what metric is constrained, and what threshold applies. This registry is the source of truth for performance budget enforcement.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Performance budgets overview: {{xref:PBP-01}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Critical paths: {{xref:PERF-02}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Route map/layout: {{xref:FE-01}} | OPTIONAL
Mobile nav map: {{xref:MOB-01}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Budget registry (budget_id list)
budget_id (stable identifier)
Surface type (web_screen/mobile_screen/api_endpoint/bundle)
Surface selector (screen_id/route_id/endpoint_id)
Metric name (LCP/TTI/startup/p95_latency)
Threshold value (ms/size)
Measurement source (CI/RUM/synthetic)
Window/aggregation (p95/median)
Fail behavior (block/warn)
Owner (team)
Telemetry requirements (budget breach metric)

Optional Fields
Critical path linkage (path_id) | OPTIONAL
Notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must map to real surface IDs; do not invent endpoint/screen IDs.
Threshold units must be explicit and consistent (ms, kb).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_budgets: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Budgets (repeat)
   Budget
   budget_id: {{budgets[0].budget_id}}
   surface_type: {{budgets[0].surface_type}}
   surface_selector: {{budgets[0].surface_selector}}
   metric: {{budgets[0].metric}}
   threshold: {{budgets[0].threshold}}
   measurement_source: {{budgets[0].measurement_source}}
   aggregation: {{budgets[0].aggregation}}
   fail_behavior: {{budgets[0].fail_behavior}}
   owner: {{budgets[0].owner}}
   critical_path_id: {{budgets[0].critical_path_id}} | OPTIONAL
   notes: {{budgets[0].notes}} | OPTIONAL
   telemetry_metric: {{budgets[0].telemetry_metric}}
   open_questions:
   {{budgets[0].open_questions[0]}} | OPTIONAL

(Repeat per budget.)

3. References
   Regression gates: {{xref:PBP-06}} | OPTIONAL
   Reporting: {{xref:PBP-07}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-06}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define budgets with selector, metric, threshold, and fail behavior.
intermediate: Required. Define measurement source, aggregation, ownership, telemetry.
advanced: Required. Add critical path linkage and strict unit consistency + review rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, meta notes, critical path id, notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If budgets[*].budget_id is UNKNOWN → block Completeness Gate.
If budgets[].surface_selector is UNKNOWN → block Completeness Gate.
If budgets[*].metric is UNKNOWN → block Completeness Gate.
If budgets[].threshold is UNKNOWN → block Completeness Gate.
If budgets[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
registry_defined == true
thresholds_defined == true
measurement_defined == true
ownership_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-03

PBP-03 — Web Budgets (Core Web Vitals)
Header Block
template_id: PBP-03
title: Web Budgets (Core Web Vitals)
type: web_performance_budgets
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-03_Web_Budgets.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PBP-02", "CPR-01", "FE-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-02", "CPR-01", "FE-01", "PERF-02", "PERF-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical web performance budgets, including Core Web Vitals and related metrics, and how they map to pages/routes and CI/RUM measurement. This template is the detailed rulebook for web budgets referenced by the registry.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Client perf budget: {{xref:CPR-01}} | OPTIONAL
Route map/layout: {{xref:FE-01}} | OPTIONAL
Critical paths: {{xref:PERF-02}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Metrics included (LCP/INP/CLS/TTFB)
Budget thresholds per metric (ms/score)
Route/page mapping rule (route_id/page_id)
Measurement sources (RUM vs lab)
Device/network assumptions (desktop/mobile)
Aggregation rules (p75/p95)
Fail behavior mapping (warn/block)
Ownership (who owns web budgets)
Telemetry requirements (CWV regressions detected)

Optional Fields
Third-party script budget rule | OPTIONAL

Image/font budget rule | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must be expressed with explicit percentiles and environments.
If RUM and lab differ, define the authoritative metric per gate.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Metrics
   metrics: {{metrics.list}}
2. Thresholds
   thresholds: {{thresholds.by_metric}}
3. Mapping
   route_mapping_rule: {{map.route_rule}}
   page_list: {{map.pages}} | OPTIONAL
4. Measurement
   rum_rule: {{measure.rum_rule}}
   lab_rule: {{measure.lab_rule}} | OPTIONAL
5. Assumptions
   device_rule: {{assume.device_rule}}
   network_rule: {{assume.network_rule}} | OPTIONAL
6. Aggregation
   aggregation: {{agg.rule}}
7. Fail Behavior
   fail_map: {{fail.map}}
8. Ownership
   owner: {{owner.team}}
9. Telemetry
   cwv_regression_metric: {{telemetry.cwv_regression_metric}}
   inp_regression_metric: {{telemetry.inp_regression_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-06}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define metrics, thresholds, aggregation, fail map.
intermediate: Required. Define mapping, measurement sources, assumptions, telemetry.
advanced: Required. Add third-party and asset budget rules and strict RUM vs lab authority rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, page list, lab rule, network rule, optional telemetry, third-party/assets rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If metrics.list is UNKNOWN → block Completeness Gate.
If thresholds.by_metric is UNKNOWN → block Completeness Gate.
If map.route_rule is UNKNOWN → block Completeness Gate.
If fail.map is UNKNOWN → block Completeness Gate.
If telemetry.cwv_regression_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
metrics_and_thresholds_defined == true
mapping_defined == true
measurement_defined == true
fail_behavior_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-04

PBP-04 — API Budgets (latency/error budgets per endpoint class)
Header Block
template_id: PBP-04
title: API Budgets (latency/error budgets per endpoint class)
type: api_performance_budgets
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-04_API_Budgets.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PERF-03", "API-01", "SLO-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-03", "API-01", "API-02", "SLO-06", "PFS-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical API performance budgets by endpoint class (read/write/auth/admin), including latency and error-rate thresholds, measurement windows, and how budgets are enforced in CI and in production monitoring.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Latency budgets: {{xref:PERF-03}} | OPTIONAL
Endpoint catalog: {{xref:API-01}} | OPTIONAL
Endpoint specs: {{xref:API-02}} | OPTIONAL
Measurement rules: {{xref:SLO-06}} | OPTIONAL
Performance constraints: {{xref:PFS-05}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Endpoint classes (class_id list)
Class definitions (what endpoints included)
Latency budgets per class (p95 ms)
Error-rate budgets per class (%/rate)
Window rules (5m/30d)
Exclusion rules (what doesn’t count)
Measurement source (server metrics, APM)
Enforcement rule (warn/block)
Ownership (api/platform team)
Telemetry requirements (budget breaches by class)

Optional Fields
Per-endpoint overrides policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must be expressed with explicit percentile and window.
Class membership must be deterministic (endpoint tags/IDs).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Classes
   classes: {{classes.list}}
2. Definitions
   definitions: {{classes.definitions}}
3. Latency Budgets
   latency_budgets: {{budgets.latency_by_class}}
4. Error Budgets
   error_budgets: {{budgets.error_by_class}}
5. Windows
   windows: {{windows.rule}}
6. Exclusions
   exclusions: {{exclude.rule}}
7. Measurement
   metric_source_rule: {{measure.source_rule}}
8. Enforcement
   enforcement_rule: {{enforce.rule}}
9. Ownership
   owner: {{owner.team}}
10. Telemetry
    api_budget_breach_metric: {{telemetry.breach_metric}}
    error_budget_breach_metric: {{telemetry.error_breach_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PERF-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-06}}, {{xref:ALRT-02}}, {{xref:SLO-03}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define classes, latency/error budgets, windows, enforcement.
intermediate: Required. Define exclusions, measurement source, telemetry.
advanced: Required. Add per-endpoint override policy and strict deterministic class membership rules + audits.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional telemetry, overrides policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If classes.list is UNKNOWN → block Completeness Gate.
If budgets.latency_by_class is UNKNOWN → block Completeness Gate.
If windows.rule is UNKNOWN → block Completeness Gate.
If enforce.rule is UNKNOWN → block Completeness Gate.
If telemetry.breach_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
classes_defined == true
budgets_defined == true
measurement_defined == true
enforcement_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-05

PBP-05 — Profiling Plan (what to measure, tools)
Header Block
template_id: PBP-05
title: Profiling Plan (what to measure, tools)
type: profiling_plan
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-05_Profiling_Plan.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PERF-02", "PERF-10", "OBS-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PERF-02", "PERF-10", "OBS-04", "PBP-02", "LOAD-08"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical profiling plan: what performance dimensions are profiled (CPU, memory, render, DB), what tools are used, when profiling is required, and how profiling results are captured as evidence. This plan supports budget enforcement and bottleneck remediation.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Critical paths: {{xref:PERF-02}} | OPTIONAL
Performance runbooks: {{xref:PERF-10}} | OPTIONAL
Trace instrumentation: {{xref:OBS-04}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Bottleneck analysis template: {{xref:LOAD-08}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Profiling targets (client/server/db)
Dimensions measured (CPU, memory, render, IO)
Tooling per target (profiler/APM/devtools)
When to profile (budget breach, before release)
Sampling strategy (how often)
Data capture rules (what artifacts stored)
Baseline comparison rule (before/after)
Ownership (who runs profiling)
Telemetry requirements (profiling runs completed)

Optional Fields
Flamegraph storage rule | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Profiling outputs must be captured as artifacts and tied to incidents/breaches when applicable.
Avoid profiling with production PII in captured payloads; follow redaction constraints.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Targets
   targets: {{targets.list}}
2. Dimensions
   dimensions: {{dims.list}}
3. Tooling
   tooling: {{tools.by_target}}
4. Triggers
   profile_when: {{triggers.when}}
5. Sampling
   sampling_rule: {{sampling.rule}}
6. Capture
   artifacts: {{capture.artifacts}}
   storage_location: {{capture.storage_location}} | OPTIONAL
7. Baselines
   baseline_rule: {{baseline.rule}}
8. Ownership
   owner: {{owner.team}}
9. Telemetry
   profiling_runs_metric: {{telemetry.profiling_runs_metric}}
   profiling_completion_metric: {{telemetry.completion_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-06}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define targets, dimensions, triggers, capture artifacts, baseline rule.
intermediate: Required. Define tooling and sampling rule and telemetry.
advanced: Required. Add flamegraph storage and strict evidence linking to breach tickets/runbooks.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, storage location, flamegraph storage, optional metrics, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

If targets.list is UNKNOWN → block Completeness Gate.
If dims.list is UNKNOWN → block Completeness Gate.
If triggers.when is UNKNOWN → block Completeness Gate.
If capture.artifacts is UNKNOWN → block Completeness Gate.
If telemetry.profiling_runs_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
profiling_plan_defined == true
capture_defined == true
baseline_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-06

PBP-06 — Perf CI Gates (bundle size, CWV, API p95)
Header Block
template_id: PBP-06
title: Perf CI Gates (bundle size, CWV, API p95)
type: performance_ci_gates
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-06_Perf_CI_Gates.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PBP-02", "PERF-09", "CICD-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-02", "PBP-03", "PBP-04", "PERF-09", "CICD-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical performance CI gating configuration: which budget checks run in CI, what thresholds are enforced, how baselines are compared, and how failures are reported and stored. This template is the CI-facing enforcement of the PBP budget registry.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Web budgets: {{xref:PBP-03}} | OPTIONAL
API budgets: {{xref:PBP-04}} | OPTIONAL
Perf regression gates: {{xref:PERF-09}} | OPTIONAL
CI pipeline: {{xref:CICD-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
CI gate registry (gate_id list)
gate_id (stable identifier)
Gate type (bundle_size/cwv/api_latency/startup)
Surface binding (route_id/endpoint_class)
Metrics checked (size, LCP, INP, p95)
Thresholds (from PBP registry)
Baseline comparison rule (vs main)
Fail behavior (block/warn)
Artifacts produced (reports, diffs)
Waiver rule (time-bound)
Telemetry requirements (gate failures, flake rate)

Optional Fields
Canary/perf lab integration | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
CI gates must be deterministic and reproducible.
Thresholds must be sourced from PBP registry; do not hardcode mismatched values.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Registry Summary
   total_ci_gates: {{meta.total}}
   notes: {{meta.notes}} | OPTIONAL
2. Gates (repeat)
   CI Gate
   gate_id: {{gates[0].gate_id}}
   type: {{gates[0].type}}
   binding: {{gates[0].binding}}
   metrics: {{gates[0].metrics}}
   thresholds: {{gates[0].thresholds}}
   baseline_rule: {{gates[0].baseline_rule}}
   fail_behavior: {{gates[0].fail_behavior}}
   artifacts: {{gates[0].artifacts}}
   waiver_rule: {{gates[0].waiver_rule}}
   telemetry_metric: {{gates[0].telemetry_metric}}
   flake_metric: {{gates[0].flake_metric}} | OPTIONAL
   canary_integration: {{gates[0].canary_integration}} | OPTIONAL
   open_questions:
   {{gates[0].open_questions[0]}} | OPTIONAL

(Repeat per gate.)

3. References
   Budget registry: {{xref:PBP-02}} | OPTIONAL
   Quality gates: {{xref:QA-07}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:REL-06}}, {{xref:QA-07}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define CI gates, thresholds, baseline rule, fail behavior, artifacts.

intermediate: Required. Define waiver rule and telemetry metrics.
advanced: Required. Add canary/lab integration and strict flake governance + quarantine rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, flake metric, canary integration, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If gates[*].gate_id is UNKNOWN → block Completeness Gate.*
*If gates[*].thresholds is UNKNOWN → block Completeness Gate.
If gates[*].baseline_rule is UNKNOWN → block Completeness Gate.*
*If gates[*].fail_behavior is UNKNOWN → block Completeness Gate.
If gates[*].artifacts is UNKNOWN → block Completeness Gate.*
*If gates[*].telemetry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
ci_gate_registry_defined == true
thresholds_defined == true
baseline_defined == true
artifacts_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-07

PBP-07 — Budget Reporting (dashboards, trends)
Header Block
template_id: PBP-07
title: Budget Reporting (dashboards, trends)
type: performance_budget_reporting
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-07_Budget_Reporting.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PBP-02", "OBS-07", "ANL-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-02", "OBS-07", "OBS-03", "ANL-07", "PERF-09"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical reporting for performance budgets: dashboards, periodic summaries, trend views, and ownership. This template ensures budgets are monitored continuously (not only in CI) and ties budget violations to actions.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Dashboards inventory: {{xref:OBS-07}} | OPTIONAL
Metrics catalog: {{xref:OBS-03}} | OPTIONAL
Funnels/KPIs (priority weighting): {{xref:ANL-07}} | OPTIONAL
Perf gates: {{xref:PERF-09}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Report types (dashboard, weekly summary)
Cadence per report type
Audience/distribution
Required dimensions (route/screen/endpoint_class)
Dashboard references (dashboard_id list)
Trend window (7d/30d)
Action rules (what happens on repeated breaches)
Ownership (who publishes/reviews)
Telemetry requirements (reports published, breaches tracked)

Optional Fields
Executive summary format | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Reporting must reference the budget registry and use consistent IDs.
Repeated violations must create actions (tickets/backlog).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Report Types
   types: {{report.types}}
2. Cadence
   cadence: {{report.cadence}}
3. Audience
   audience: {{audience.list}}
4. Dimensions
   dimensions: {{dims.list}}
5. Dashboards
   dashboard_ids: {{dashboards.ids}}
6. Trends
   trend_window: {{trend.window}}
   trend_rule: {{trend.rule}} | OPTIONAL
7. Actions
   action_rule: {{actions.rule}}
8. Ownership
   owner: {{owner.team}}
9. Telemetry
   reports_published_metric: {{telemetry.published_metric}}
   budget_breach_tracked_metric: {{telemetry.breach_tracked_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PERF-10}}, {{xref:REL-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define types, cadence, dashboards, trend window, action rule.
intermediate: Required. Define audience/dimensions and telemetry metrics.
advanced: Required. Add exec summary format and strict breach-to-ticket automation linkage.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, trend rule, optional metrics, exec summary, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If report.types is UNKNOWN → block Completeness Gate.

If report.cadence is UNKNOWN → block Completeness Gate.
If dashboards.ids is UNKNOWN → block Completeness Gate.
If trend.window is UNKNOWN → block Completeness Gate.
If actions.rule is UNKNOWN → block Completeness Gate.
If telemetry.published_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
reporting_defined == true
cadence_defined == true
dashboard_linkage_defined == true
actions_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-08

PBP-08 — Budget Exceptions & Waivers (temporary overrides)
Header Block
template_id: PBP-08
title: Budget Exceptions & Waivers (temporary overrides)
type: performance_budget_exceptions_waivers
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-08_Budget_Exceptions_Waivers.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["COMP-08", "PBP-06", "REL-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "COMP-08", "PBP-06", "PBP-02", "REL-02", "AUDIT-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical process for temporary performance budget exceptions/waivers: when and how budgets can be temporarily exceeded, how waivers are approved, how they expire, and how they are audited and remediated.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Exceptions/waivers framework: {{xref:COMP-08}} | OPTIONAL
Perf CI gates: {{xref:PBP-06}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Change management workflow: {{xref:REL-02}} | OPTIONAL
Privileged change audit: {{xref:AUDIT-07}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Waiver types (temporary breach, measurement gap)
Request required fields (budget_id, reason, duration)
Approver roles (perf owner, release owner)
Max duration rule
Expiry/renewal rule
Compensating actions rule (ticket required)
Audit logging rule (who approved)
Reversion rule (auto remove waiver)
Telemetry requirements (active waivers, expired)

Optional Fields
Emergency waiver rule (incident mode) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
No waiver without expiry and a remediation plan.
Waivers must reference budget_id from PBP-02 and be time-bounded.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Waiver Types
   types: {{types.list}}
2. Request
   required_fields: {{request.fields}}
3. Approvals
   approver_roles: {{approve.roles}}
   workflow_ref: {{xref:REL-02}} | OPTIONAL
4. Duration
   max_days: {{duration.max_days}}
   renewal_rule: {{duration.renewal_rule}} | OPTIONAL
5. Compensating Actions
   action_rule: {{actions.rule}}
   ticket_ref_rule: {{actions.ticket_ref_rule}} | OPTIONAL
6. Audit
   audit_required: {{audit.required}}
   audit_ref: {{xref:AUDIT-07}} | OPTIONAL
7. Reversion
   revert_rule: {{revert.rule}}
8. Telemetry
   active_waivers_metric: {{telemetry.active_metric}}
   expired_waivers_metric: {{telemetry.expired_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-06}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PERF-09}}, {{xref:REL-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define types, required fields, max days, revert rule.
intermediate: Required. Define approvals, compensating actions, audit, telemetry.
advanced: Required. Add emergency waiver rule and stricter renewal governance + breach tracking.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, workflow ref, renewal rule, audit ref, optional metric, emergency waiver, open_questions
If any Required Field is UNKNOWN, allow only if:

{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If types.list is UNKNOWN → block Completeness Gate.
If request.fields is UNKNOWN → block Completeness Gate.
If duration.max_days is UNKNOWN → block Completeness Gate.
If revert.rule is UNKNOWN → block Completeness Gate.
If telemetry.active_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
waiver_process_defined == true
expiry_and_reversion_defined == true
audit_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-09

PBP-09 — Bundle Size & Asset Budgets (web/mobile)
Header Block
template_id: PBP-09
title: Bundle Size & Asset Budgets (web/mobile)
type: bundle_size_asset_budgets
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-09_Bundle_Size_Asset_Budgets.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["PBP-02", "CPR-03", "MOB-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "PBP-02", "CPR-03", "MOB-05", "PERF-09", "CICD-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical bundle size and asset budgets for web and mobile: JS bundle limits, chunking expectations, image/font budgets, and mobile binary size budgets. This template supports CI gating and prevents performance regressions from asset bloat.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Asset loading strategy: {{xref:CPR-03}} | OPTIONAL
Mobile release/build rules: {{xref:MOB-05}} | OPTIONAL
Perf regression gates: {{xref:PERF-09}} | OPTIONAL
CI pipeline: {{xref:CICD-03}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Budget categories (js_bundle, css, images, fonts, mobile_binary)
Thresholds per category (kb/mb)
Measurement method (build output, analyzer)
Scope mapping (global vs per-route)
Chunking/code-splitting rule
Third-party dependency budget rule
Fail behavior (warn/block)
Ownership (frontend/mobile)
Telemetry requirements (bundle regressions count)

Optional Fields
Asset optimization rules (compression) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
Budgets must use consistent units and measurement method.
Budget thresholds should be enforced in CI via PBP-06 gates.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Categories
   categories: {{cats.list}}
2. Thresholds
   thresholds: {{thresholds.by_category}}
3. Measurement
   method: {{measure.method}}
   tooling: {{measure.tooling}} | OPTIONAL
4. Scope
   scope_rule: {{scope.rule}}
5. Chunking
   chunking_rule: {{chunk.rule}}
6. Third-Party
   third_party_rule: {{third.rule}}
7. Fail Behavior
   fail_behavior: {{fail.behavior}}
8. Ownership
   owner: {{owner.team}}
9. Telemetry
   bundle_regression_metric: {{telemetry.bundle_regression_metric}}

Cross-References
Upstream: {{xref:PBP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-06}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define categories, thresholds, measurement method, fail behavior.
intermediate: Required. Define scope rule, chunking rule, telemetry.
advanced: Required. Add third-party dependency governance and asset optimization rules + stricter CI enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, tooling, asset optimization rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If cats.list is UNKNOWN → block Completeness Gate.

If thresholds.by_category is UNKNOWN → block Completeness Gate.
If measure.method is UNKNOWN → block Completeness Gate.
If fail.behavior is UNKNOWN → block Completeness Gate.
If telemetry.bundle_regression_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
categories_defined == true
thresholds_defined == true
measurement_defined == true
fail_behavior_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

PBP-10

PBP-10 — Budget Change Control (who can change thresholds)
Header Block
template_id: PBP-10
title: Budget Change Control (who can change thresholds)
type: performance_budget_change_control
template_version: 1.0.0
output_path: 10_ops/perf_budgets/PBP-10_Budget_Change_Control.md
compliance_gate_id: TMP-05.PRIMARY.PBP
upstream_dependencies: ["REL-02", "PBP-02", "AUDIT-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "REL-02", "PBP-02", "PBP-06", "AUDIT-07", "COMP-08"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical governance and change control for performance budgets: who can change thresholds, how changes are reviewed, how changes are tested/validated, and how changes are audited. This prevents “quiet” budget inflation that hides regressions.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
Change management workflow: {{xref:REL-02}} | OPTIONAL
Budget registry: {{xref:PBP-02}} | OPTIONAL
Perf CI gates: {{xref:PBP-06}} | OPTIONAL
Privileged change audit: {{xref:AUDIT-07}} | OPTIONAL
Exception management: {{xref:COMP-08}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Roles allowed to edit budgets
Review/approval requirements (PR/RFC)
Validation requirement (run perf checks)
Rollback rule (restore prior thresholds)
Audit logging rule (who changed what, why)
Emergency change process (incident mode)
Change window rules (release windows)
Telemetry requirements (budget changes count, drift)
Exception/waiver linkage rule (PBP-08)

Optional Fields
Budget-as-code notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-INCIDENT-OPS]}} | OPTIONAL
Budget changes must be reviewed and reversible.
Emergency changes must be audited and followed by post-incident review.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.

Output Format

1. Roles
   edit_roles: {{roles.edit_roles}}
   approve_roles: {{roles.approve_roles}} | OPTIONAL
2. Review
   workflow_ref: {{xref:REL-02}} | OPTIONAL
   review_rule: {{review.rule}}
3. Validation
   validation_rule: {{validate.rule}}
   ci_gate_ref: {{xref:PBP-06}} | OPTIONAL
4. Rollback
   rollback_rule: {{rollback.rule}}
5. Audit
   audit_required: {{audit.required}}
   audit_ref: {{xref:AUDIT-07}} | OPTIONAL
6. Emergency
   emergency_rule: {{emergency.rule}}
   post_review_rule: {{emergency.post_review_rule}} | OPTIONAL
7. Windows
   window_rule: {{windows.rule}}
8. Exceptions
   waiver_link_rule: {{exceptions.waiver_link_rule}}
   waiver_ref: {{xref:PBP-08}} | OPTIONAL
9. Telemetry
   budget_change_metric: {{telemetry.budget_change_metric}}
   budget_drift_metric: {{telemetry.budget_drift_metric}} | OPTIONAL

Cross-References
Upstream: {{xref:PBP-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:PBP-08}}, {{xref:PERF-09}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define edit roles, review rule, validation rule, rollback rule.
intermediate: Required. Define emergency, audit, windows, telemetry.
advanced: Required. Add budget-as-code notes and strict approval matrices + drift enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, approve roles, workflow ref, ci gate ref, audit ref, post review rule, optional metric, budget-as-code, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If roles.edit_roles is UNKNOWN → block Completeness Gate.
If review.rule is UNKNOWN → block Completeness Gate.
If validate.rule is UNKNOWN → block Completeness Gate.
If rollback.rule is UNKNOWN → block Completeness Gate.
If telemetry.budget_change_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.PBP
Pass conditions:
required_fields_present == true
roles_and_review_defined == true
validation_defined == true
rollback_defined == true
audit_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true